



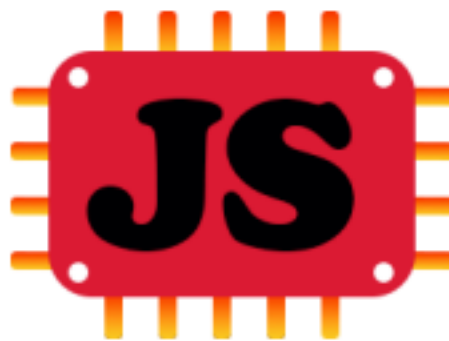
Electronics (BJT)

Lecture 10

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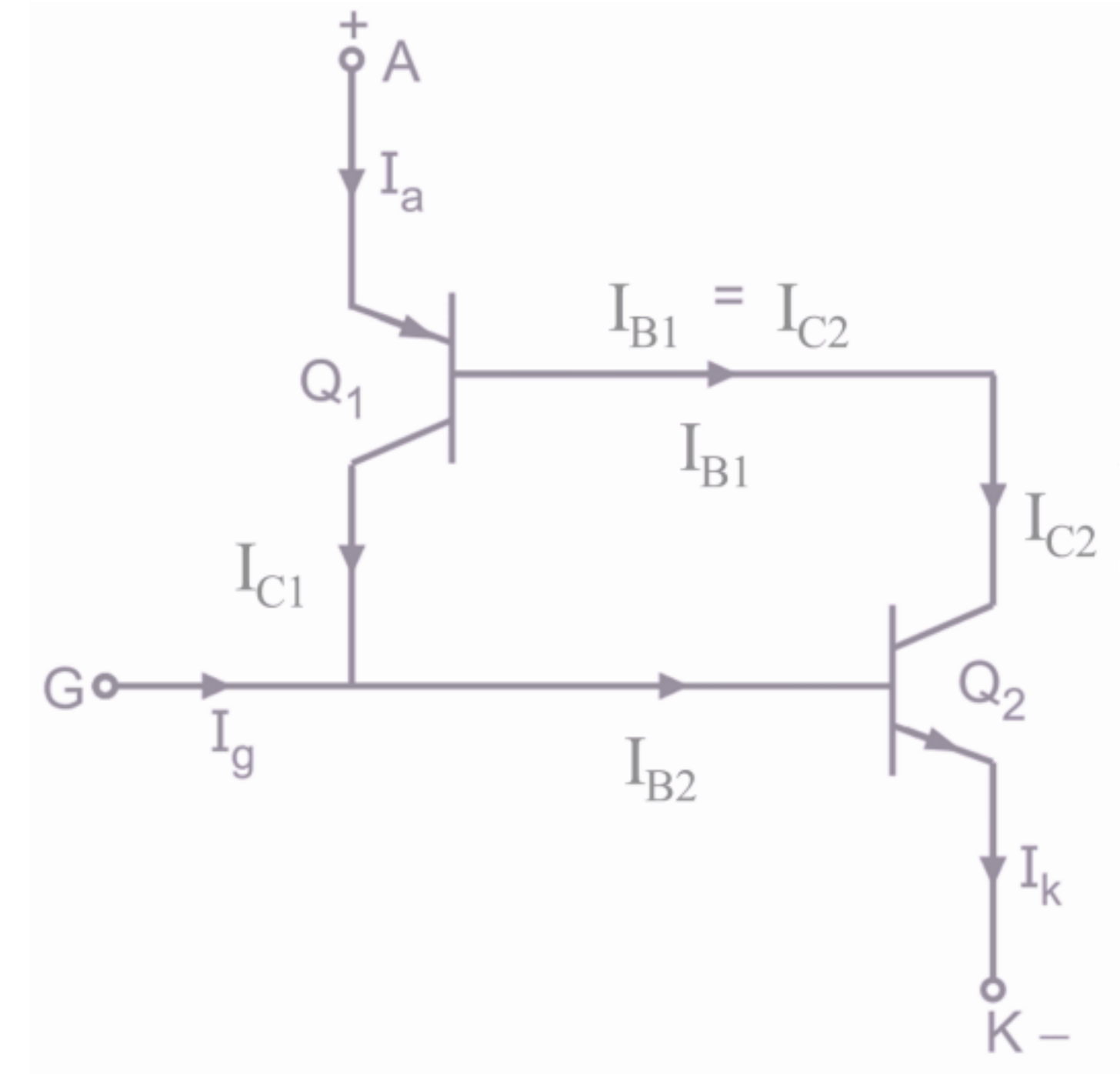


Circuit analysis with 2 transistors



Steps:

1. Assume circuit parameters
2. Assess which transistor is on
3. Assume state of the on-transistor (active/saturation)
4. Solve and evaluate if the assumed state is correct



Evaluate all node voltages and branch currents ($\beta = 100$)

- Q_1 : $V_{BE} = 4 - 0 = 4 > 0.7 \therefore \text{ON} \rightarrow V_{BC} = 4 - 5 < 0.4$ (active)
 - Q_2 : $V_{EB} = 0 - 4 < 0.7 \therefore \text{OFF}$
- *** $(4.62 - 5 < 0.4) \therefore \text{connect}$

KVL from (a) $\rightarrow$ (b): $5 - 0 = 10I_B + V_{BE} + 1 \cdot I_E$

$$\Rightarrow I_B = \frac{5 - 0.7}{10 + 1 \cdot (\beta + 1)} = \frac{V_{BB} - V_{BE}}{R_B + (\beta + 1)R_E}$$

$$= 0.038 \text{ mA}$$

$$I_C = \beta I_B = 3.87 \text{ mA}$$

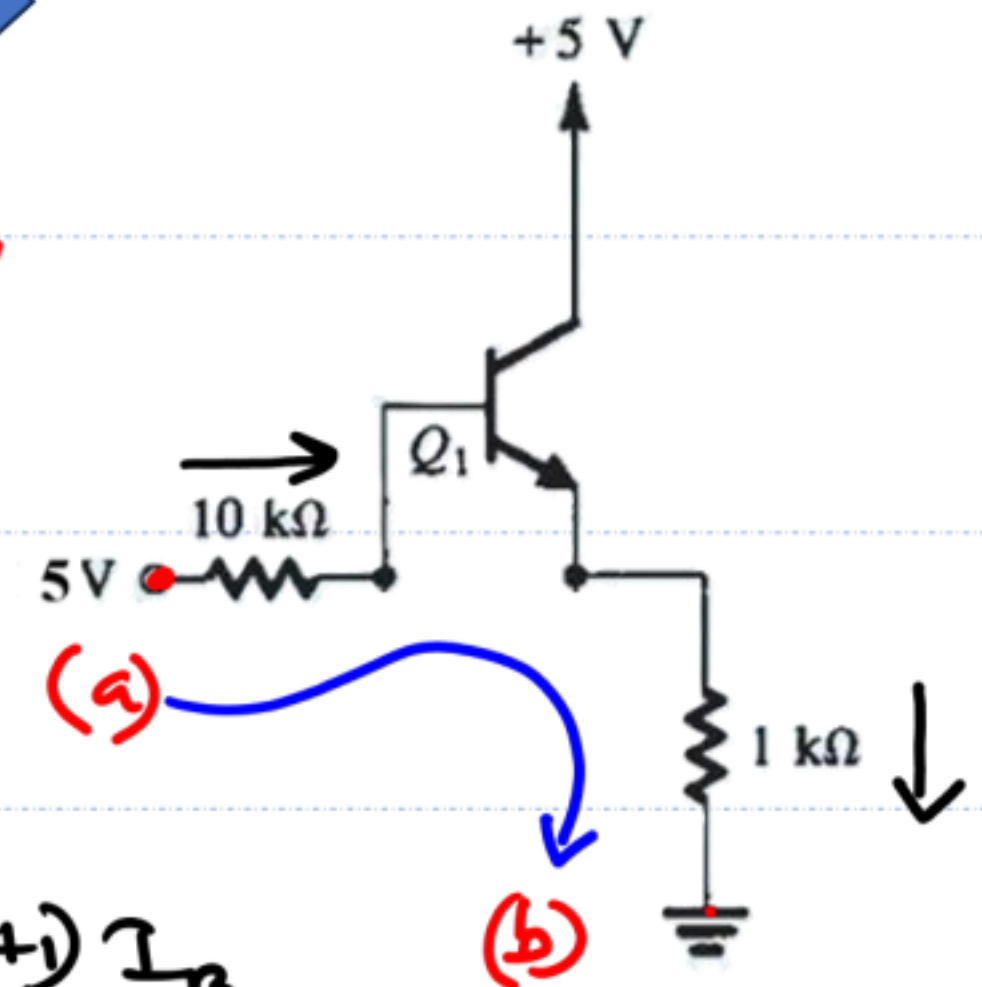
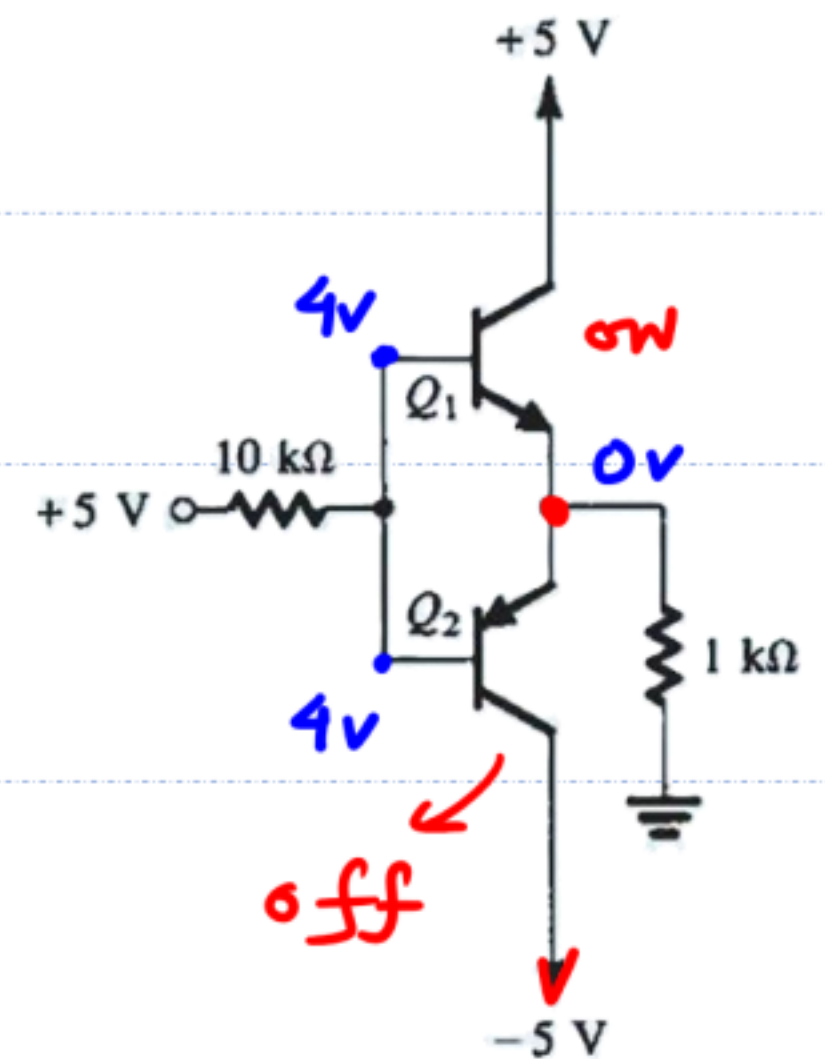
$$I_E = 3.91 \text{ mA}$$

$$V_{C1} = 5 \text{ V} \quad V_{E1} = 1 \cdot I_E = 3.91 \text{ V}$$

$$V_{E2} = -5 \text{ V} = V_{E1}$$

$$\therefore V_{B1} = 5 - 10I_B = 4.62 \text{ V} = V_{B2}$$

$$I_E = (\beta + 1)I_B$$



Evaluate all node voltages and branch currents ($\beta = 100$)

- Q_1 : $V_{BE} = -5 - 0 < 0.7 \therefore \text{OFF}$
- Q_2 : $V_{EB} = 0 - (-5) = 5 > 0.7 \therefore \text{ON} \rightarrow V_{CB} = -5 - (-5) = 0 < 0.4 \therefore \text{active.}$

KVL from (b) $\rightarrow$ (a) :

$$0 - (-5) = 1 \cdot I_E + V_{EB} + 10 I_B \quad [I_E = (\beta + 1) I_B]$$

$$\Rightarrow I_{B_2} = \frac{5 - 0.7}{10 + (\beta + 1) \cdot 1} = 0.0387 \text{ mA}$$

$$I_{C_2} = \beta I_B = 3.87 \text{ mA}$$

$$I_{E_2} = 3.91 \text{ mA}$$

$$V_{C_2} = -5 \text{ V}$$

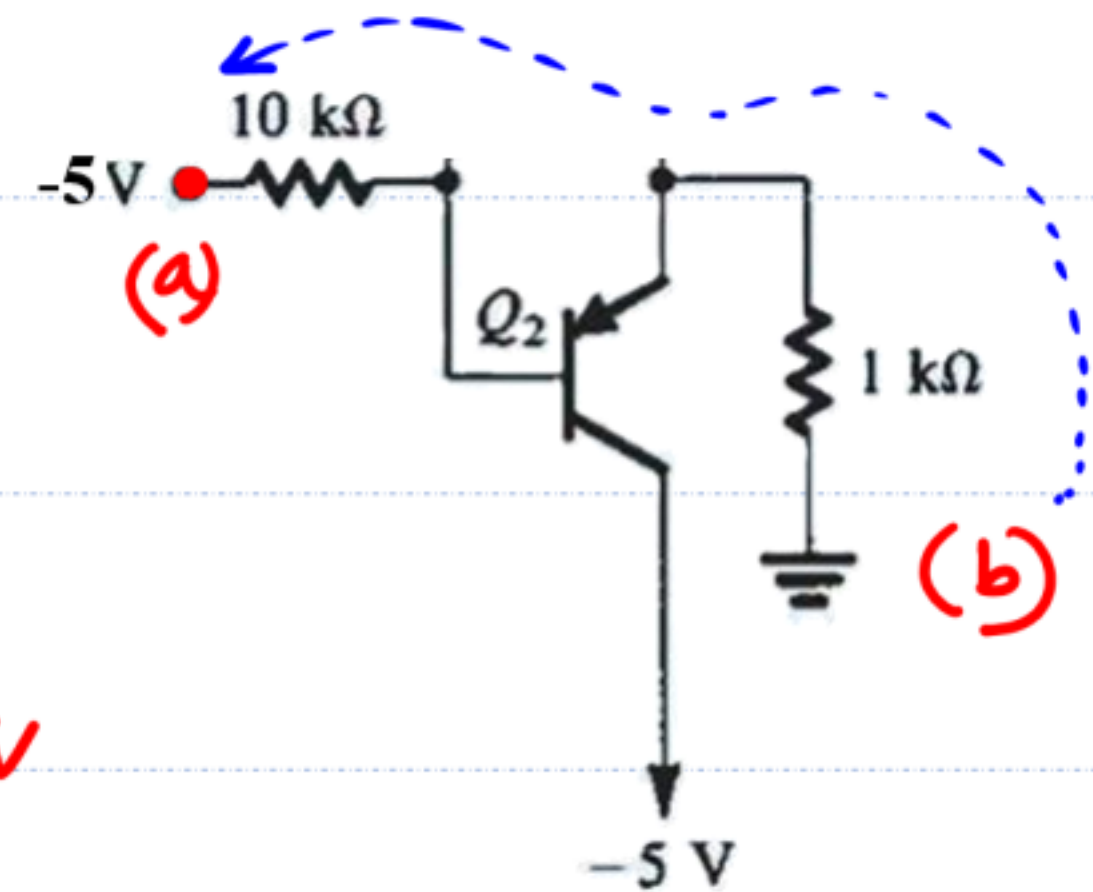
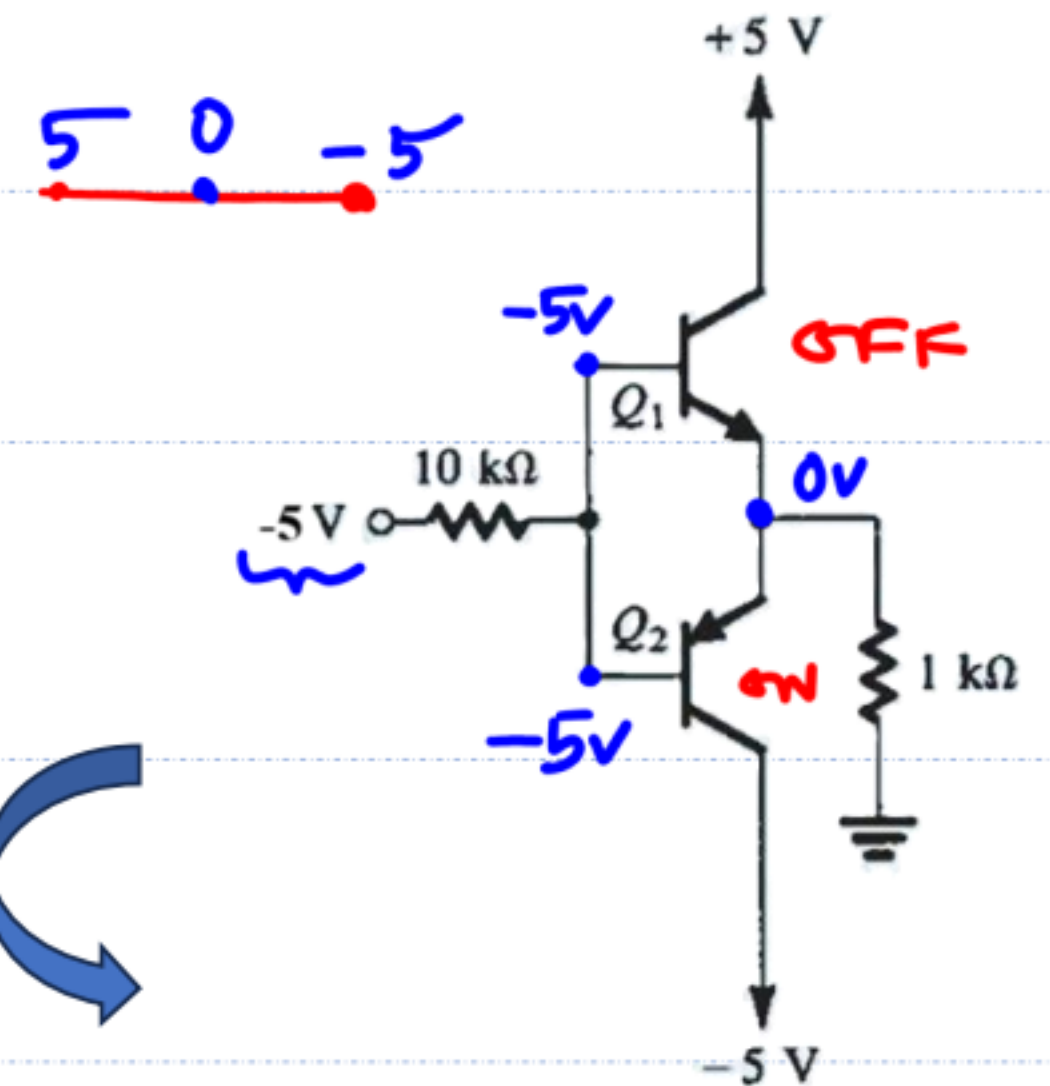
$$(\text{check}) V_{CB_2} = -5 - (-4.62) < 0.4$$

$\therefore$ indeed active

$$\therefore V_{B_2} = -5 + 10 I_B = -4.62 \text{ V}$$

$$\therefore V_{E_2} = 0 - 1 I_E = -3.91 \text{ V}$$

$$\text{or } V_{B_1} = V_{B_2} \quad V_{C_1} = 5 \text{ V} \\ V_{E_1} = V_{E_2}$$



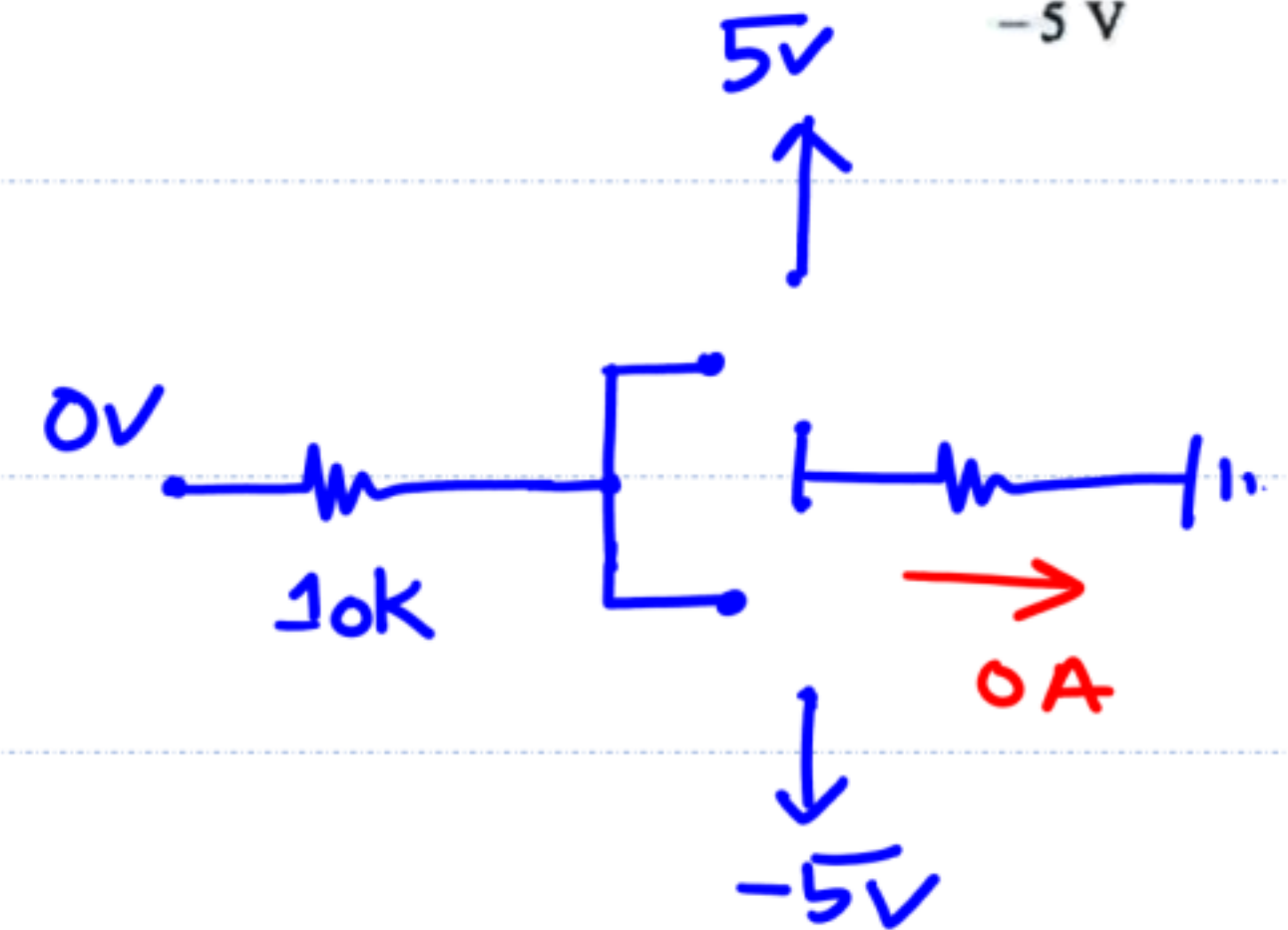
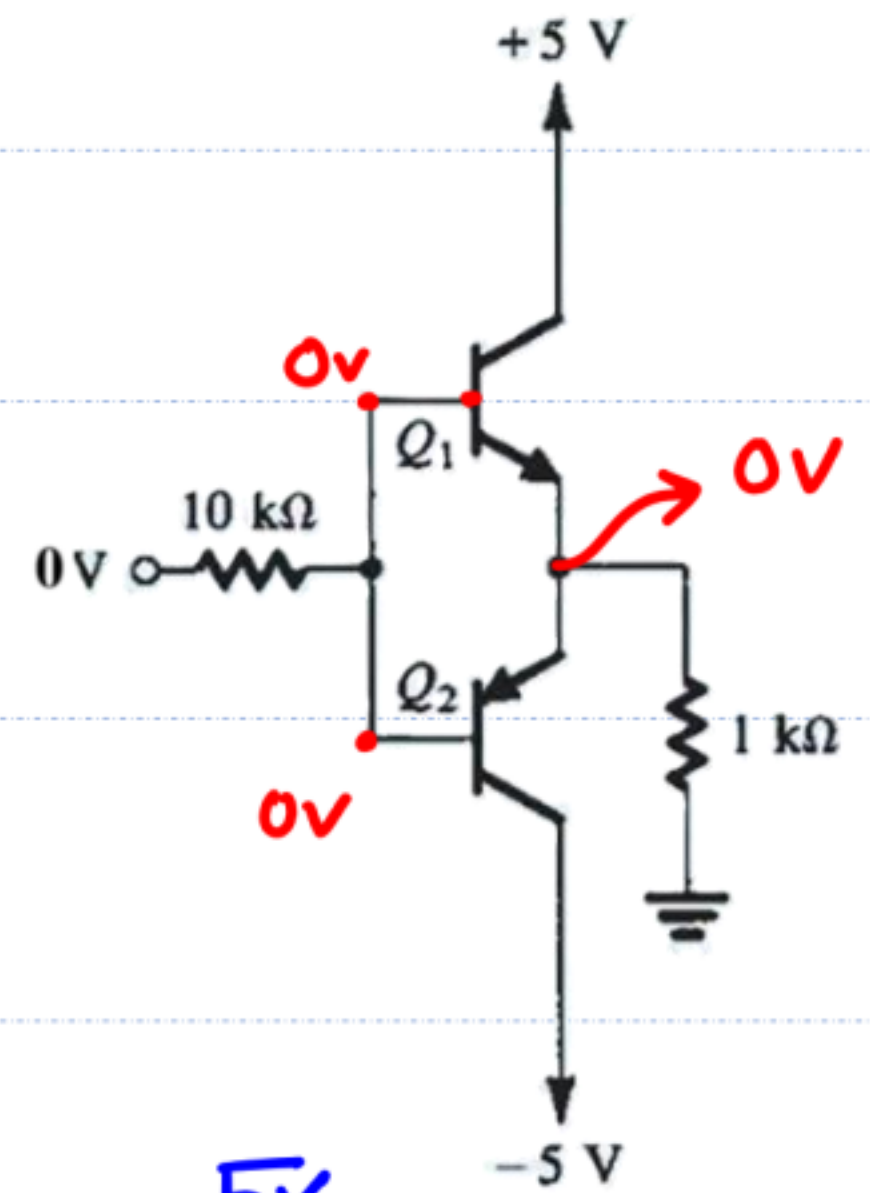
Evaluate all node voltages and branch currents ($\beta = 100$)

$$\underline{Q1} : V_{BE} = 0 - 0 < 0.7 \therefore \text{OFF}$$

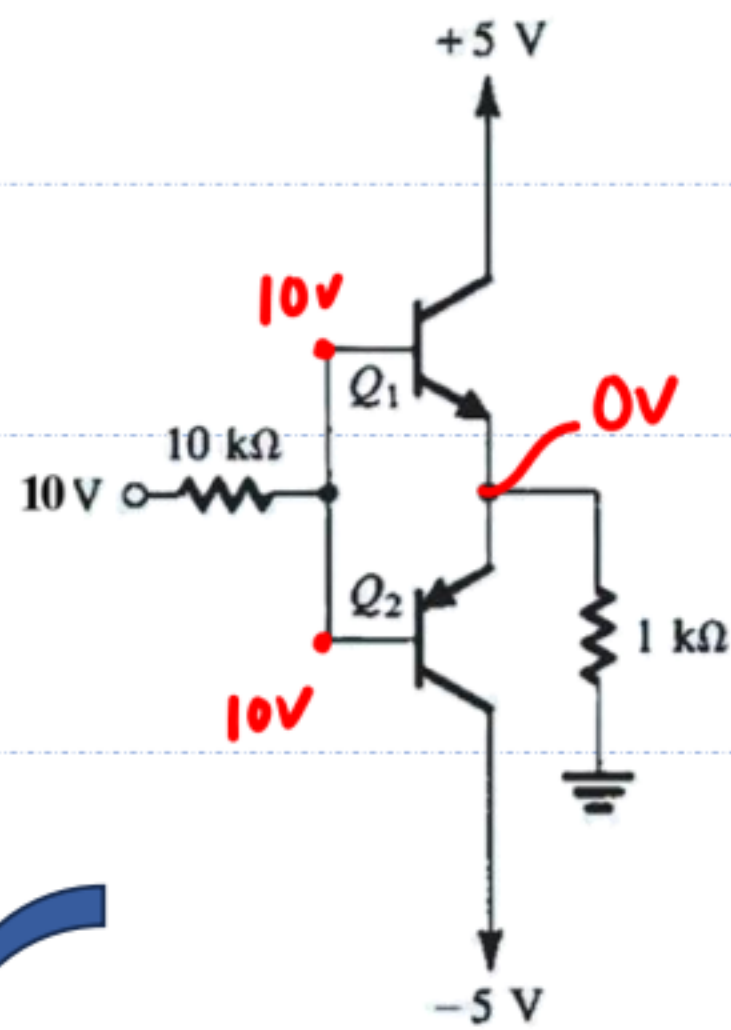
$$\underline{Q2} : V_{EB} = 0 - 0 < 0.7 \therefore \text{OFF}$$

$$V_{B1} = V_{B2} = 0V \quad V_{C2} = -5V$$
$$V_{C1} = 5V \quad V_{E1} = V_{E2} = 0V$$

$$I_{E1} = I_{E2} = I_{B1} = I_{B2} = I_{C1} = I_{C2} = 0$$



Evaluate all node voltages and branch currents ($\beta_{min} = 30$) ← **



Q1: $V_{BE} = 10 - 0 > 0.7V \therefore \text{ON} \rightarrow V_{CE} = 10 - 5 = 5$
 $V_{CE} = 5V > 0.4V \rightarrow \text{saturation}$



Q2: $V_{EB} = 0 - 10 < 0.7 \therefore \text{OFF}$

$V_{CE1} = 0.2V$

$\therefore V_{E1} = 4.8V$

$I_{E1} = \frac{4.8}{1K} = 4.8mA$

$\therefore V_{B1} = V_{E1} + 0.7 = 5.5V$

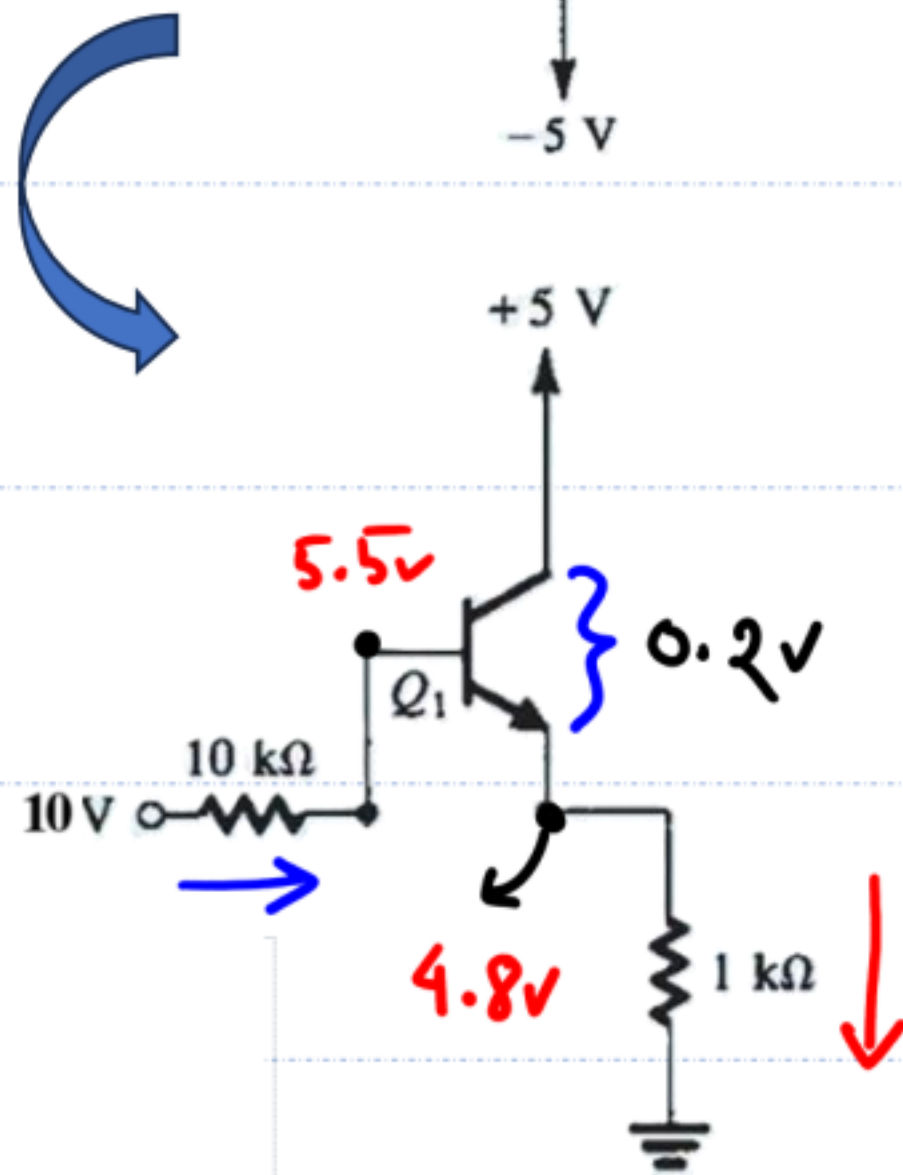
$I_{B1} = \frac{10 - 5.5}{10K} = 0.45mA$

$\therefore I_C = I_{E1} - I_{B1} = 4.35mA$

$\therefore \beta_{forced} = \frac{I_C(s)}{I_B(s)} = \frac{4.35}{0.45} = 9.7$

$\beta_f < \beta_{min}$

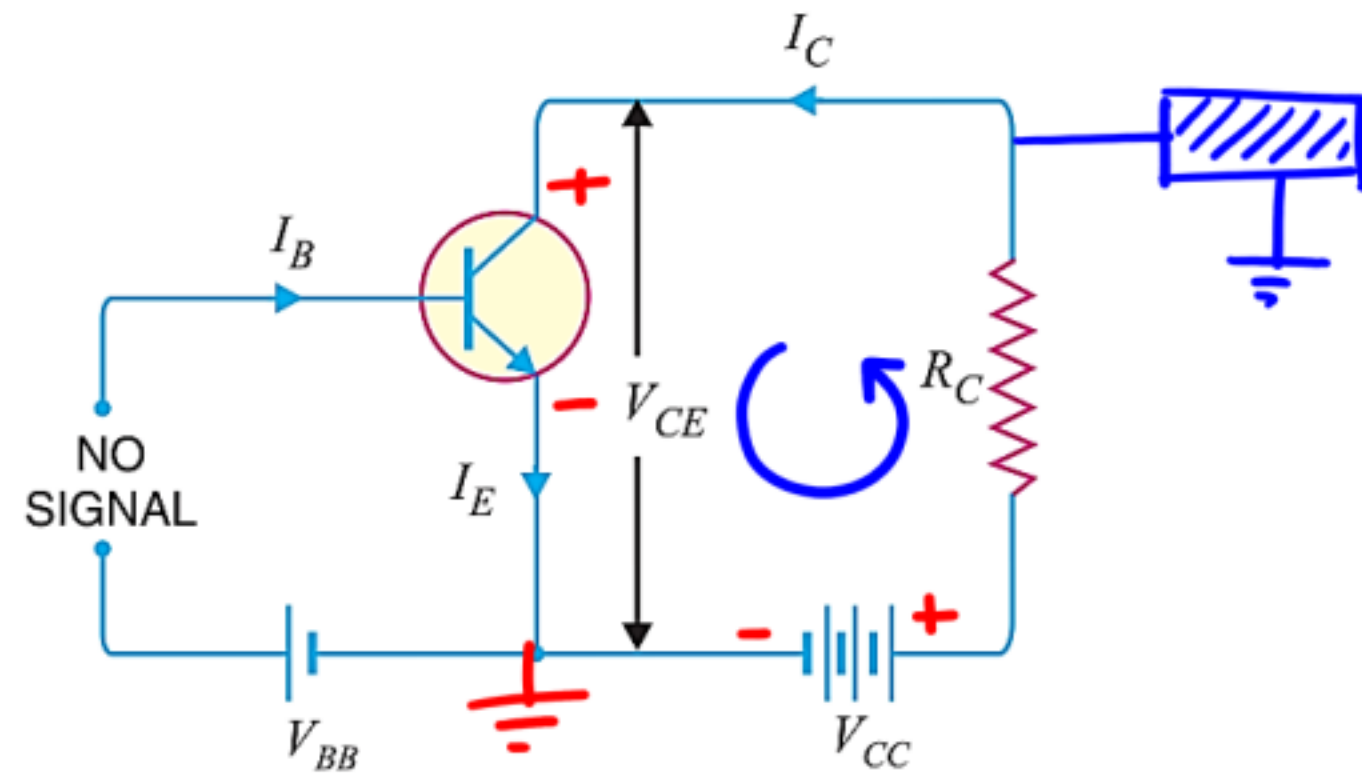
$\therefore Q1 \rightarrow \text{sat.} \checkmark$



Transistor DC Load Line Analysis

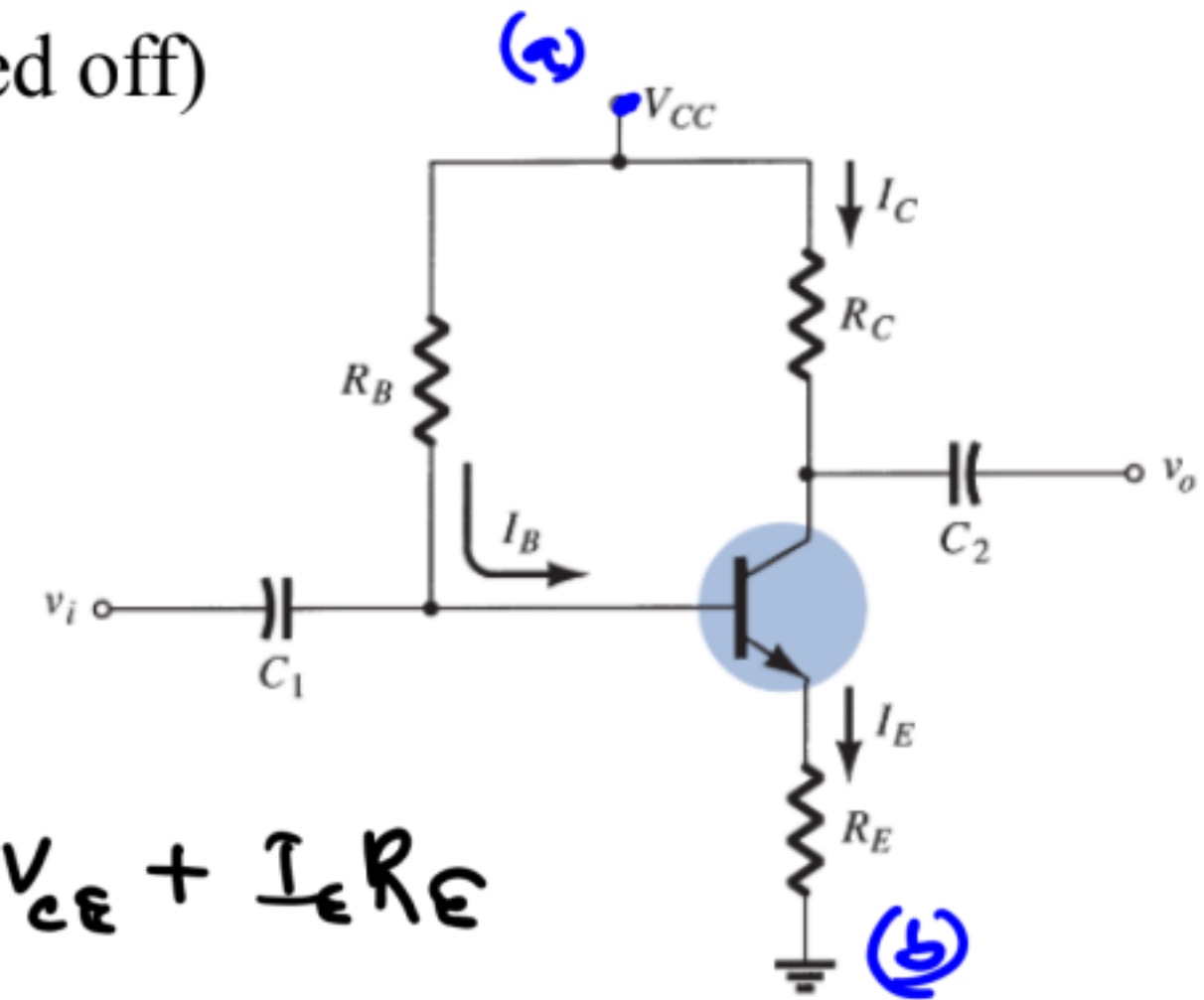


- The DC load line of a transistor shows the relationship between the DC current (I_C) and DC voltage (V_{CE}) of the circuit under consideration for a given load.
- Only DC conditions prevail (all AC sources and reactive elements are turned off)



$$-V_{CE} + I_C R_C + V_{CE} = 0$$

$$\therefore V_{CE} = V_{CC} - I_C R_C \quad \text{--- (a)}$$



$$V_{CC} - 0 = I_C R_C + V_{CE} + I_E R_E$$

$$V_{CC} = V_{CE} + I_C (R_C + R_E)$$

$$\therefore V_{CE} = V_{CC} - I_C (R_C + R_E)$$

$$I_C \approx I_E \quad (\beta \text{ high})$$

Transistor DC Load Line Analysis

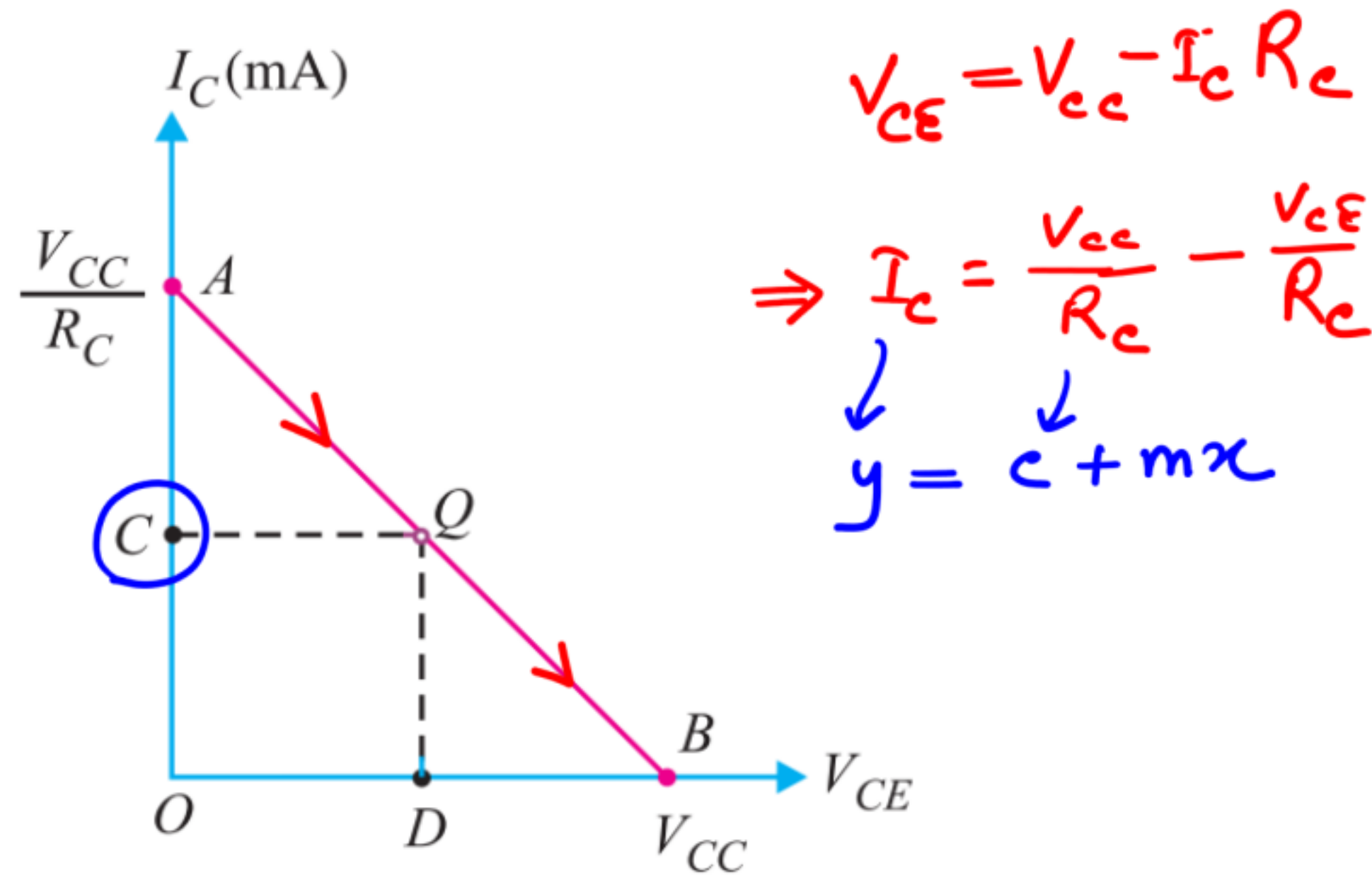


Why only current I_C and voltage V_{CE} are chosen for DC load-line?

1. The load is typically connected to the collector. That's why, the output characteristics need to include collector current and voltage.
2. Consider V_{BE} : It is almost constant (0.7V) in active mode. But, while keeping $V_{BE} = 0.7V$, the transistor can still go to saturation. Thus, considering V_{BE} for load-line analysis is pointless.
3. Consider V_{BC} : While the base-collector voltage affects the Transistor's mode of operation, it isn't taken for load-line analysis because, the load **directly** relates to the voltage V_{CE} only.
4. The base current I_B is not considered either because, although $I_B = \beta I_C$ & thus affects I_C , but its value is small & doesn't directly affect V_{CE}



- We plot this equation in I_C vs V_{CE} graph. This is known as the **DC LOAD LINE**

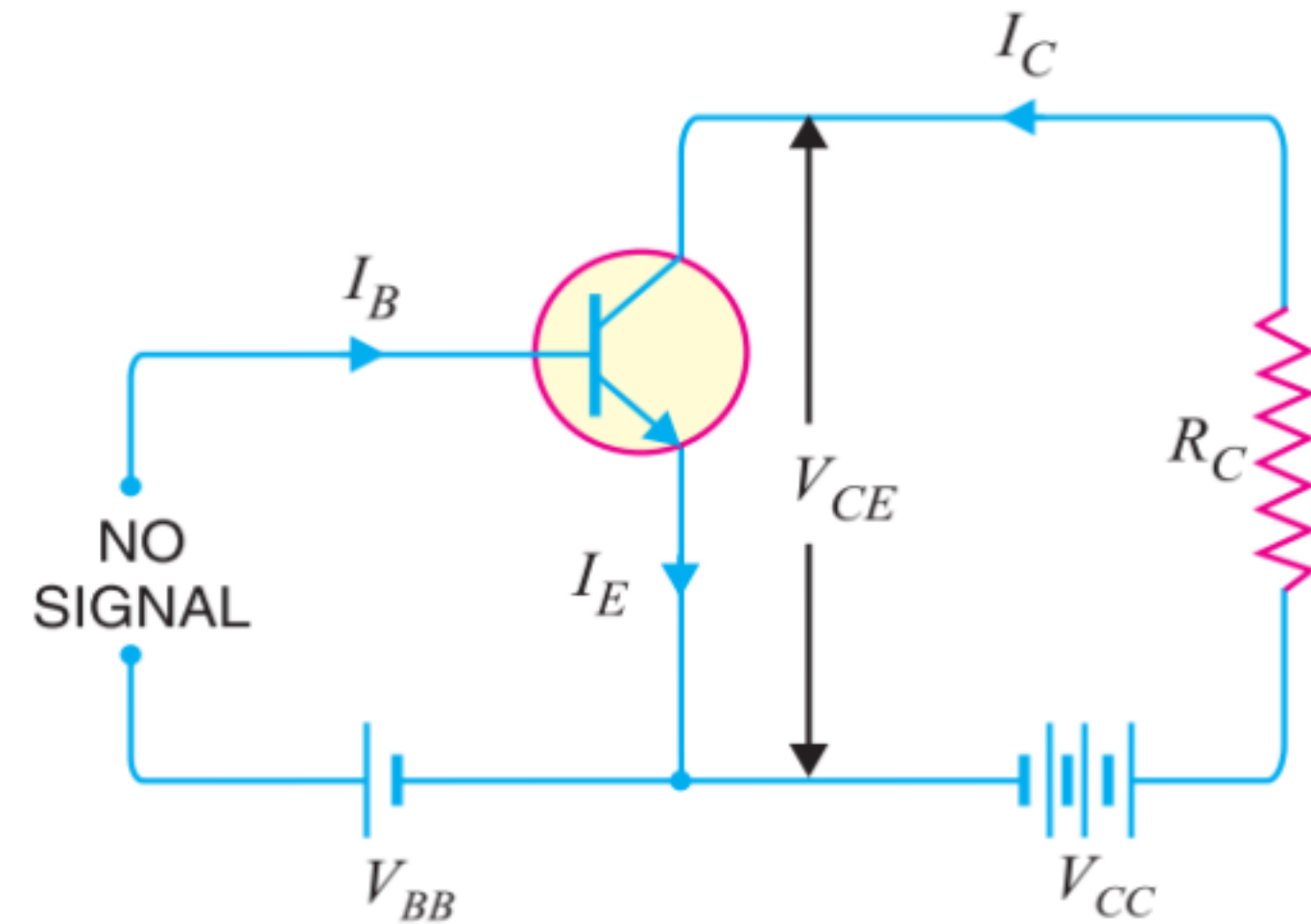


$$V_{CE} = V_{CC} - I_C R_C$$

$$\Rightarrow I_C = \frac{V_{CC}}{R_C} - \frac{V_{CE}}{R_C}$$

$$y = c + mx$$

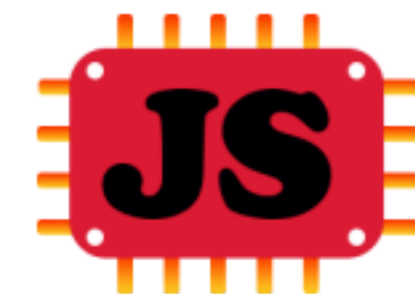
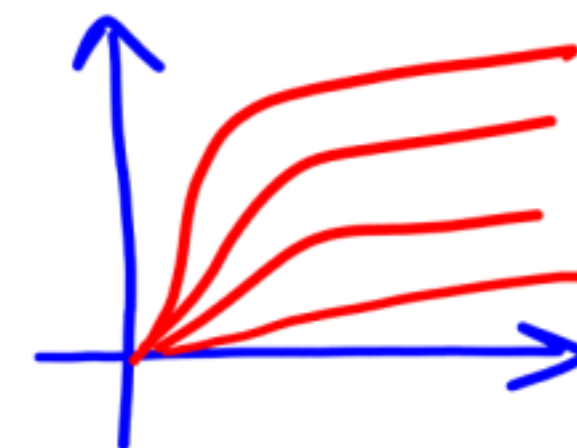
$$\left(m = -\frac{1}{R_C} \right)$$



Why DC load line?



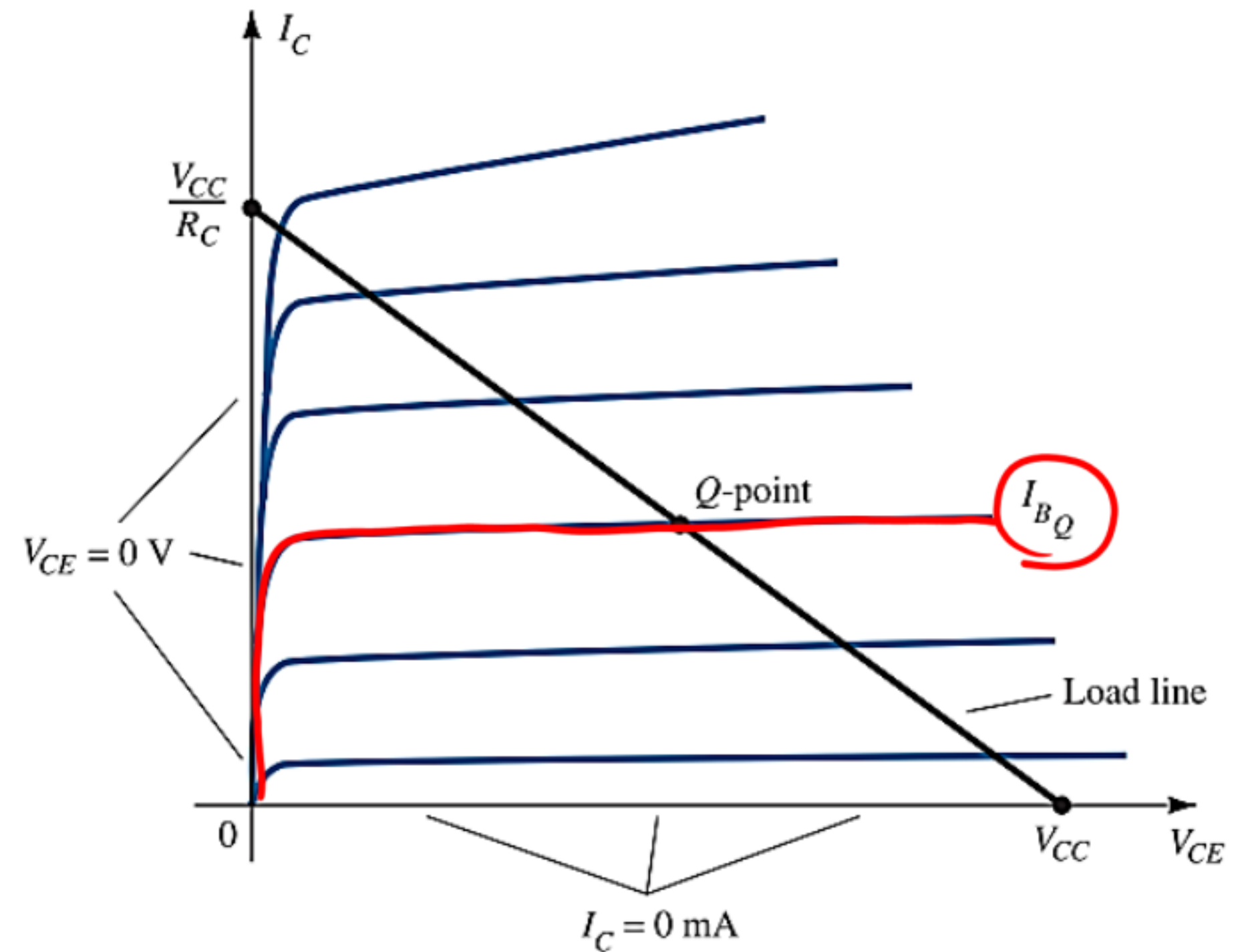
- It is easier and more convenient to find I_C for a particular V_{CE} using DC load line than by using the characteristic curve of I_C vs V_{CE} !
- The Line is called Load Line, because resistance R_C is the load resistance for the BJT circuit
- Load line equation for NPN transistor: $V_{CE} = V_{CC} - I_C R_C$
- Load line equation for PNP transistor: $V_{EC} = V_{CC} - I_C R_C$



Operating point, Q (1)



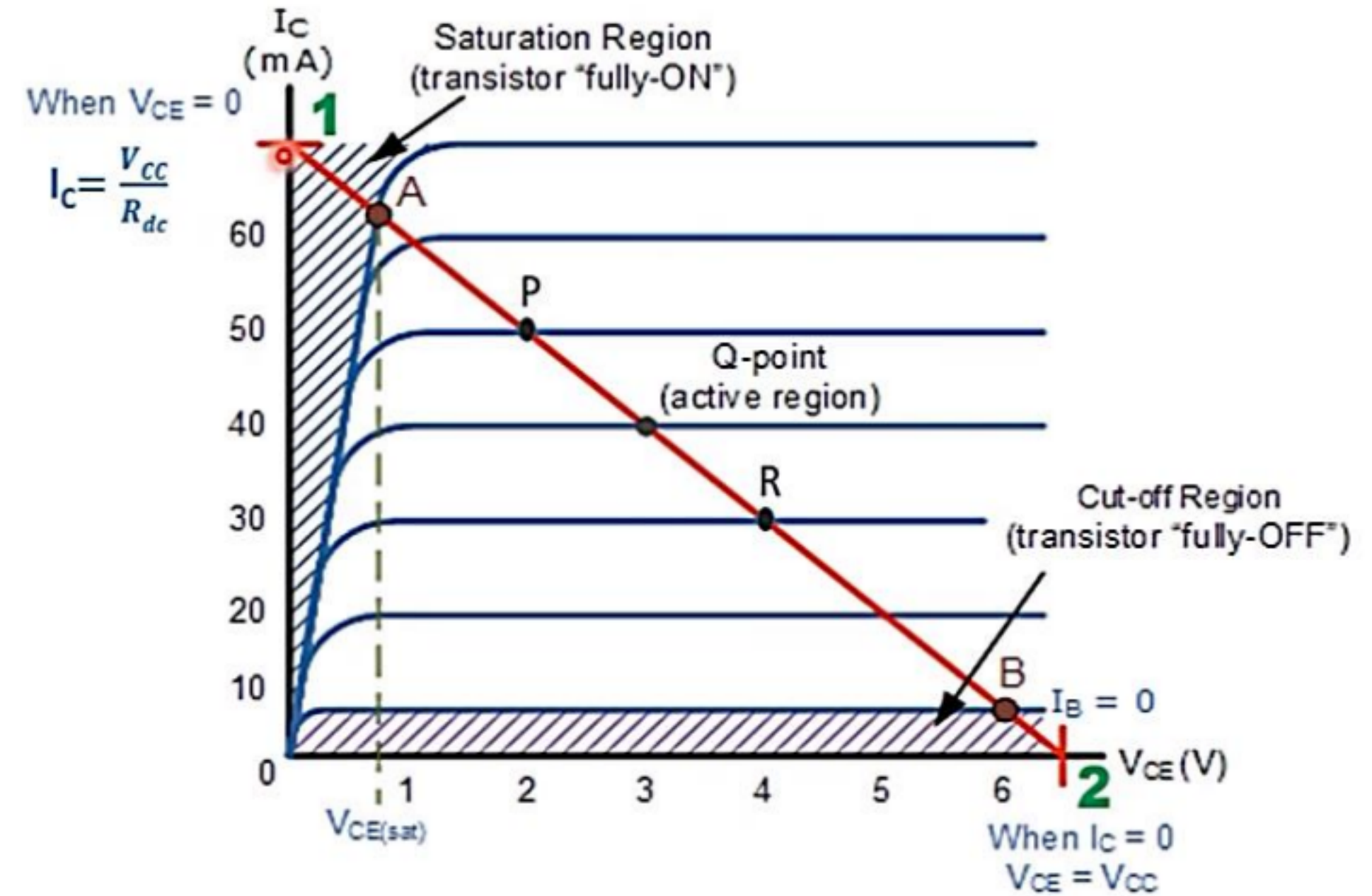
- Superimpose the DC Load Line onto the characteristic curve
- The intersecting point is called Q-point.
- Quiescent (silent) point $\Rightarrow$ Q-point
- It's called Q-point because at this point there's no AC signal in the circuit.
- Co-ordinate of Q $\equiv (V_{CE}, I_C)$



Location of the Operating point, Q (1)



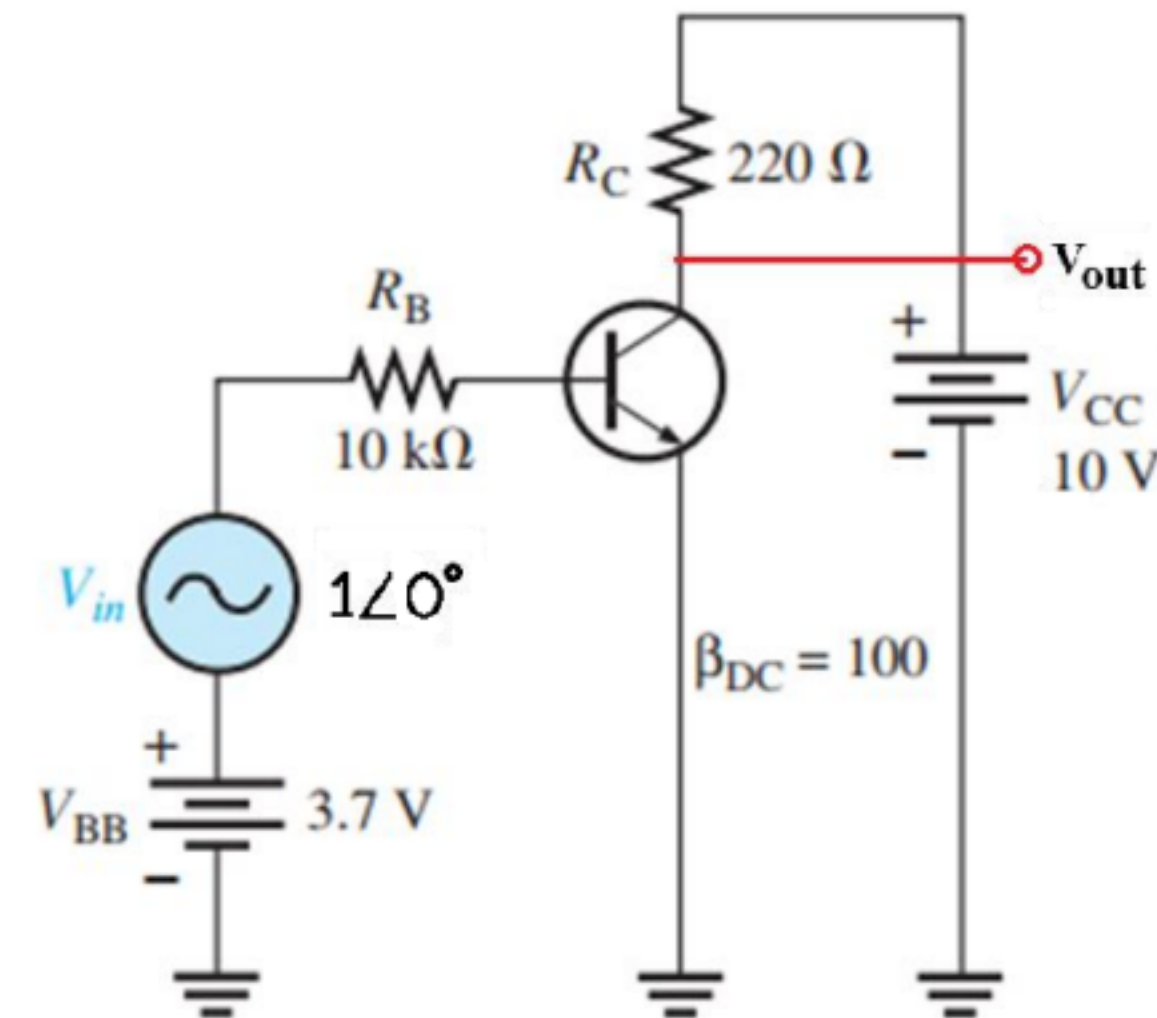
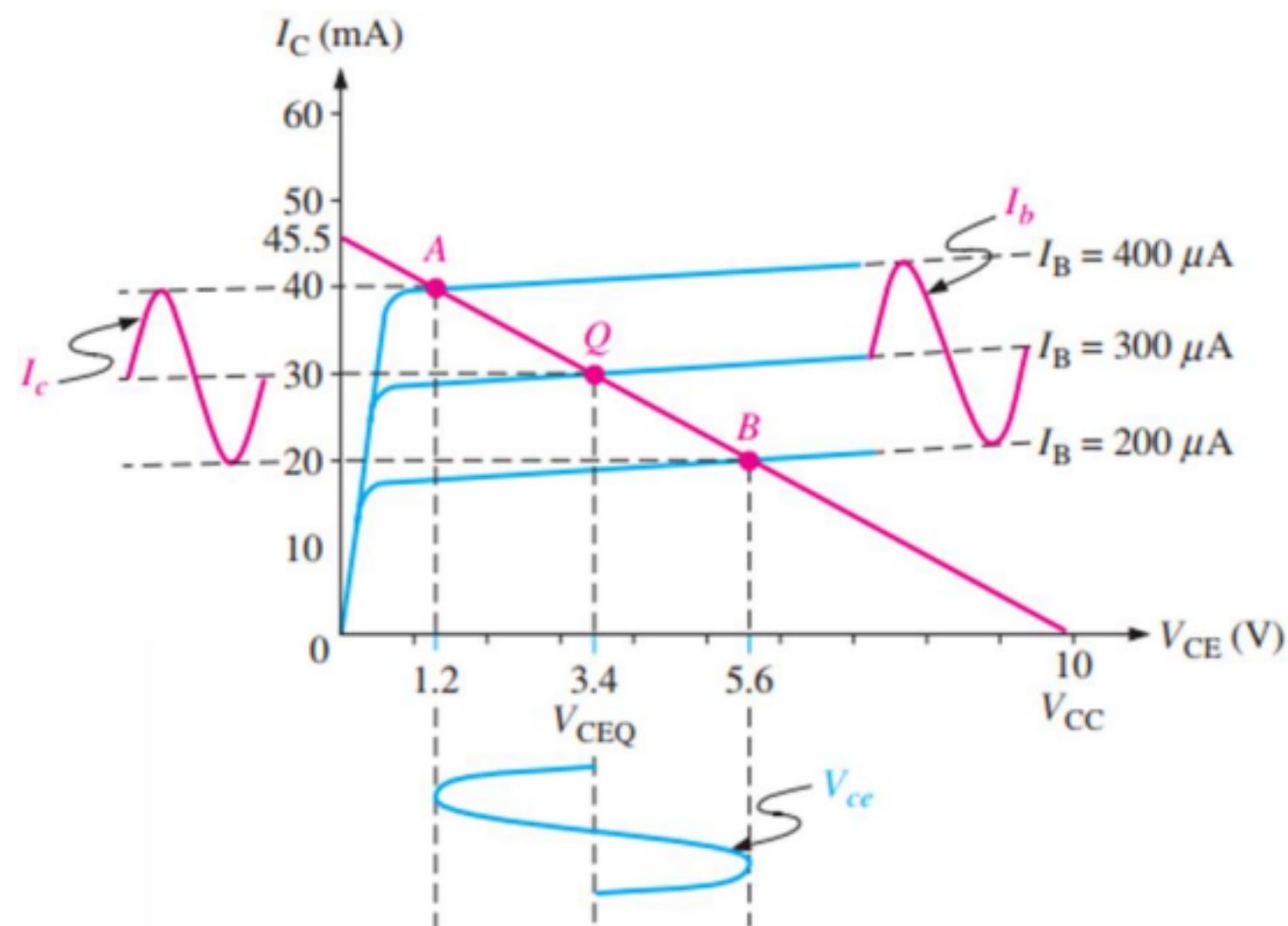
- BJT is mainly used as an amplifier.
- To use BJT as an amplifier, the BJT must be in the active region.
- Thus, the operating point, Q should be in the active region.



Location of the Operating point, Q (2)



- Q-point should be in the **middle of the DC-load line, or the Active region.**
- This is because, choosing the mid-point of the DC-load line allows for **maximum possible amplification** of the input sinusoidal signal without any distortion in the positive or negative half cycle! We'll explain this in sha ALLAAH in the **Amplifier lecture** 😊

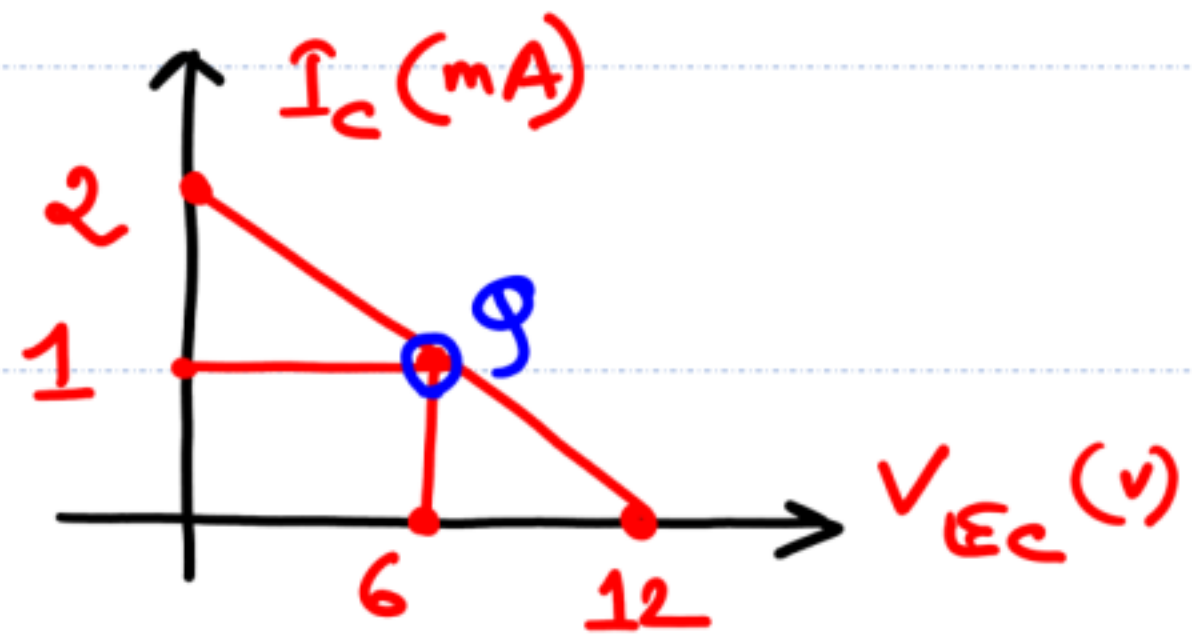


In the circuit, if $V_{CC} = 12V$ & $R_C = 6k\ \text{ohm}$, draw the DC load line. What will be the Q-point if zero signal base current is $20\ \mu A$ & $\beta = 50$

KVL at loop ① :

$$-V_{ce} + V_{Ee} + I_c R_c = 0$$

$$\therefore V_{Ee} = V_{ce} - I_c R_c = 12 - 6I_c \text{ --- ①}$$



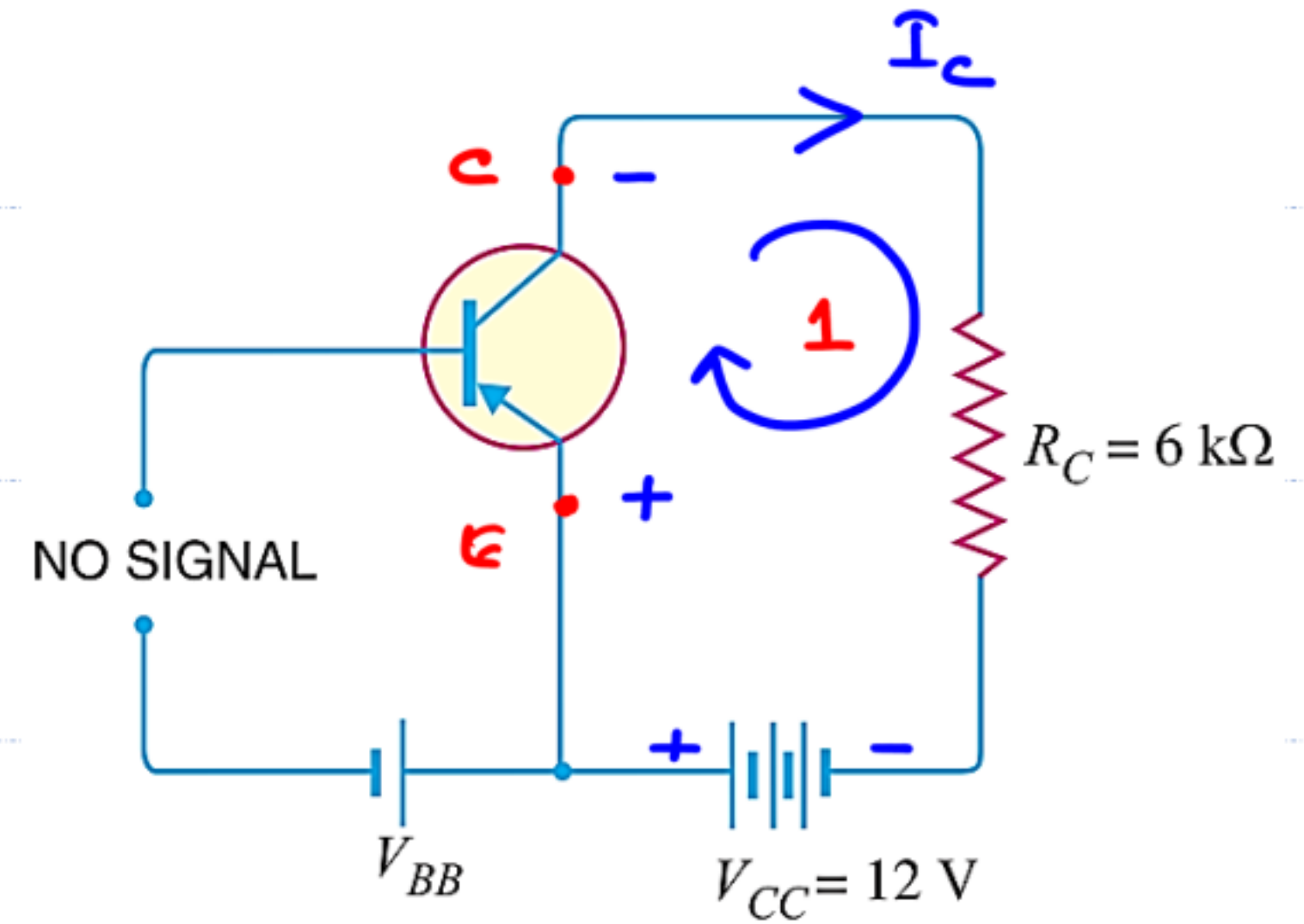
$$\begin{cases} V_{Ee} = V_{ce} = 12 \\ I_c = \frac{V_{ce}}{R_c} = 2\text{mA} \end{cases}$$

Q-pt:

$$I_{BQ} = 20\ \mu A$$

$$I_{cQ} = \beta I_{BQ} = 50 I_{BQ} = 1\text{mA}$$

$$V_{EeQ} = 12 - 6 \times (1) = 6V \quad Q = (6, 1)$$

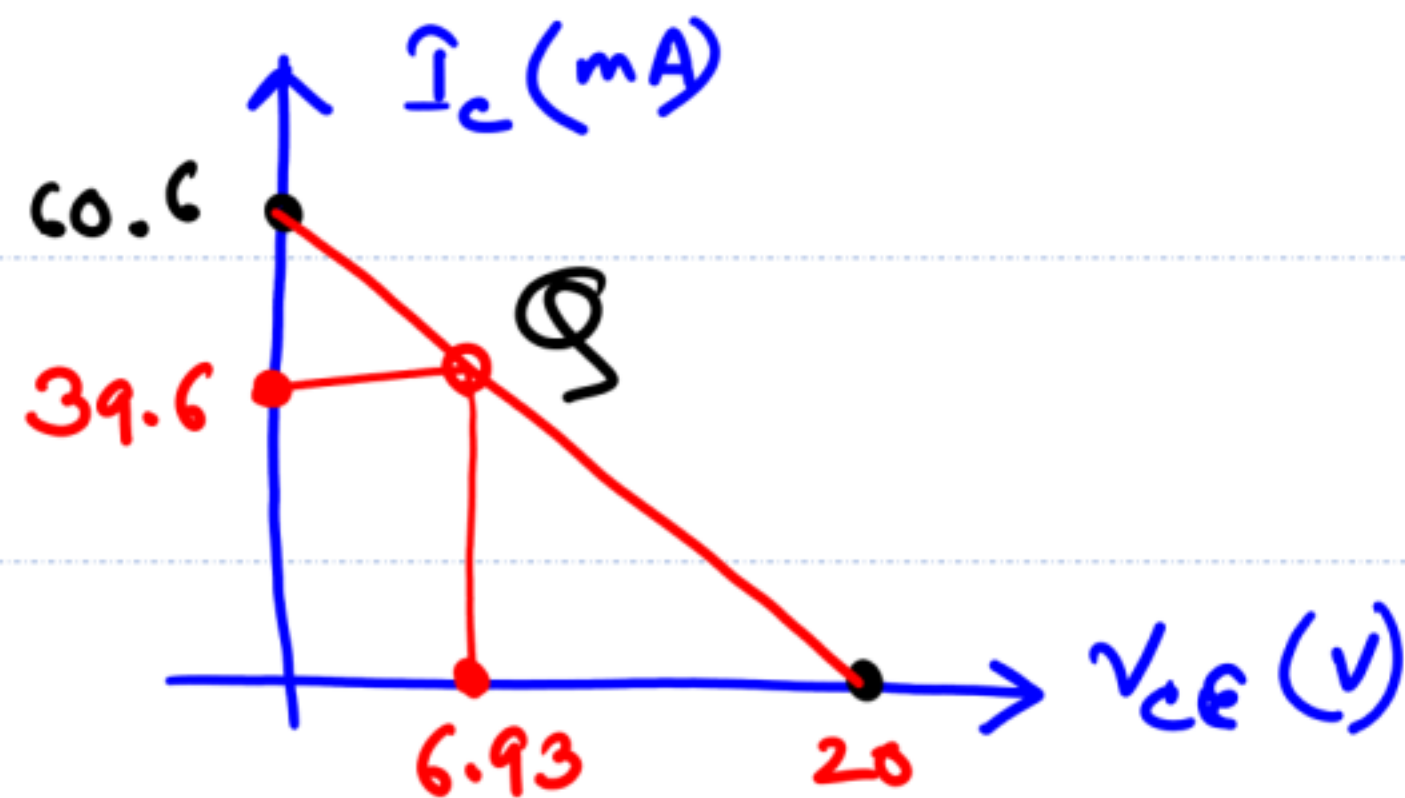


Determine the Q point of the transistor circuit. And draw the DC load line ($\beta = 200$)

KVL at loop ① :

$$-V_{CE} + I_C R_C + V_{CE} = 0$$

$$\therefore V_{CE} = V_{CE} - I_C R_C = 20 - 330 I_C$$



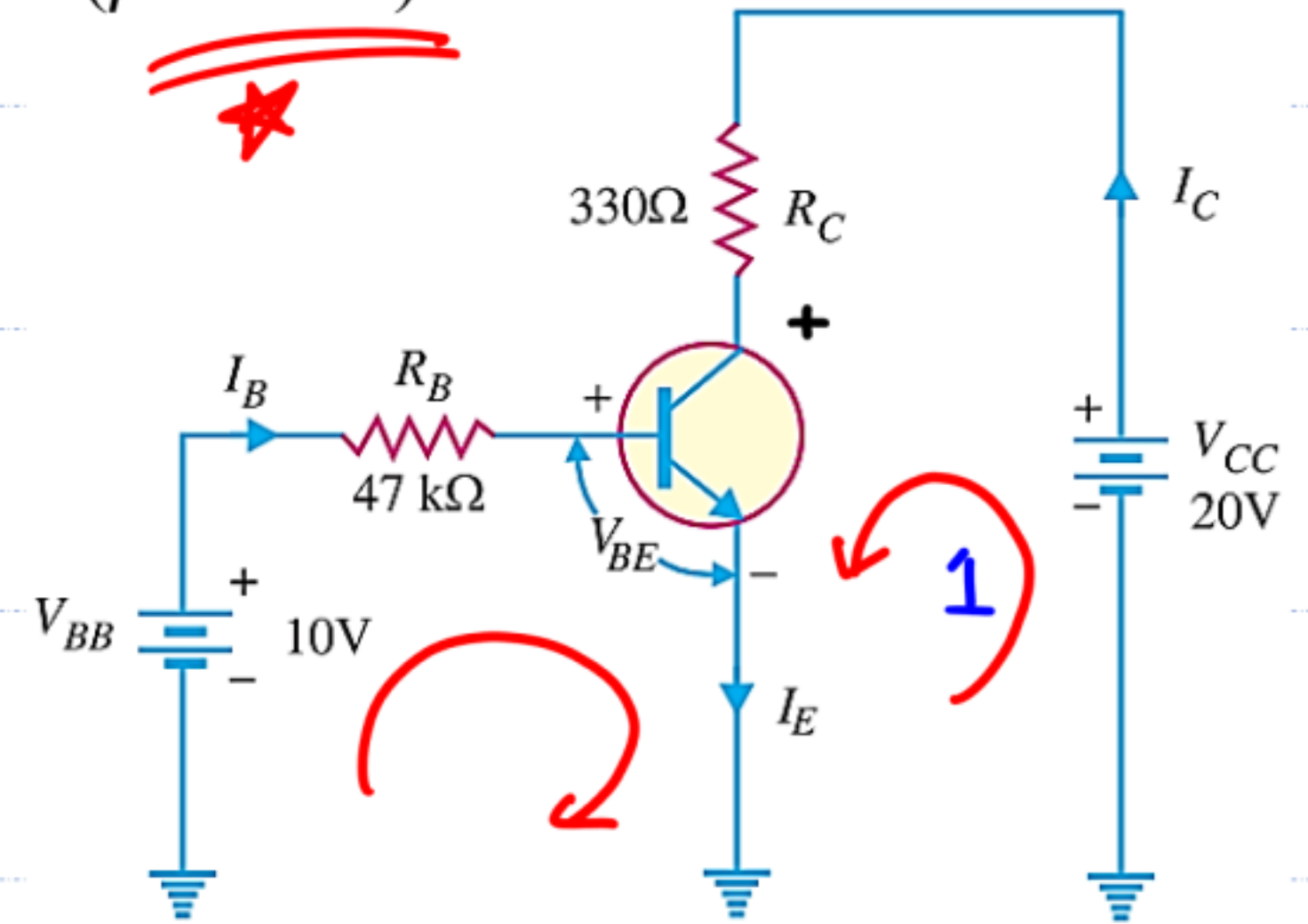
Q-point:

$$I_{BQ} = \frac{V_{BB} - V_{BE}}{R_B} = \frac{10 - 0.7}{47} = 0.2 \text{ mA}$$

$$I_{CQ} = \beta I_{BQ} = 39.6 \text{ mA}$$

$$V_{CEQ} = 20 - 330 (39.6 \text{ m}) = 6.93 \text{ V}$$

$$Q \equiv (6.93, 39.6 \text{ mA})$$



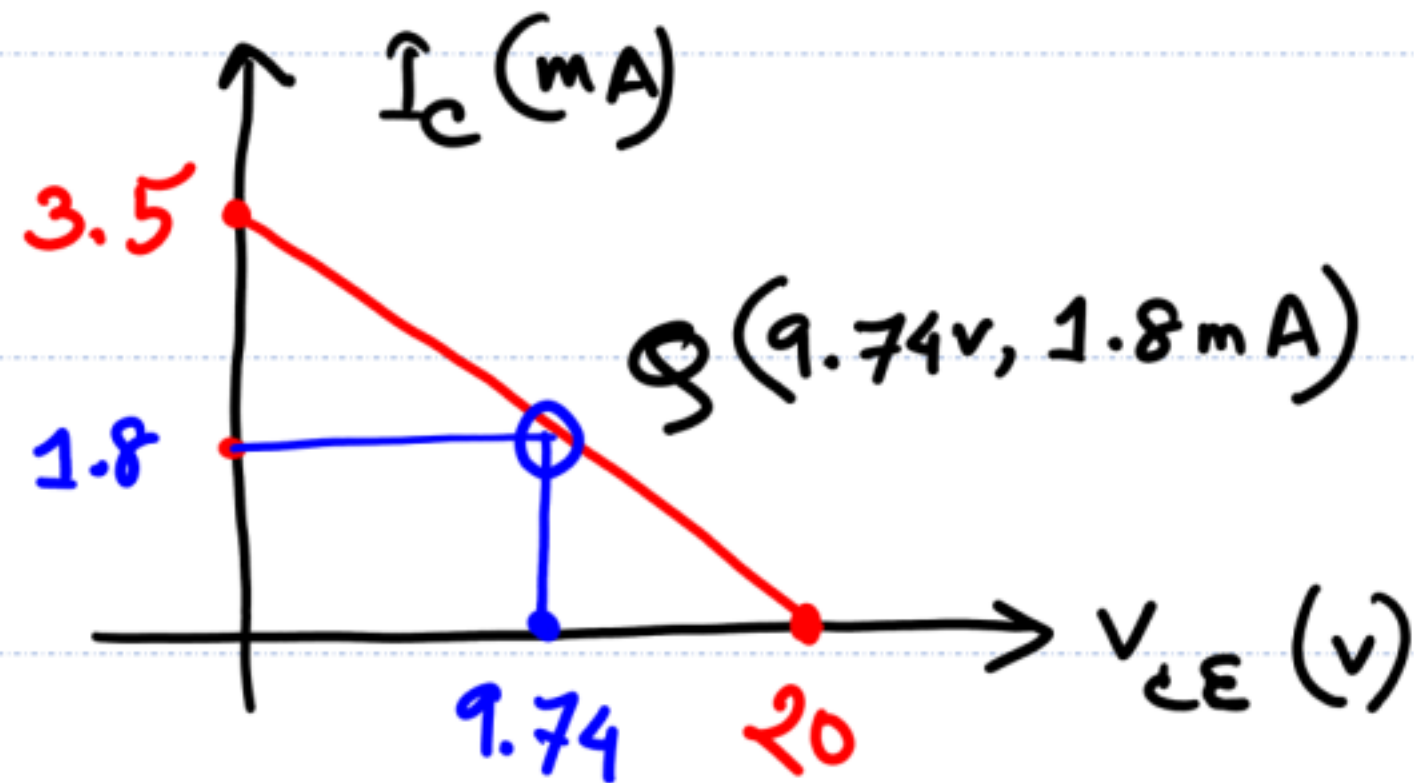
Determine the Q point of the transistor circuit. And draw the DC load line ($\beta = 100$)

KVL from (a) to (b) :

$$10 - (-10) = 20 = I_C R_C + V_{CE} + I_E R_E$$

$$\Rightarrow V_{CE} = 20 - I_C (R_C + R_E) \quad \text{--- (1)} \quad (I_E \approx I_C) \quad \beta \text{ high}$$

$$(V_{CE} = 20 - 5.7 I_C) \quad \checkmark$$



$$\begin{cases} V_{CE} = 20 \\ I_C = 20/5.7 \end{cases}$$

Q-pt

KVL at (1) :

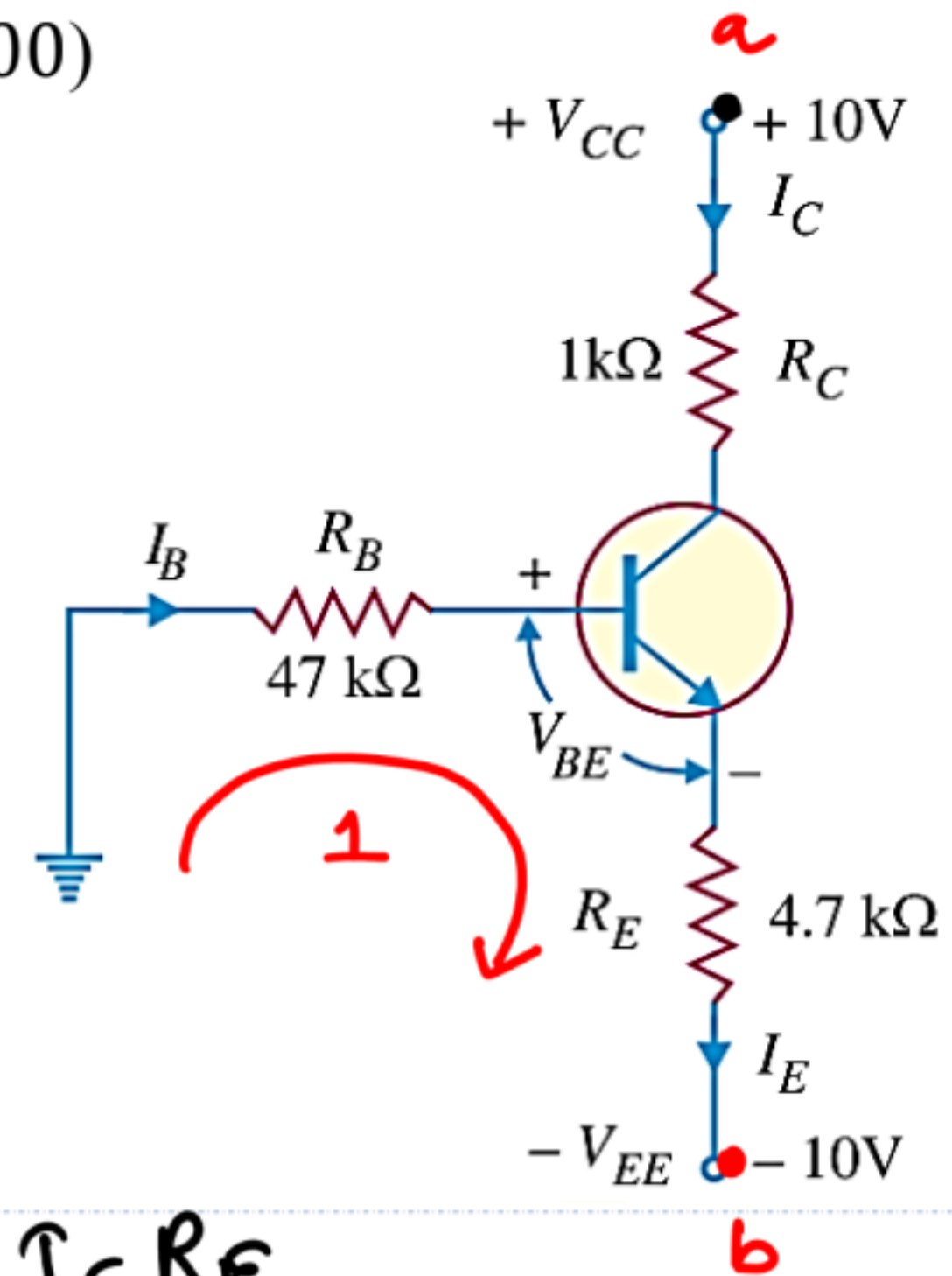
$$0 - (-10) = I_B R_B + V_{BE} + I_E R_E$$

$$\therefore I_{BQ} = \frac{10 - 0.7}{R_B + (\beta + 1) R_E}$$

$$= 0.018 \text{ mA}$$

$$\therefore I_{CQ} = \beta I_{BQ} = 1.8 \text{ mA}$$

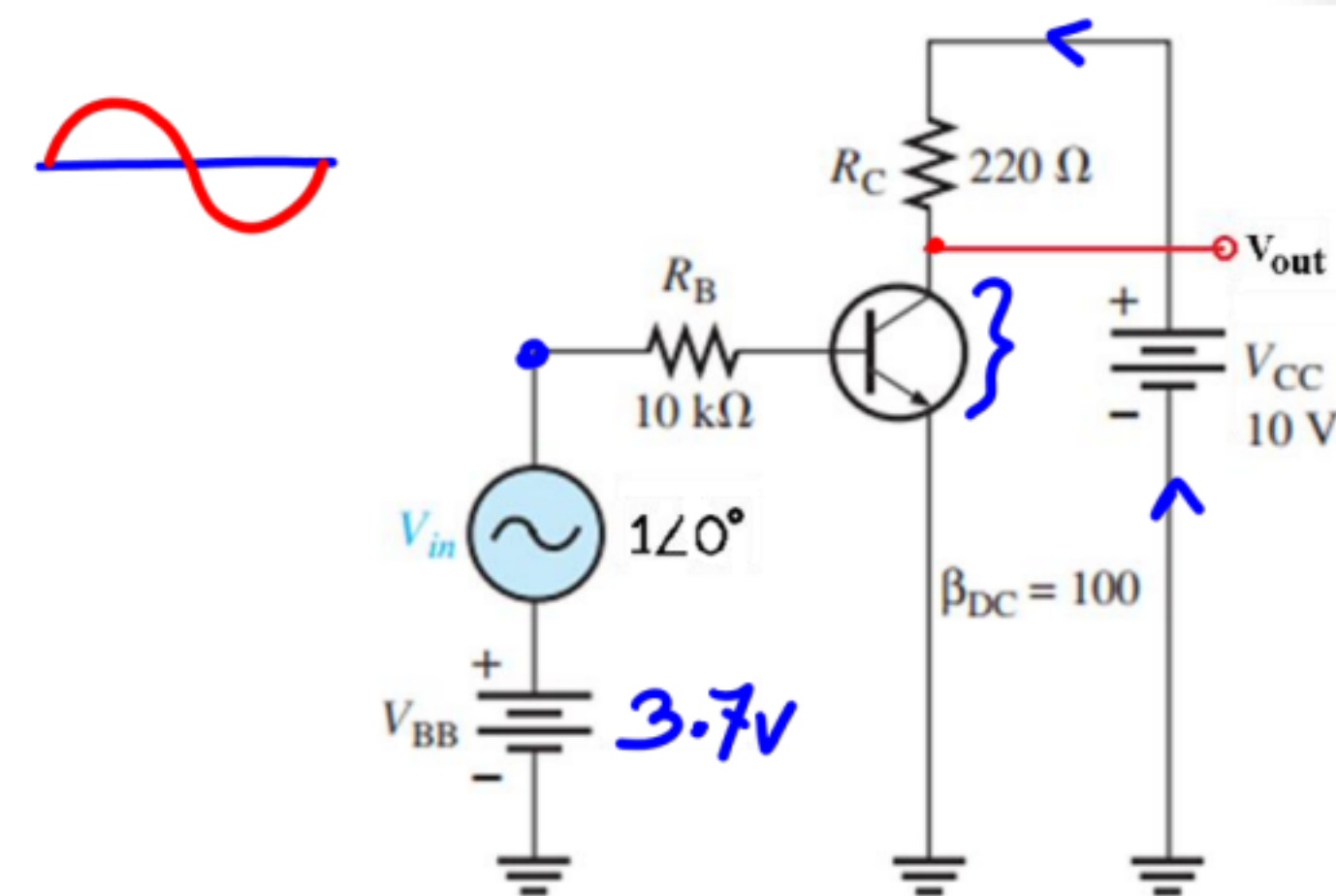
$$V_{CEQ} = 9.74 \text{ V}$$



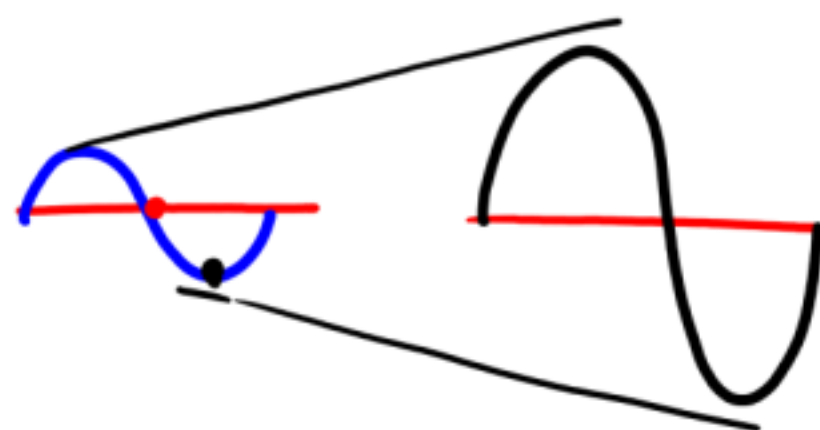
Biasing of Transistors



- Transistors are mainly used for amplification.
- In case of amplifiers, small input AC voltages are amplified to larger AC voltages.
- AC sinusoidal signals have both positive and negative values.
- Negative voltage value at the base will turn the BJT off! (cut-off). Thus, no amplification
- Thus, BJTs are forced to be in the active mode by external DC voltages by following the conditions of $V_{BC} < 0.4v$ & $V_{BE} \geq 0.7v$
- Thus, applying DC voltages to the transistor to keep it at our desired mode is called Biasing.



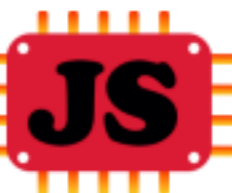
v_{in}	BJT state	v_{out}
-1	off	10
-0.5	off	10
0	off	10
0.5	off	10
1	active	< 10



BJT Biasing Configurations



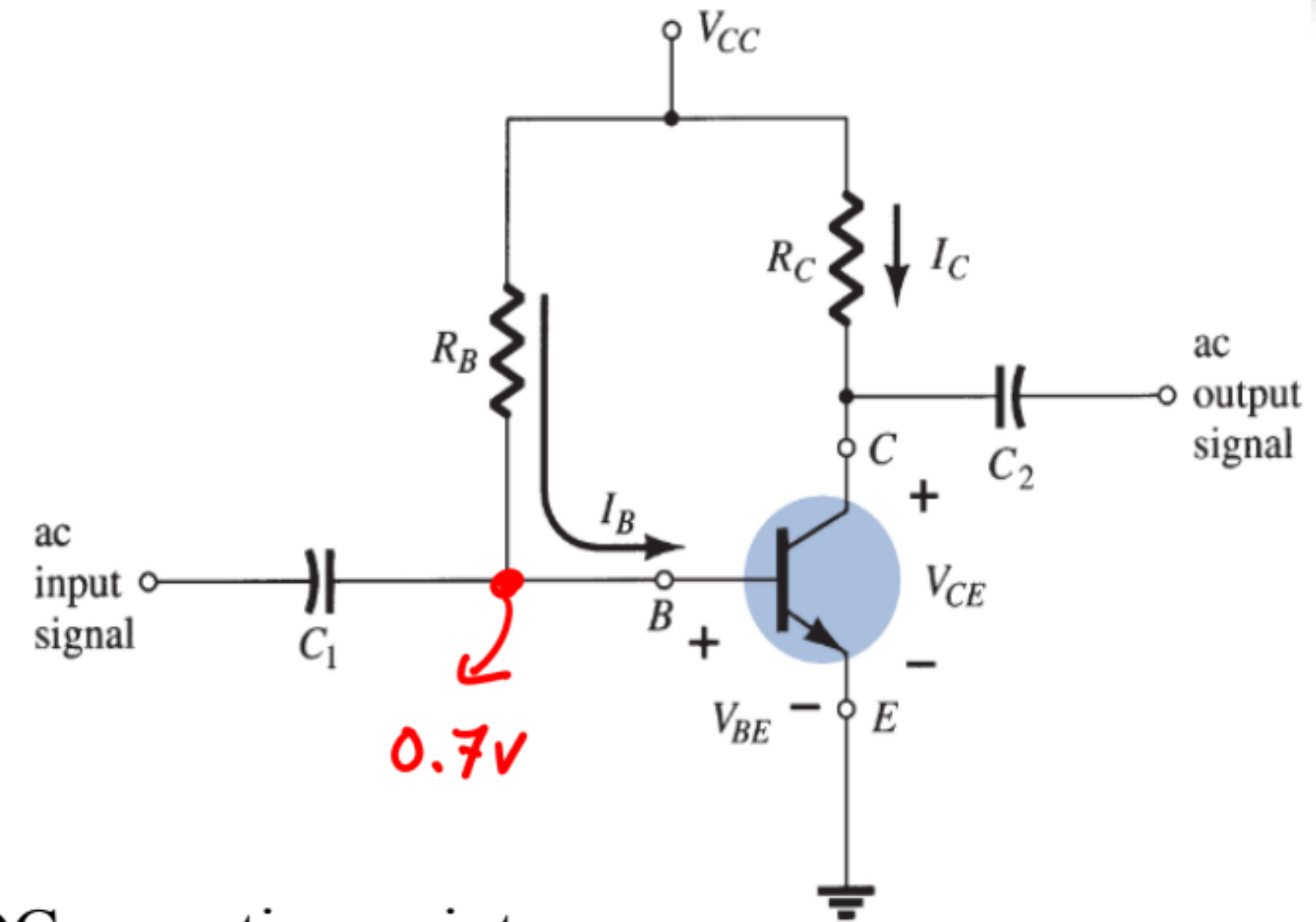
- Fixed-Bias Configuration
- Emitter-Bias Configuration
- Voltage-Divider Bias Configuration
- DC Biasing Configuration with Collector Feedback
- Emitter-Follower Configuration
- Common-Base Configuration



Fixed Bias Configuration



- Also known as base bias or fixed resistance biasing.
- Here, the base current I_B remains constant because the DC voltage source V_{CC} and the resistor R_B are constant. This means that the transistor's operating point also remains fixed.
- This simple arrangement is thermally unstable.
- If the transistor's temperature increases, the collector current will increase, which can move the DC operating point away from its desired position. This can affect the amplifier gain and result in distortion.



$$I_B = \frac{V_{CC} - 0.7}{R_B}$$



Determine the following: I_{BQ} , I_{CQ} , V_{CEQ} , V_B , V_C , V_{BC}

$$I_{BQ} = \frac{V_{CC} - 0.7}{R_B} = 0.047 \text{ mA}$$

$$I_{CQ} = \beta I_{BQ} = 2.35 \text{ mA}$$

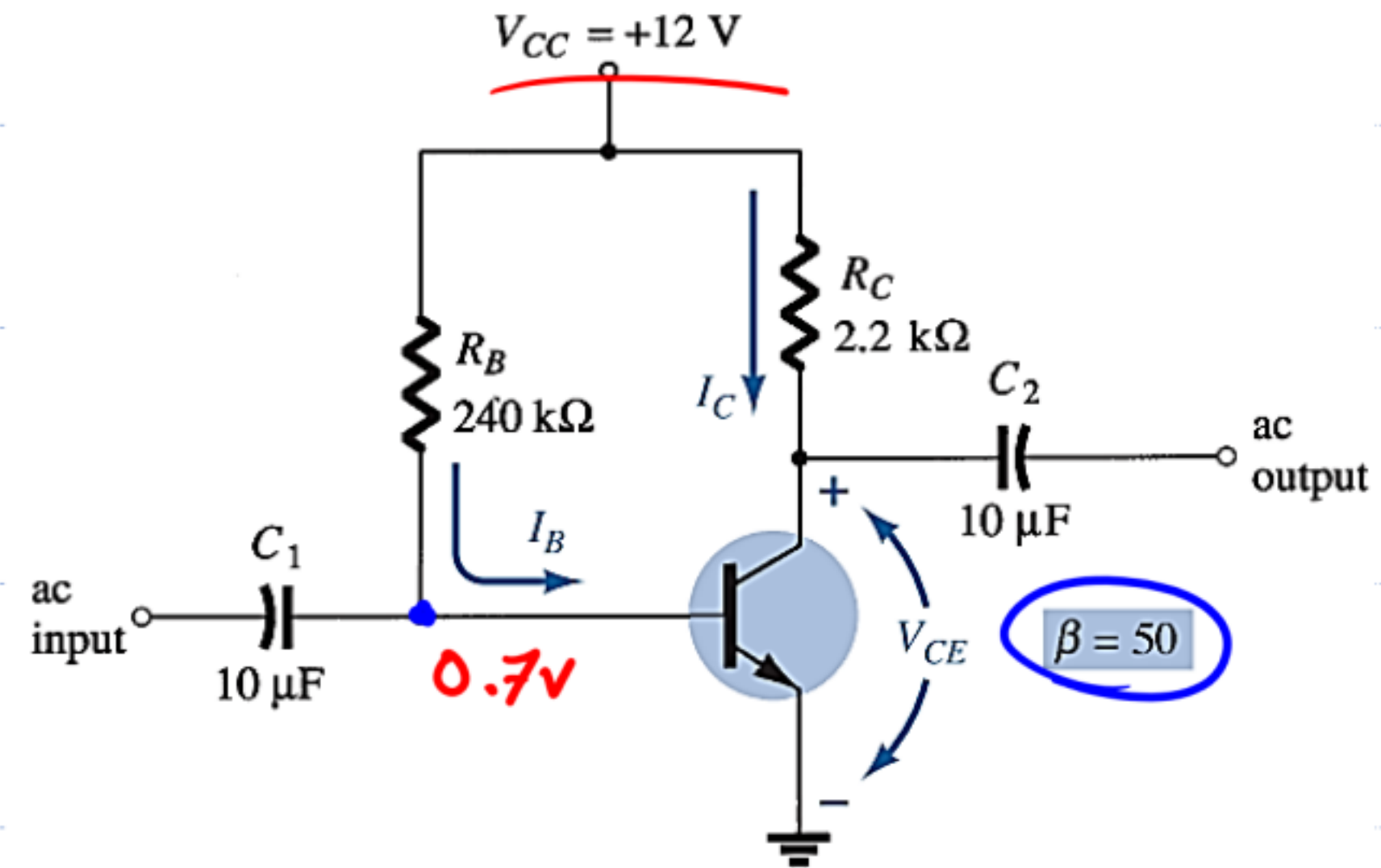
$$I_{EQ} = I_{CQ} + I_{BQ}$$

$$V_{CEQ} = V_C = V_{CC} - I_C R_C = 6.83 \text{ V}$$

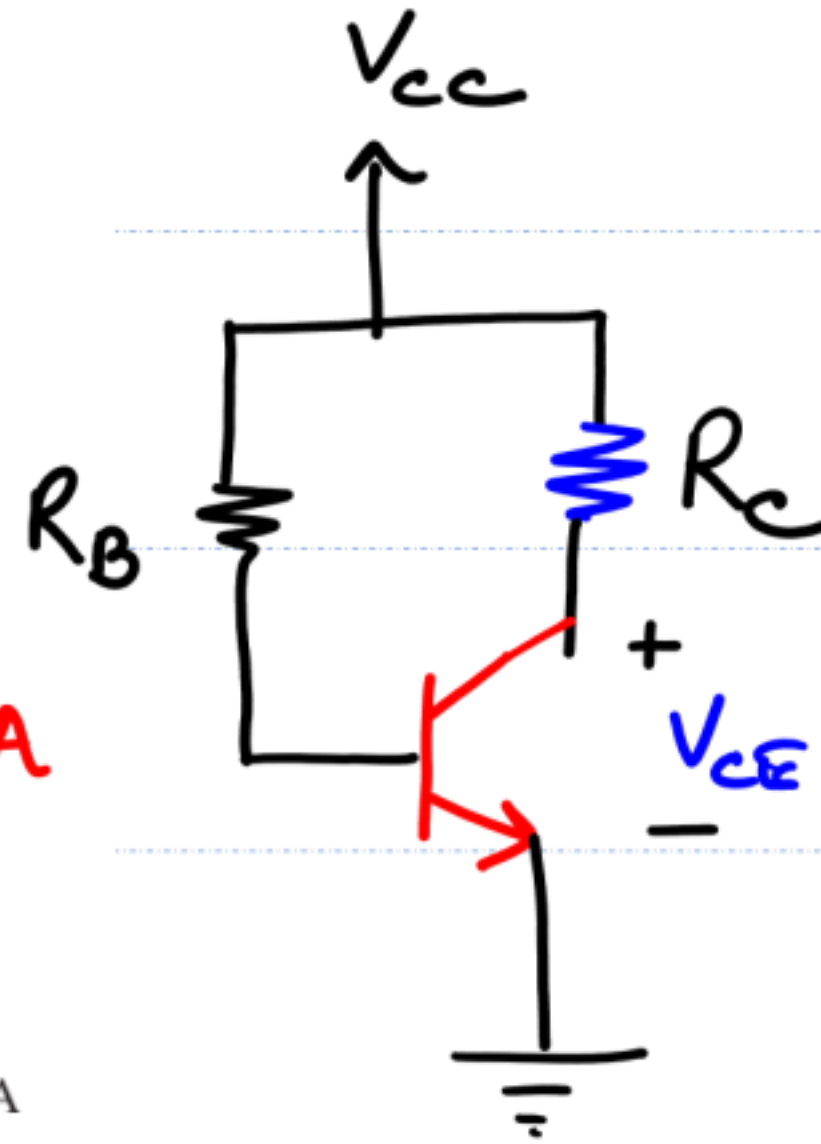
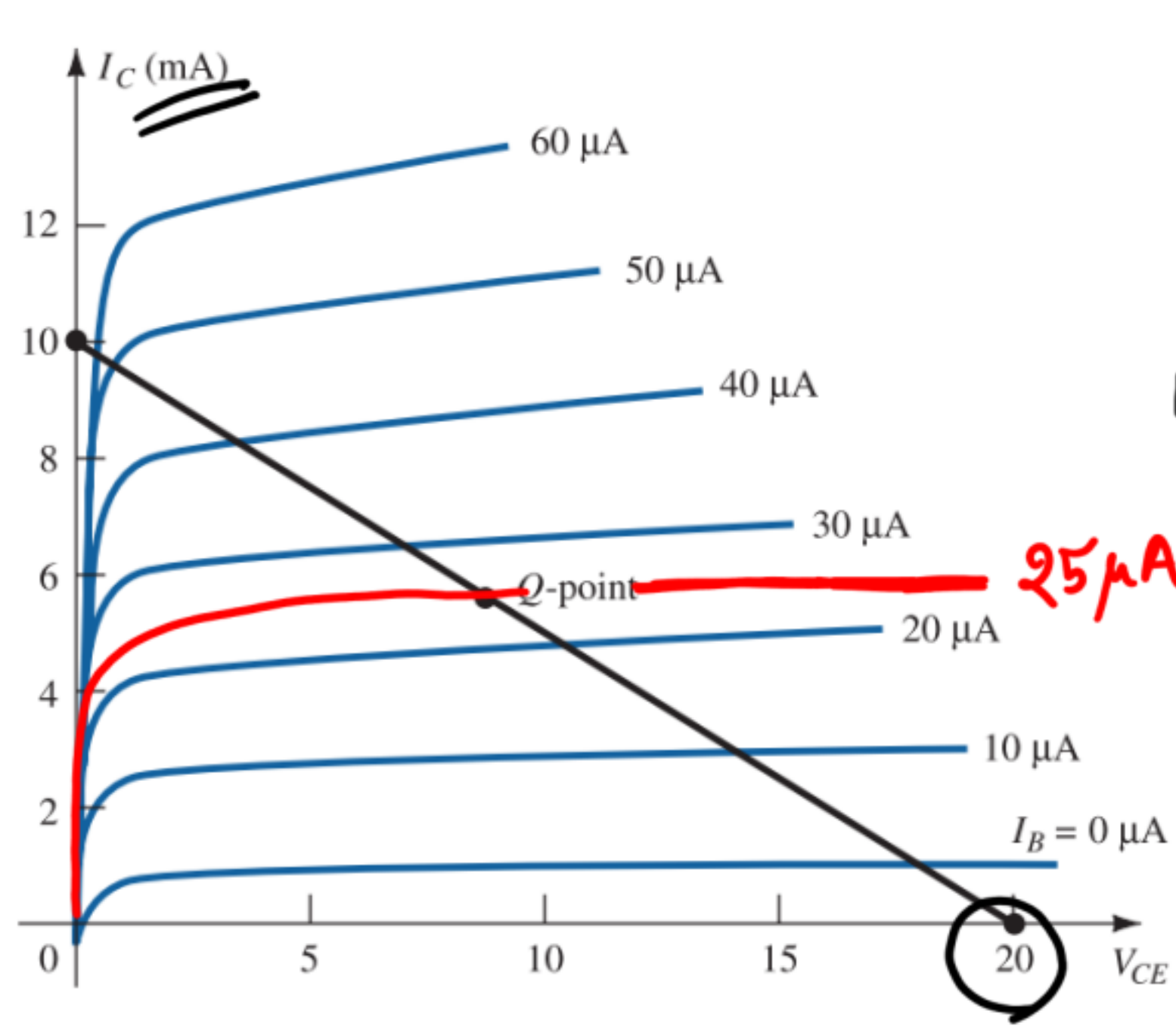
$$V_B = 0.7 \text{ V}$$

$$V_{BE} = V_B - V_C = -6.13 \text{ V} < 0.4 \text{ V}$$

$\therefore$ active mode correct



Given the load line below & the defined Q-point, determine the required values of V_{CC} , R_C , R_B for a fixed bias configuration



$$V_{CE} = V_{CC} - I_C R_C \quad \text{--- ①}$$

$$\begin{cases} I_C = 0 \therefore V_{CE} = V_{CC} = 20 \text{ V} \\ V_{CE} = 0 \therefore I_C = \frac{V_{CC}}{R_C} = 10 \end{cases}$$

$$\begin{aligned} \therefore R_C &= \frac{V_{CC}}{10} \\ &= \frac{20}{10} \\ &= 2 \text{ k}\Omega \end{aligned}$$

$$\begin{aligned} R_B &= \frac{V_{CC} - V_{BE}}{I_B} \\ &= \frac{20 - 0.7}{25 \mu} = 772 \text{ k}\Omega \end{aligned}$$

$$\therefore V_{CC} = 20 \text{ V}$$

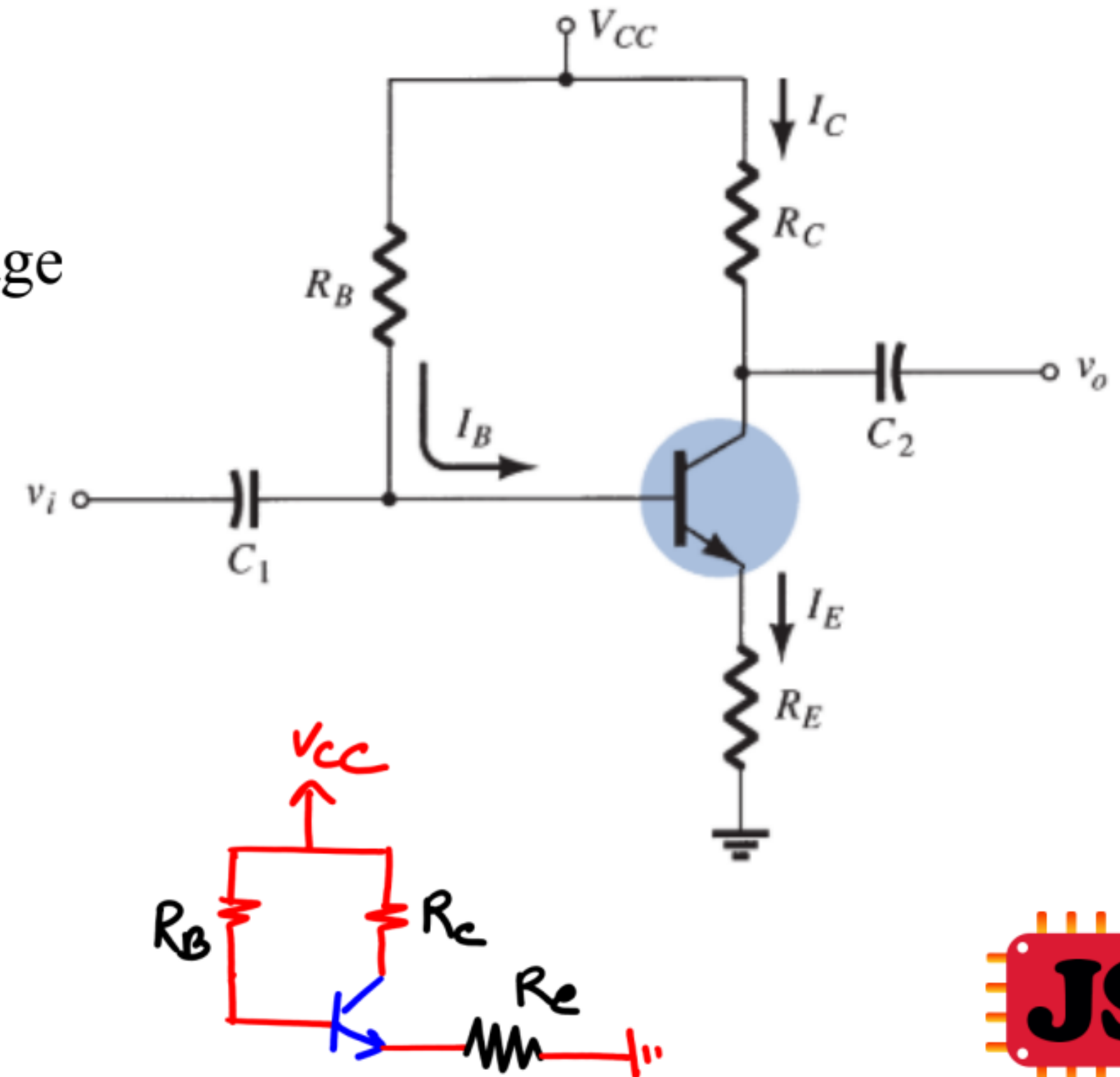
$$\therefore R_C = 2 \text{ k}\Omega$$



Emitter-Bias Configuration



- The emitter resistor R_E helps stabilize the collector current I_C against temperature fluctuations by creating negative feedback. If I_C increases due to temperature, V_E also rises, reducing the base-emitter voltage V_{BE} and countering the increase in I_C .
- The emitter bias configuration is commonly used in amplifier circuits where stability over a range of temperatures and operating conditions is essential.



Determine I_B , I_C , V_{CE} , V_C , V_E , V_B , V_{BC}

KVL from $a \rightarrow b \rightarrow c$:

$$I_E = (\beta + 1) I_B$$

$$V_{CC} - 0 = I_B R_B + V_{BE} + I_E R_E$$

$$\Rightarrow I_B = \frac{V_{CC} - V_{BE}}{R_B + (\beta + 1) R_E} = 0.04 \text{ mA}$$

$$I_C = \beta I_B = 2 \text{ mA}$$

$$I_E = 2.04 \text{ mA}$$

$$\therefore V_C = V_{CC} - I_C R_C = 16 \text{ V}$$

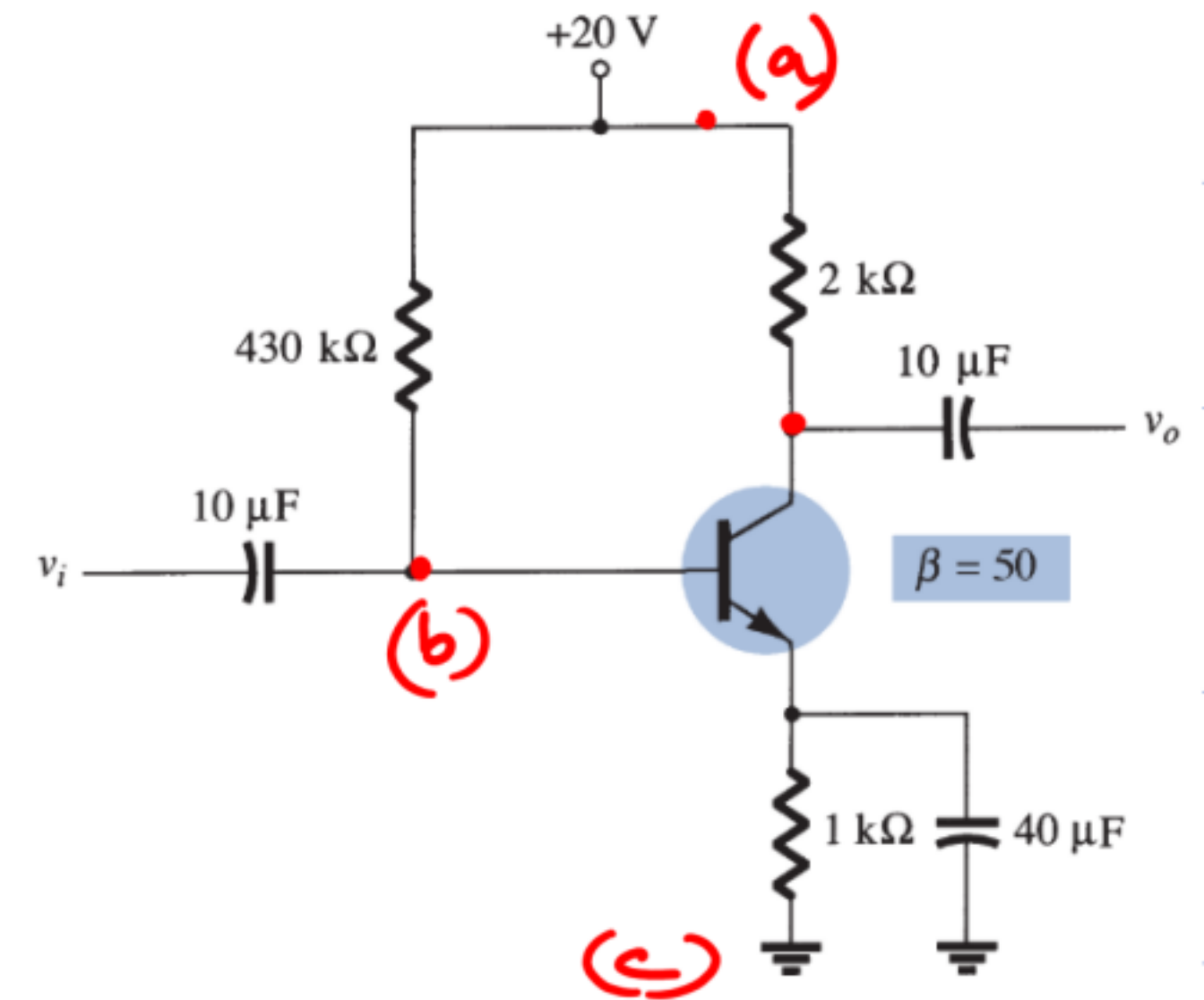
$$\therefore V_E = I_E R_E = 2.04 \text{ V}$$

$$\therefore V_B = V_{CC} - I_B R_B = 2.8 \text{ V}$$

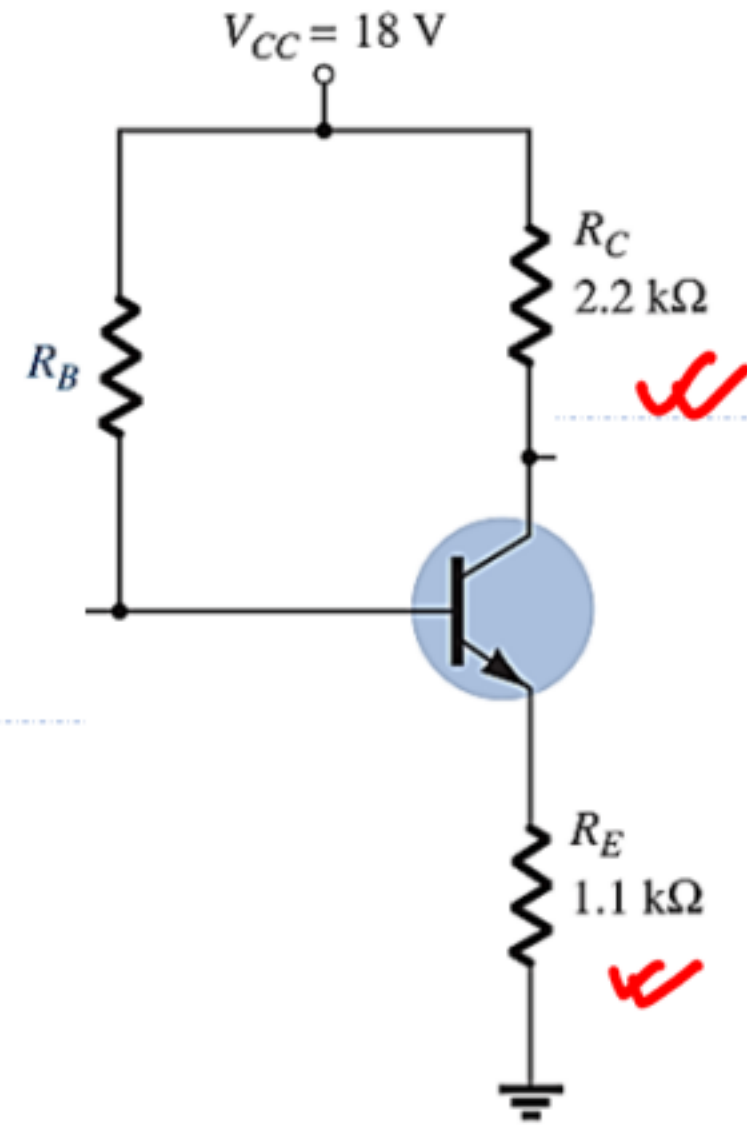
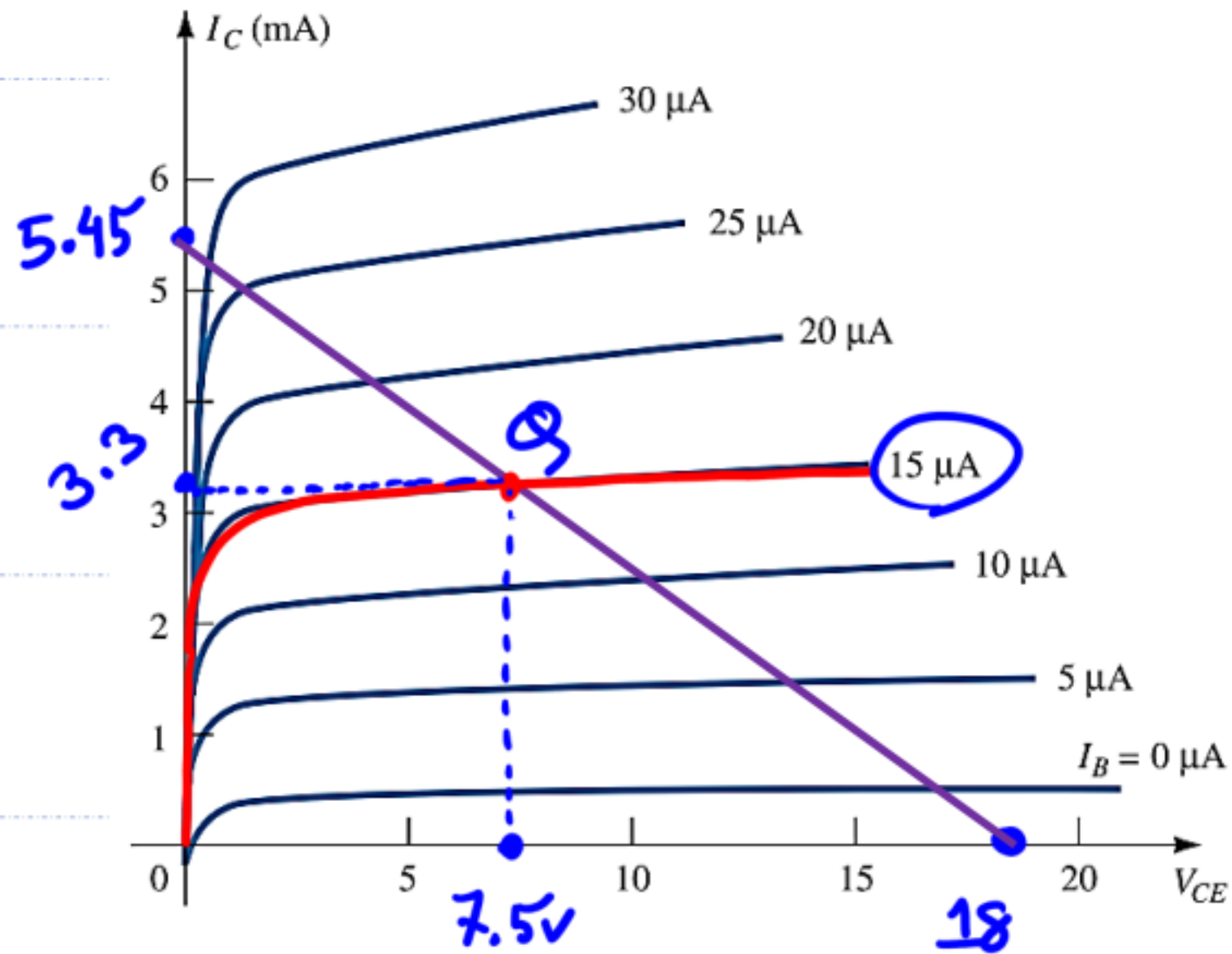
$$\begin{aligned} \therefore V_{CE} &= V_C - V_E \\ &= 13.96 \text{ V} \end{aligned}$$

$$\therefore V_{BC} = -13.2 < 0.4$$

active ✓



- ✓ Draw the load line for the network of Fig. 26a on the characteristics for the transistor appearing in Fig. 26b.
- ✓ For a Q -point at the intersection of the load line with a base current of $15 \mu\text{A}$, find the values of I_{CQ} and V_{CEQ} .
- c. Determine the dc beta at the Q -point.
- d. Using the beta for the network determined in part c, calculate the required value of R_B



From ①:

$$V_{CE} = V_{CC} - I_C(R_C + R_E) \quad \text{--- ①}$$

$$\begin{cases} I_C = 0 \rightarrow V_{CE} = V_{CE} = 18 \\ V_{CE} = 0 \rightarrow I_C = \frac{V_{CC}}{R_C + R_E} \end{cases}$$

$$= \frac{18}{3.3} = 5.45 \text{ mA}$$

from graph:

$$V_{CEQ} = 7.5 \text{ V}$$

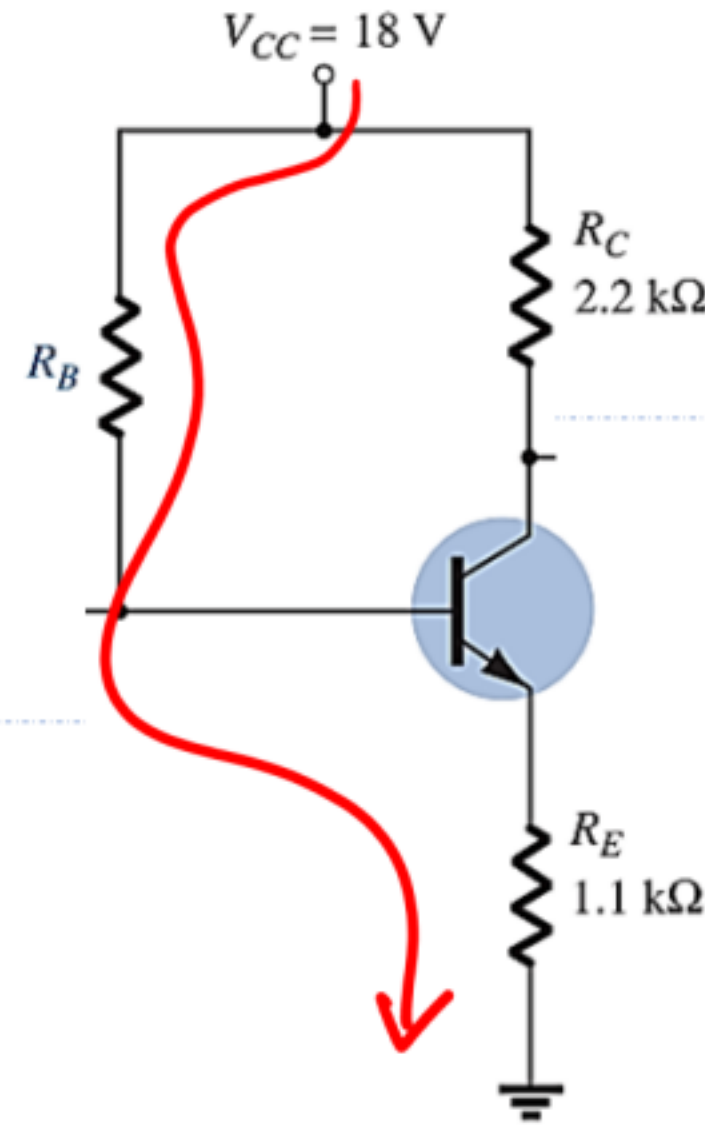
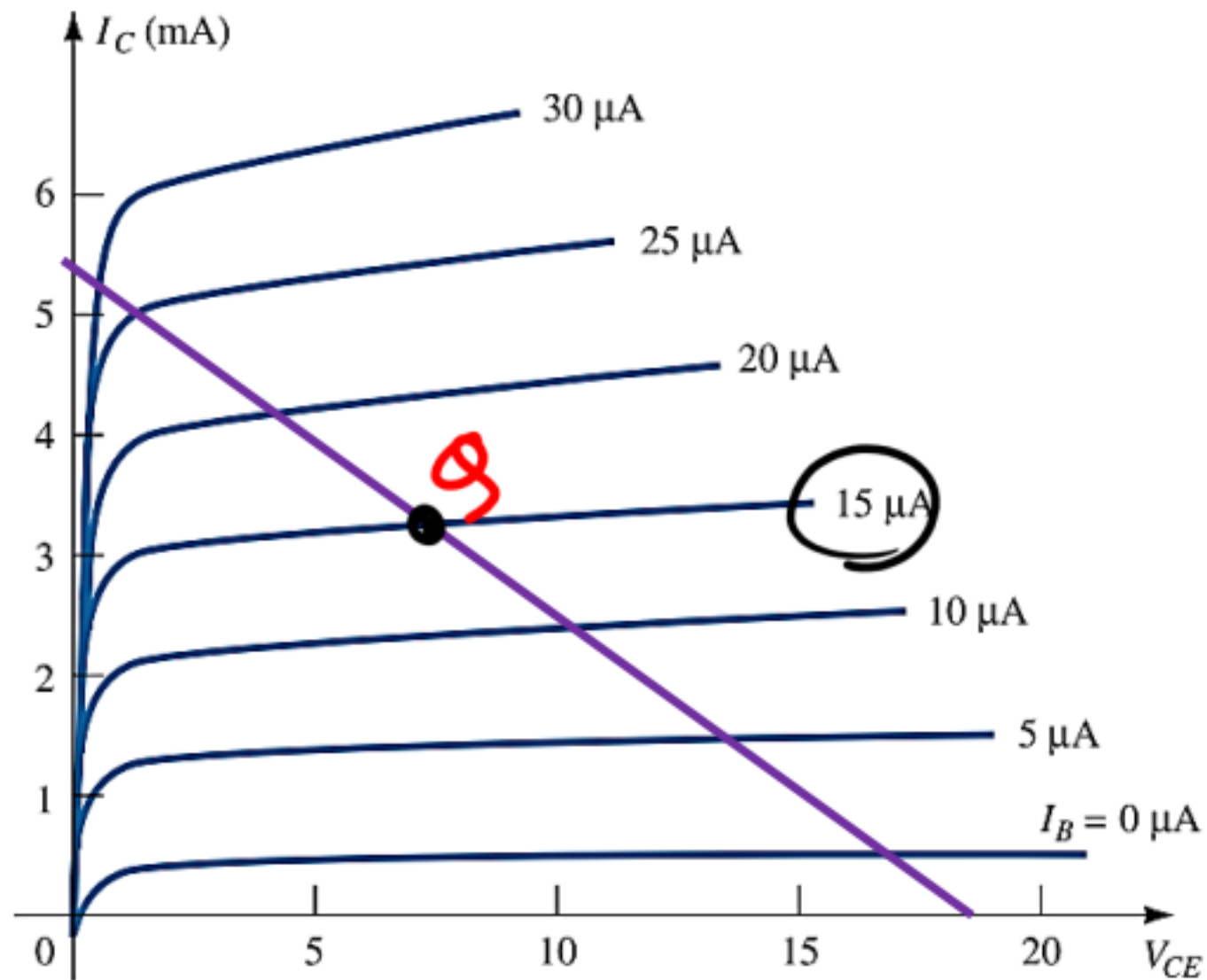
$$I_{CQ} = 3.3 \text{ mA}$$

$$\therefore \beta_{dc} = \frac{I_C}{I_B} = \frac{3.3 \text{ mA}}{15 \mu\text{A}}$$

$$= 220$$

$$V_{CC} - 0 = I_C R_C + V_{CE} + I_E R_E \quad \text{--- ①}$$

- Draw the load line for the network of Fig. 26a on the characteristics for the transistor appearing in Fig. 26b.
- For a Q -point at the intersection of the load line with a base current of $15\ \mu\text{A}$, find the values of I_{CQ} and V_{CEQ} .
- Determine the dc beta at the Q -point.
- Using the beta for the network determined in part c, calculate the required value of R_B .



$$V_{CC} = I_B R_B + V_{BE} + I_E R_E$$

$$\Rightarrow R_B = \frac{V_{CC} - V_{BE}}{I_B} - \frac{I_E R_E}{I_B}$$

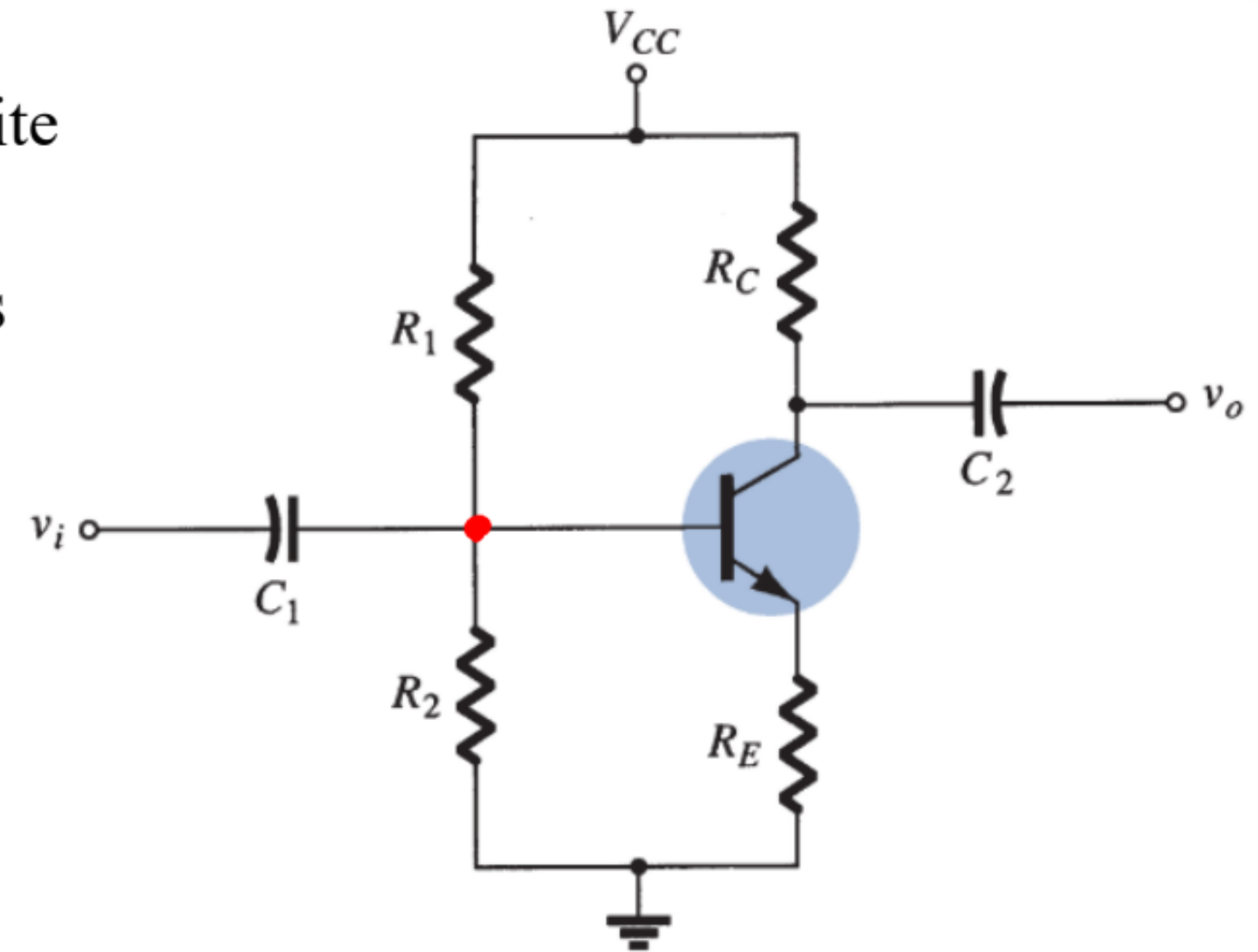
$$= \frac{18 - 0.7}{15\ \mu} - (\beta + 1) R_E$$

$$= 910\ \text{k}\Omega$$

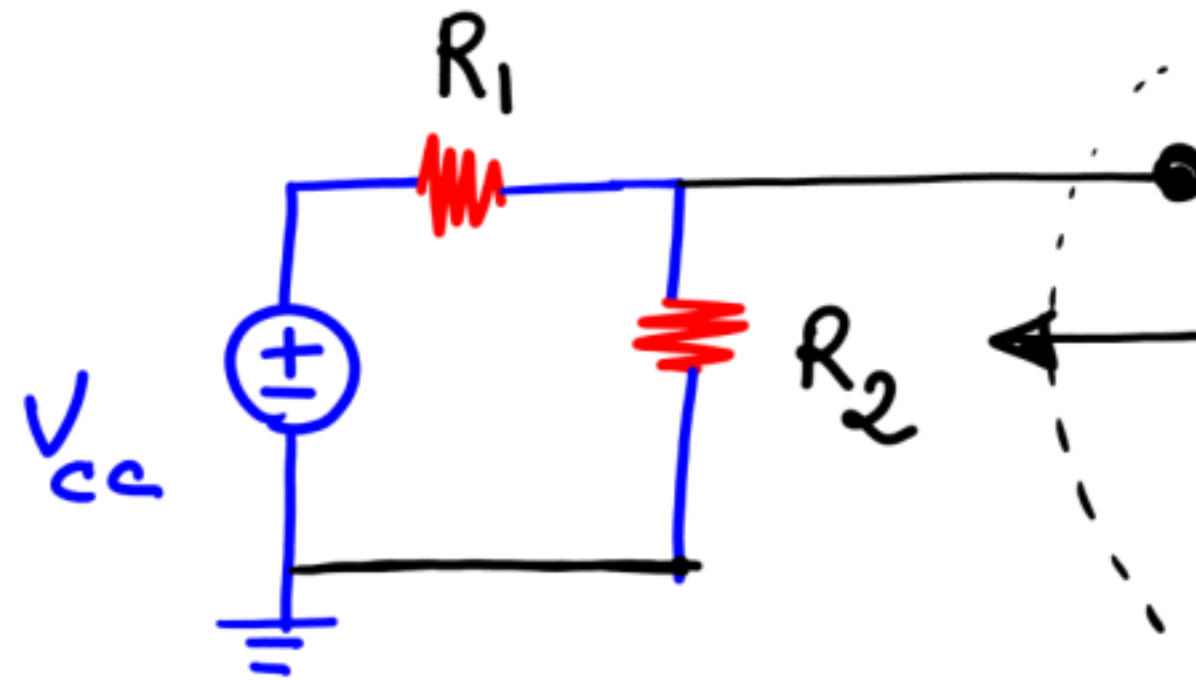
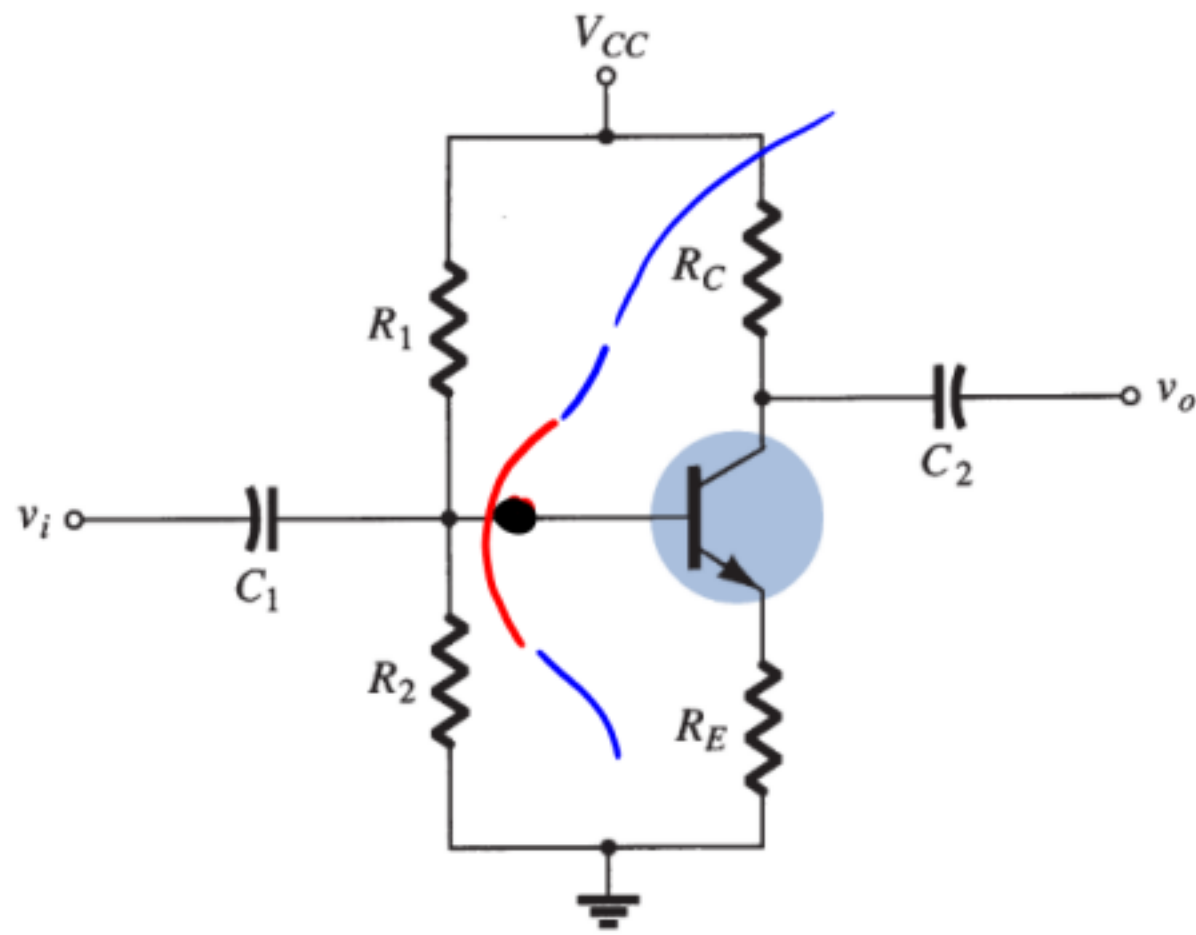
Voltage-Divider Bias Configuration



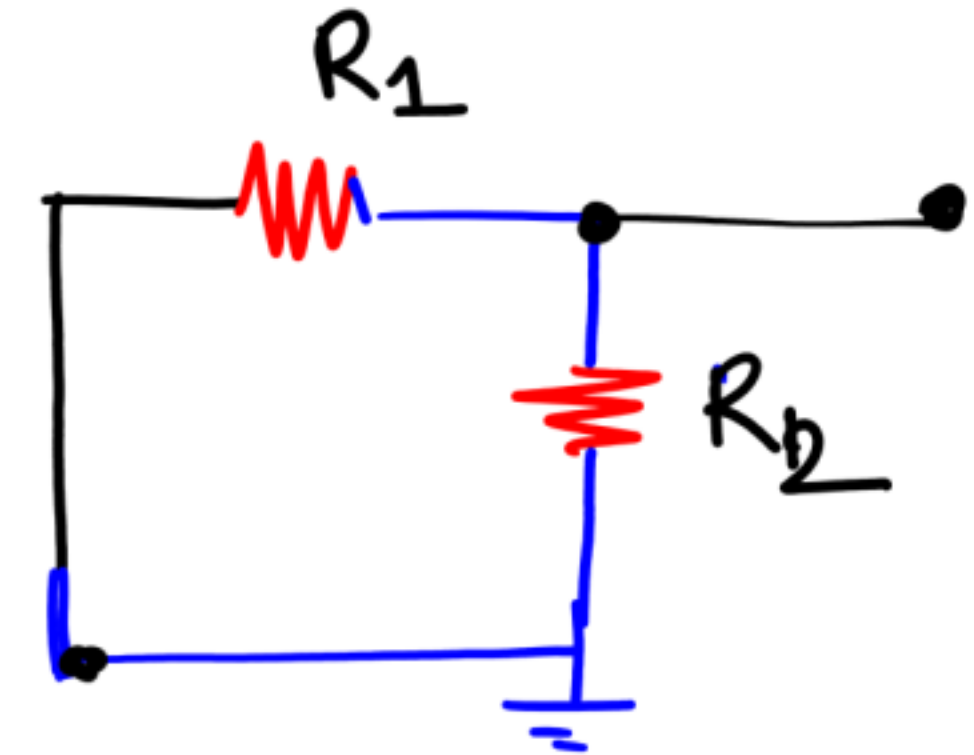
- It relies on a pair of resistors to set the base voltage, helping to maintain a stable collector current I_C despite changes in temperature or transistor properties.
- This configuration is widely used because it provides a high level of stability with a single power supply.
- **Technique:** Find Thevenin's Voltage and Resistance at the base.



Thevenin's voltage and Resistance at the Base of a BJT



$$V_{th} = V_{R_2} = V_{ce} \frac{R_2}{R_1 + R_2} \quad \text{--- ①}$$



$$R_{th} = R_1 \parallel R_2 \\ = \frac{R_1 R_2}{R_1 + R_2} \quad \text{--- ②}$$



Evaluate all node voltages and branch currents ($\beta = 100$)

$$I_B = \frac{(5 - 0) - 0.7}{R_{th} + (\beta + 1)R_E} = 0.012 \text{ mA}$$

$$I_c = \beta I_B = 1.2 \text{ mA}$$

$$I_E = 1.212 \text{ mA}$$

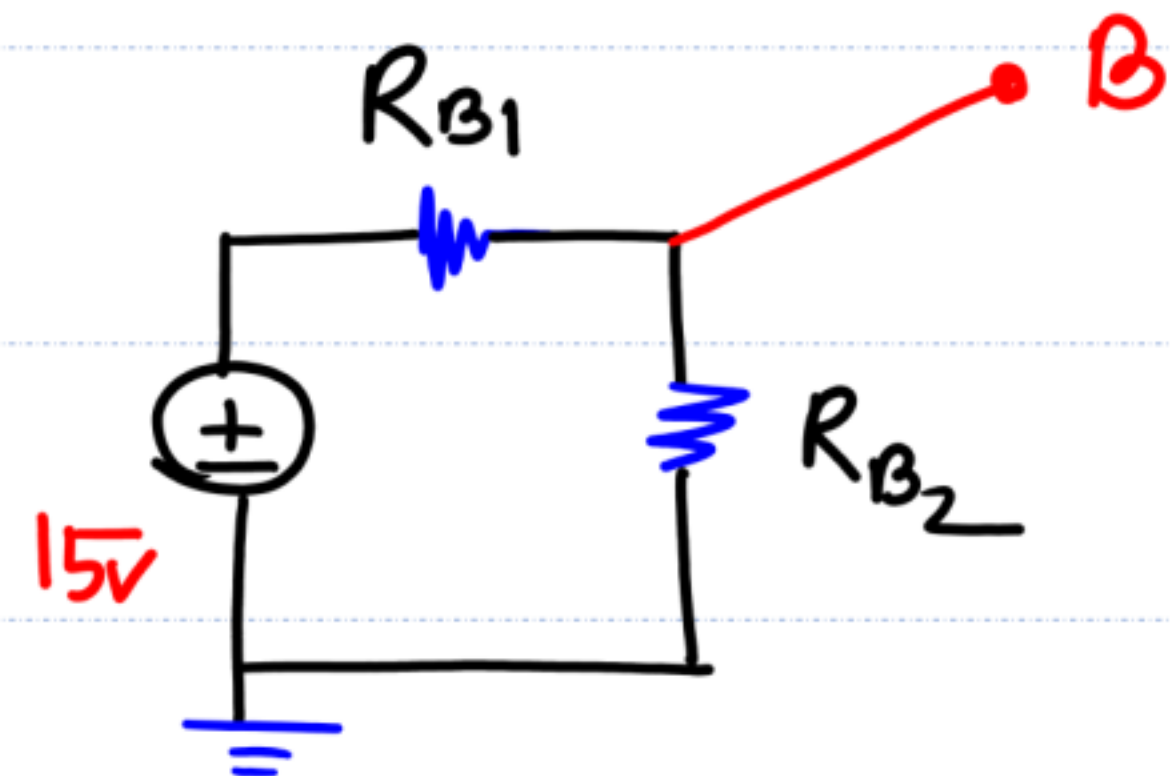
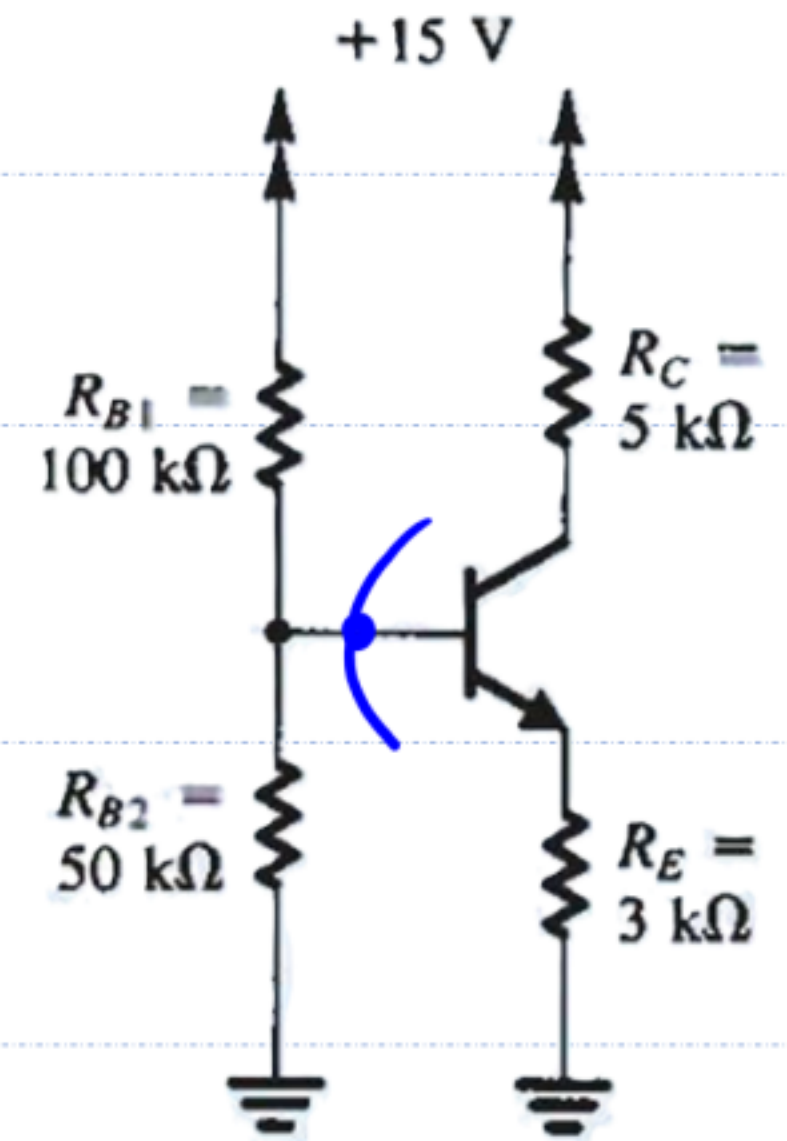
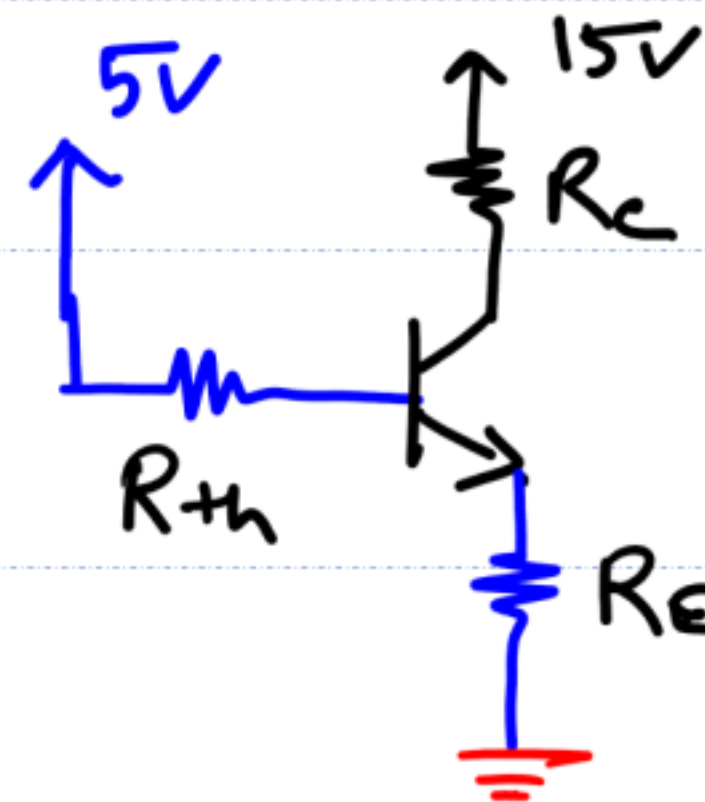
$$V_B = 5 - I_B R_{th} = 4.6 \text{ V}$$

$$V_C = 15 - I_C R_C = 9 \text{ V}$$

$$V_E = I_E R_E = 3.64 \text{ V}$$

$$R_{th} = R_{B1} \parallel R_{B2} = 33.3 \text{ k}\Omega$$

$$V_{th} = 15 \frac{R_{B2}}{R_{B1} + R_{B2}} = 5 \text{ V}$$



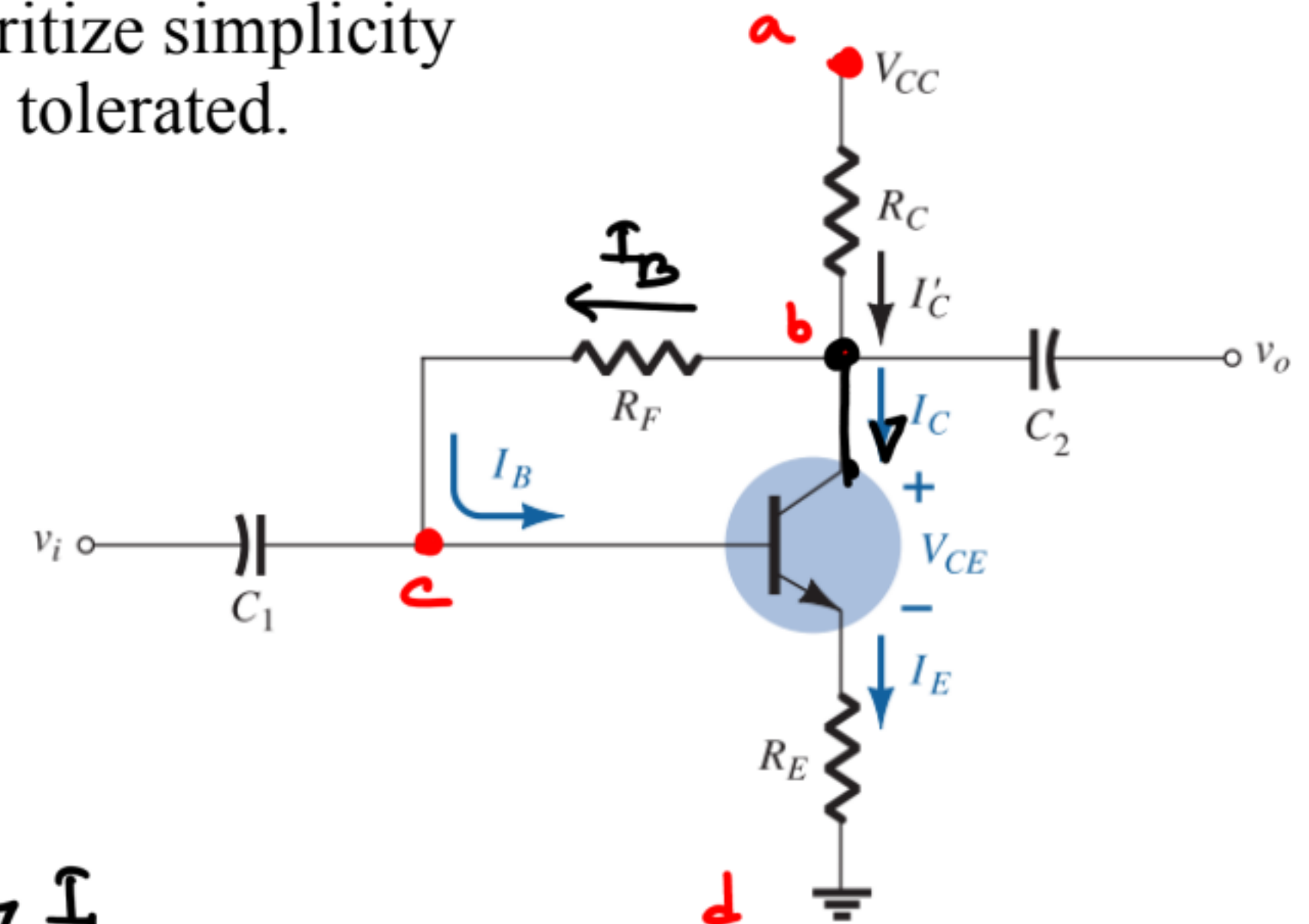
Collector Feedback Configuration



- In this configuration, the base resistor is connected to the collector rather than directly to the power supply, creating a form of negative feedback that helps stabilize the (Q-point).
- The collector feedback bias configuration is commonly used in amplifier circuits, especially in applications that prioritize simplicity and where minor variations in operating point can be tolerated.

(a → b → c → d) →

$$V_{cc} - 0 = V_{ce} = I_E R_c + I_B R_f + V_{BE} + I_E R_E$$



$$I_E = I_B + I_C$$

Determine the quiescent levels of I_{C_Q} and V_{CE_Q} for the network

KVL at $a \rightarrow b \rightarrow c \rightarrow d$:

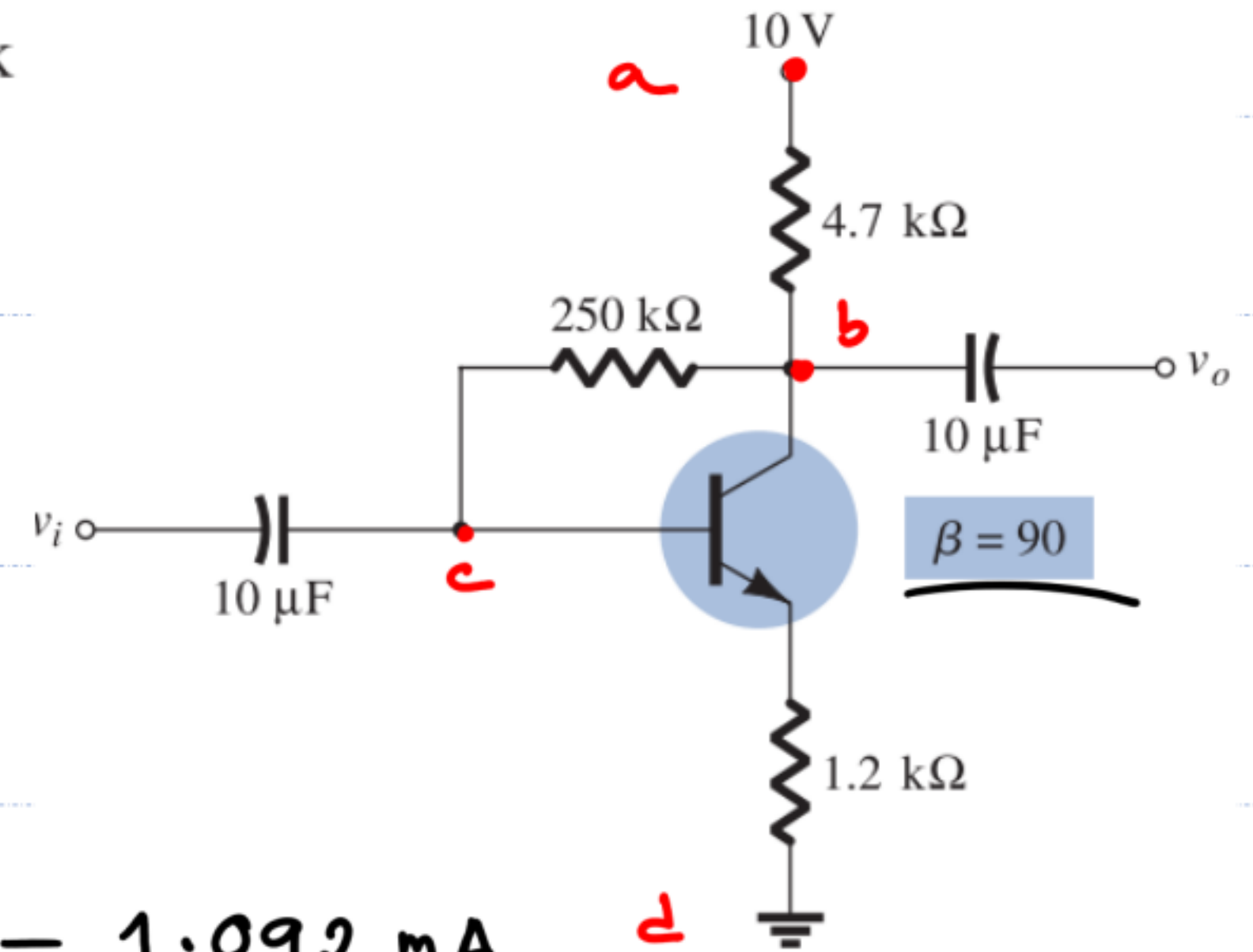
$$10 - 0 = 4.7 I_E + 250 I_B + 0.7 + 1.2 I_E$$

$$9.3 = 5.9 (\beta + 1) I_B + 250 I_B$$

$$\therefore I_B = 0.012 \text{ mA} \rightarrow \text{condition}$$

$$I_{C_Q} = \beta I_{B_Q} = 1.08 \text{ mA}$$

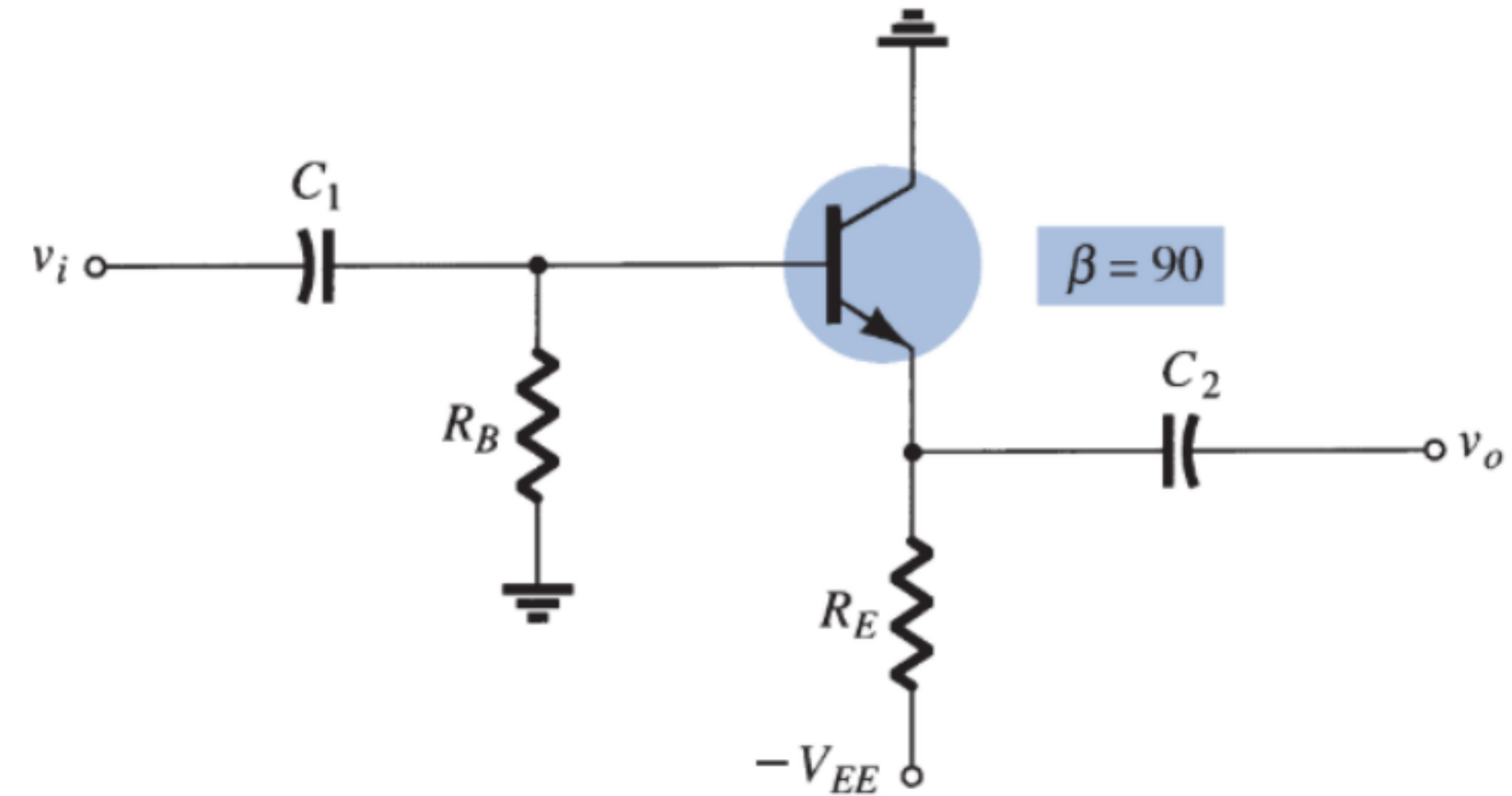
$$\begin{aligned} \therefore V_{CE_Q} &= V_C - V_E = (10 - 4.7 I_E) - (1.2 I_E) \\ &= 3.56 \text{ V} \end{aligned}$$



Emitter-Follower Configuration



- It's a widely used configuration of BJT that provides high input impedance, low output impedance, and a voltage gain close to unity (1)
- It is called an "emitter follower" because the output voltage at the emitter "follows" the input voltage applied to the base.



Determine V_{CEQ} and I_{EQ} for the network of Fig. 48.

KVL along $a \rightarrow b \rightarrow d$:

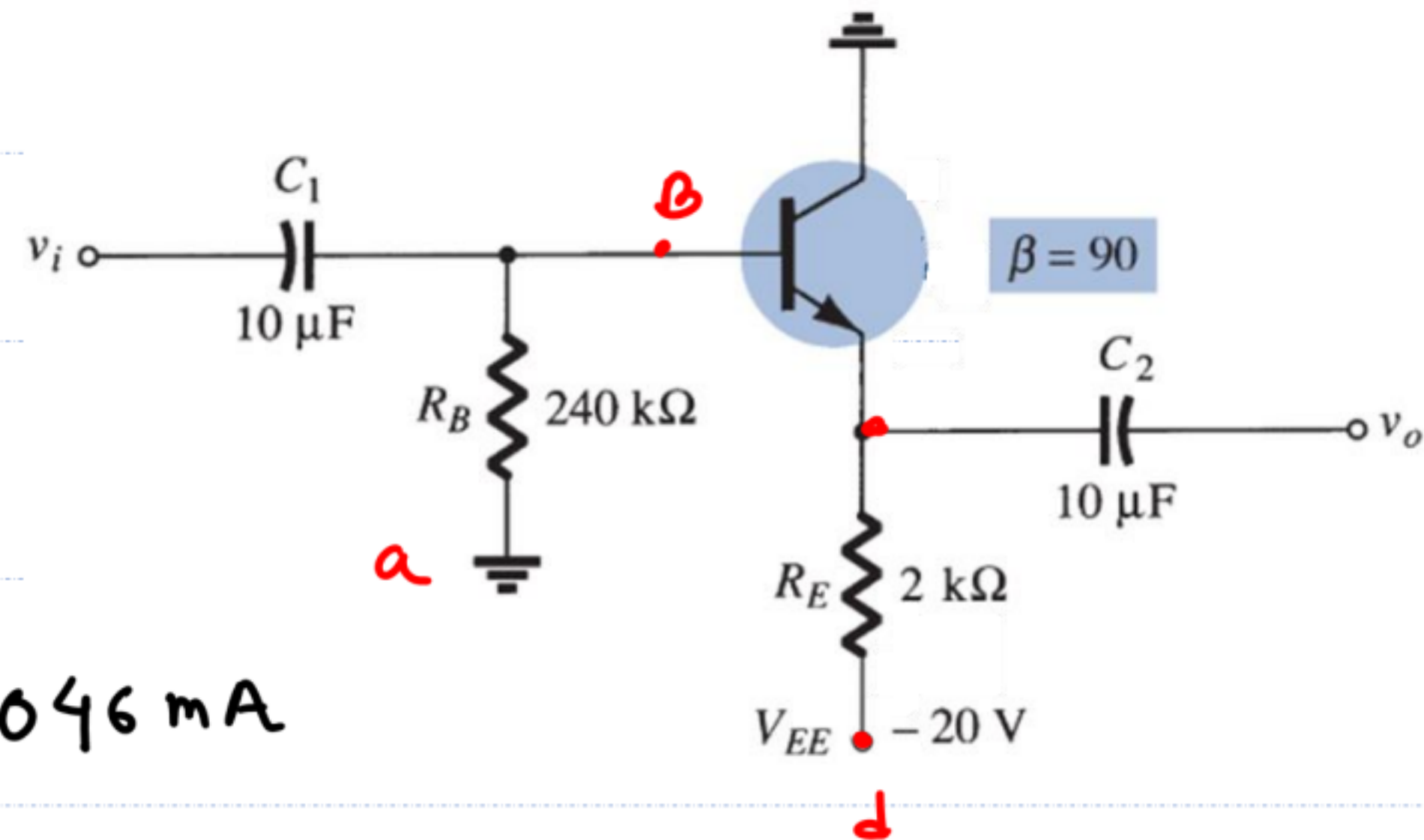
$$0 - (-20) = 20 = I_B R_B + V_{BE} + I_E R_E$$

$$\therefore I_{BQ} = \frac{20 - 0.7}{R_B + (\beta + 1) R_E} = 0.046 \text{ mA}$$

$$I_{CQ} = \beta I_{BQ} = 4.116 \text{ mA}$$

$$\therefore I_{EQ} = 4.162 \text{ mA}$$

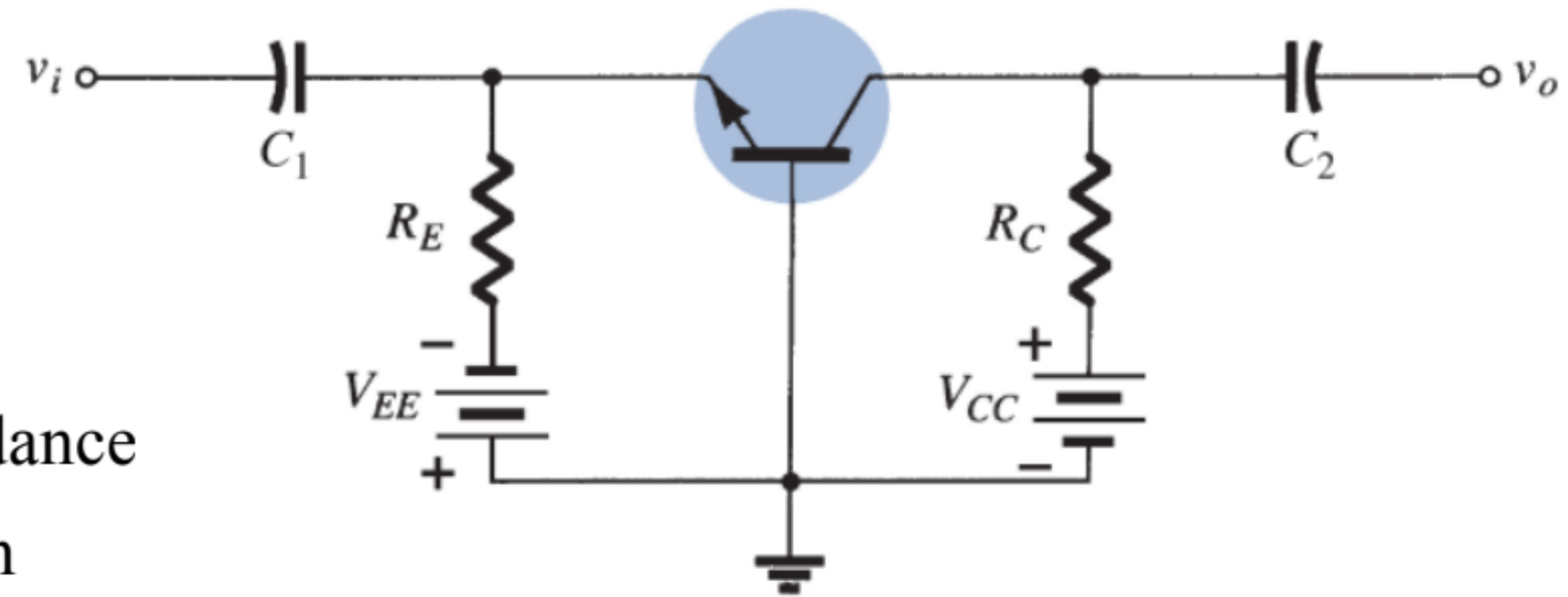
$$\begin{aligned} V_{CEQ} &= 0 - V_{EQ} = -(-20 + I_E R_E) \\ &= 20 - I_E R_E \\ &= 11.68 \text{ V} \end{aligned}$$



Common-Base Configuration



- The base is grounded & acts as the common terminal for both the input and output, which the input signal is applied to the emitter & the output is taken from the collector.
- ✓ Has low input impedance, High output impedance
- Provides high voltage gain & stable operation at high frequencies.



B E C →
i



Determine I_E , I_B , V_{CE} , V_{CB} for the given circuit.

• KVL at loop ① :

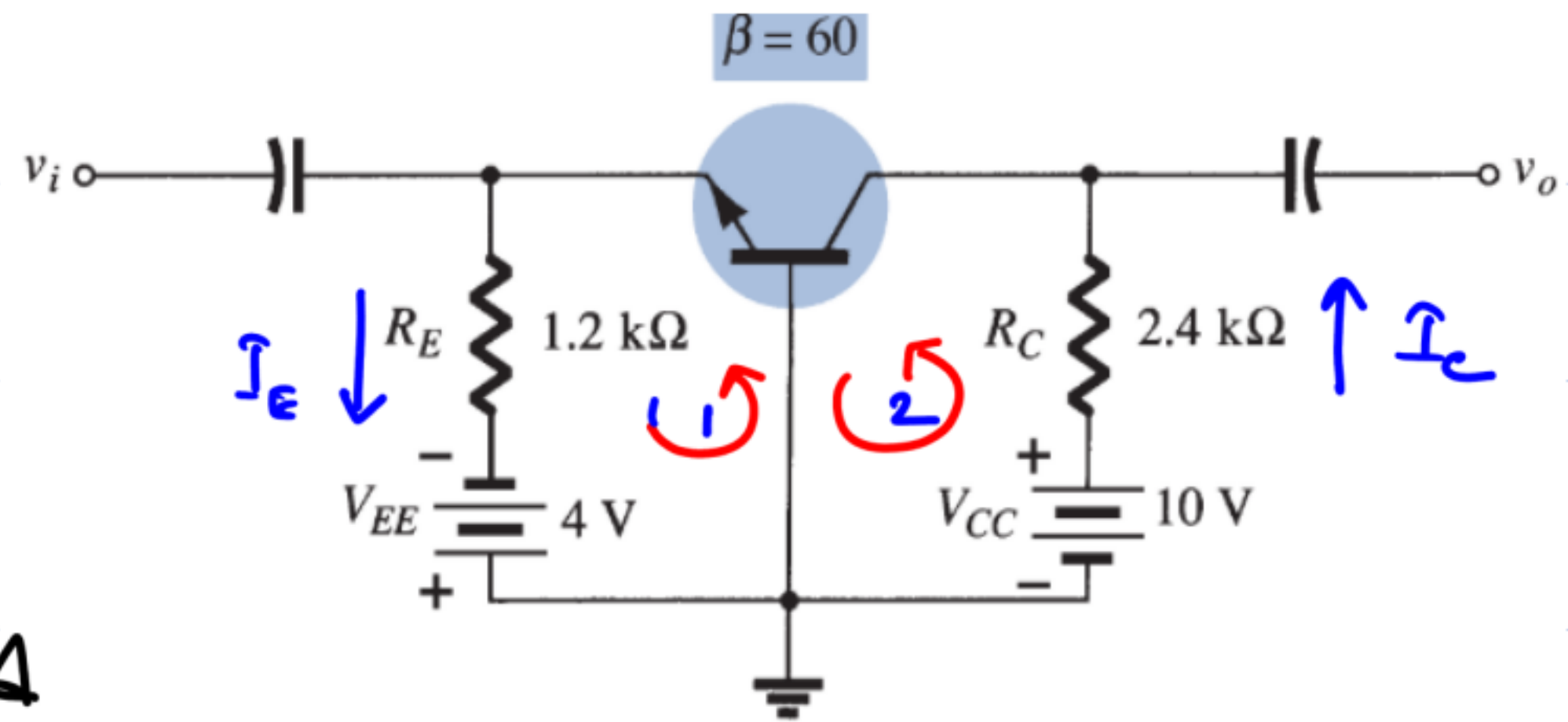
$$-V_{EE} + V_{BE} + I_E R_E = 0$$

$$\therefore I_E = \frac{4 - 0.7}{1.2} = 2.75 \text{ mA}$$

• KVL at loop ② : $-V_{CC} + I_C R_C + V_{CB} = 0$

$$\therefore V_{CB} = 10 - 2.4 \times 2.705 \\ = 3.51 \text{ V}$$

$$\therefore V_{CE} = V_{CB} + V_{BE} = 4.21 \text{ V}$$

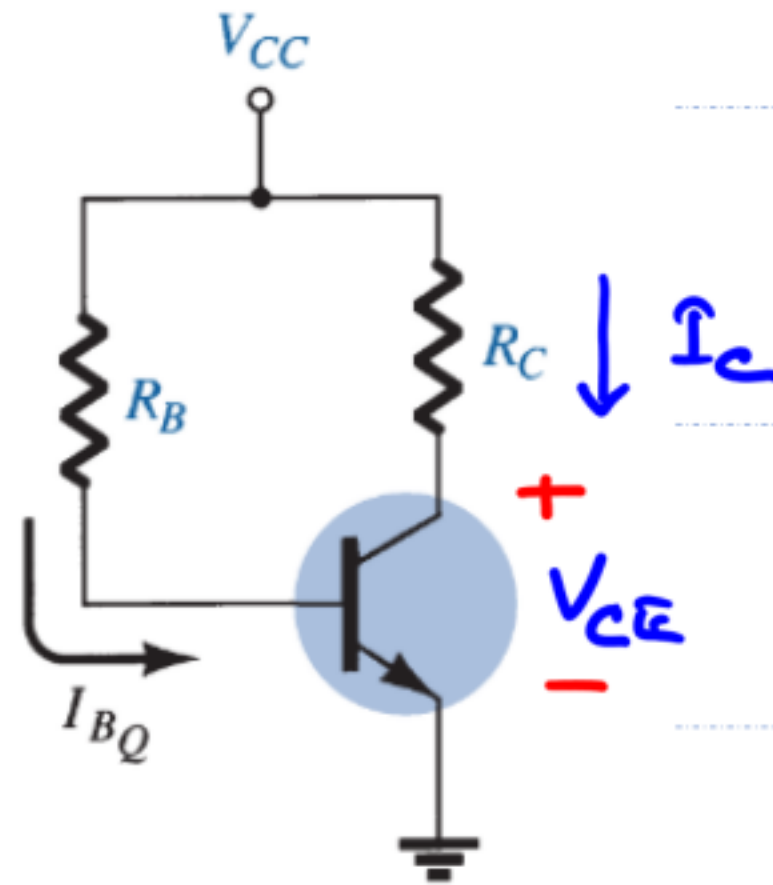
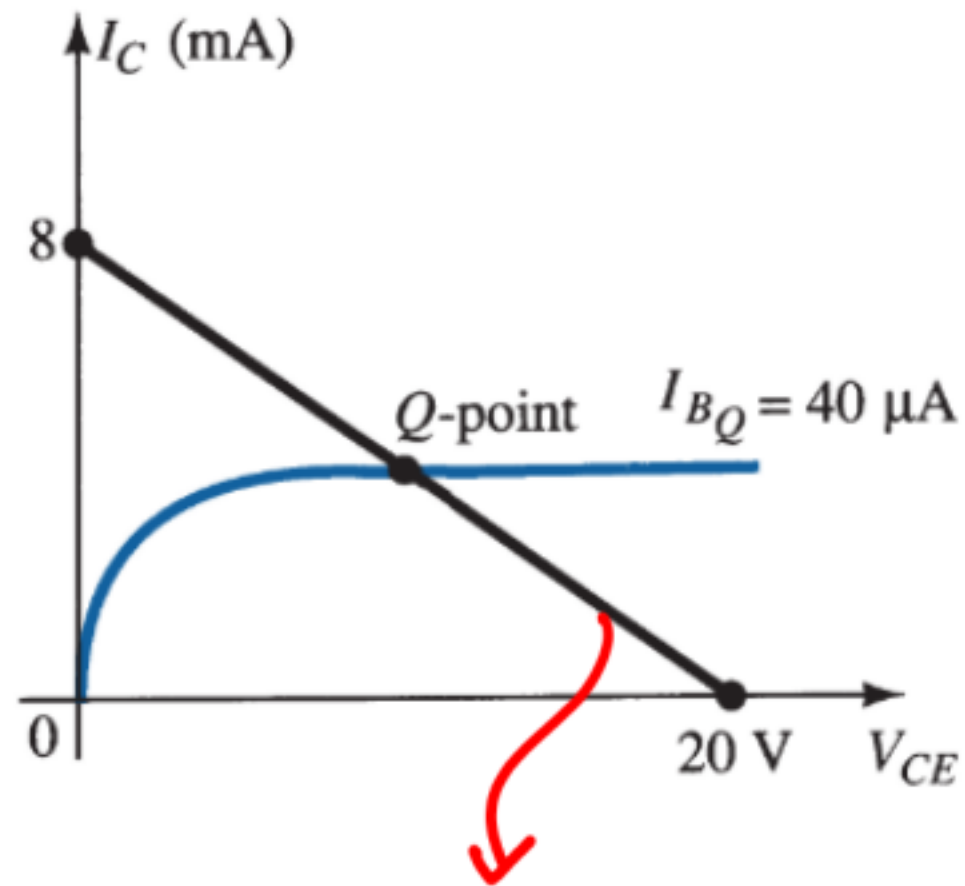


$$\frac{I_C}{I_E} = \alpha$$

$$\therefore I_C = \left(\frac{\beta}{1+\beta} \right) I_E \\ = 2.705 \text{ mA}$$



EXAMPLE 21 Given the device characteristics of Fig. 59a, determine V_{CC} , R_B , and R_C for the fixed-bias configuration of Fig. 59b.



$$R_B = \frac{V_{CC} - V_{BE}}{I_B}$$

$$= \frac{20 - 0.7}{0.040} \text{ k}\Omega$$

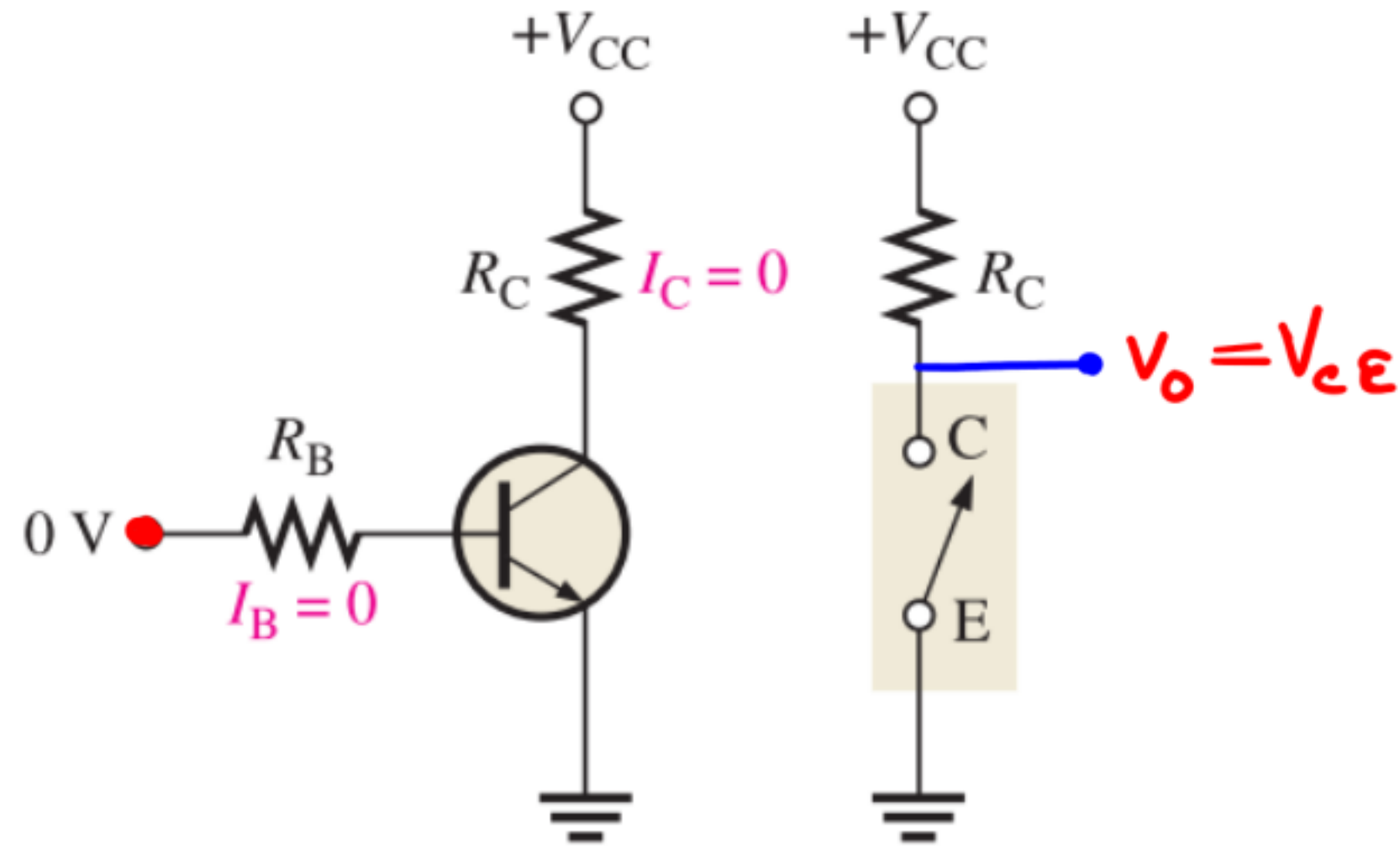
$$= 482.5 \text{ k}\Omega$$

$$V_{CE} = V_{CC} - I_C R_C$$

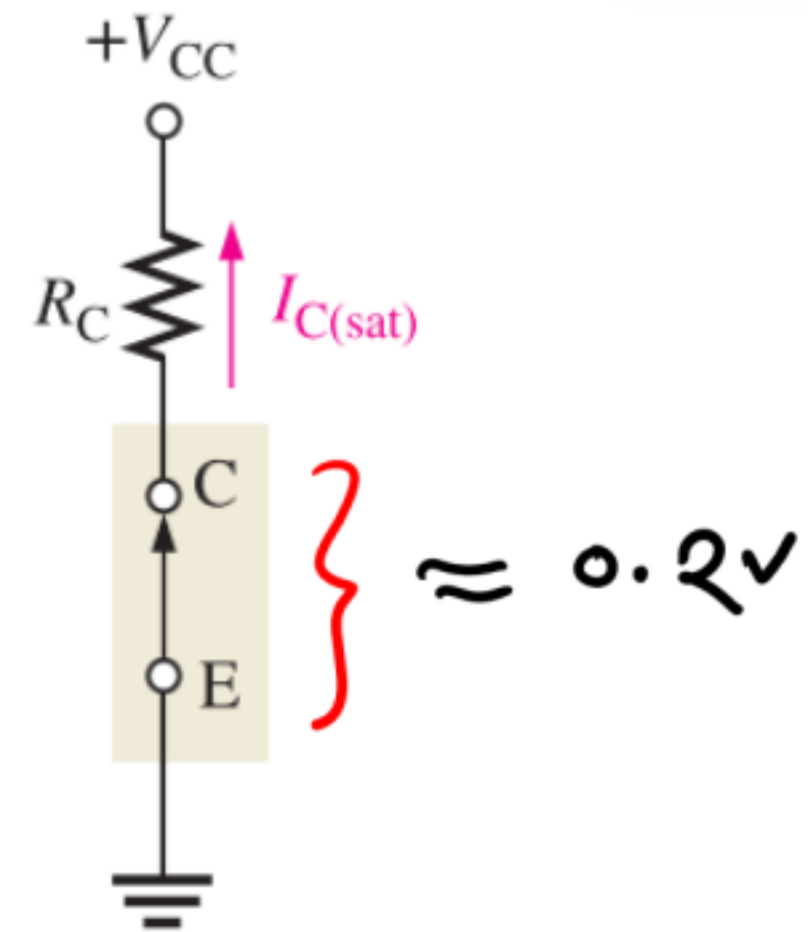
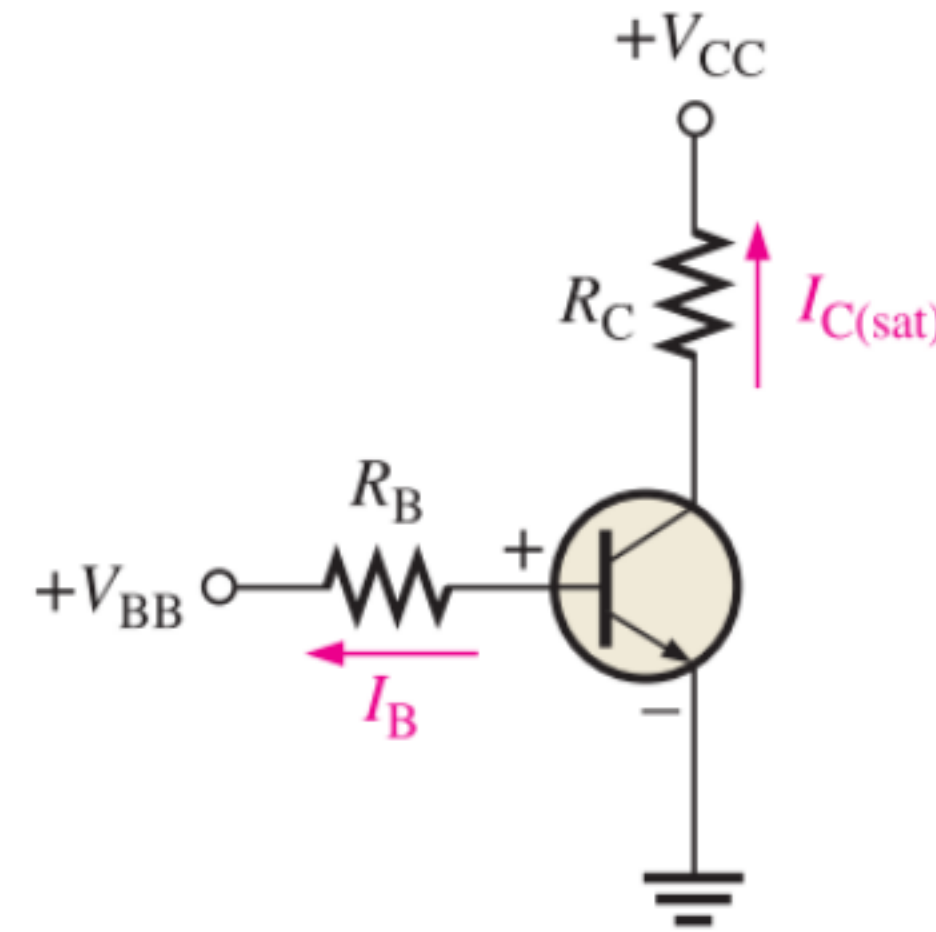
$$\begin{cases} I_C = 0 \therefore V_{CE} = V_{CE} = 20 \text{ V} \text{ ——— } \checkmark \\ V_{CE} = 0 \therefore I_C = \frac{V_{CC}}{R_C} = \frac{20}{R_C} = 8 \end{cases} \rightarrow R_C = \frac{20}{8}$$

$$= 2.5 \text{ k}\Omega$$

Transistor as a switch

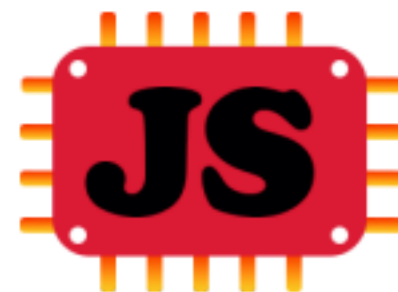


$\{ V_{BE} < 0.7 \}$
cut-off

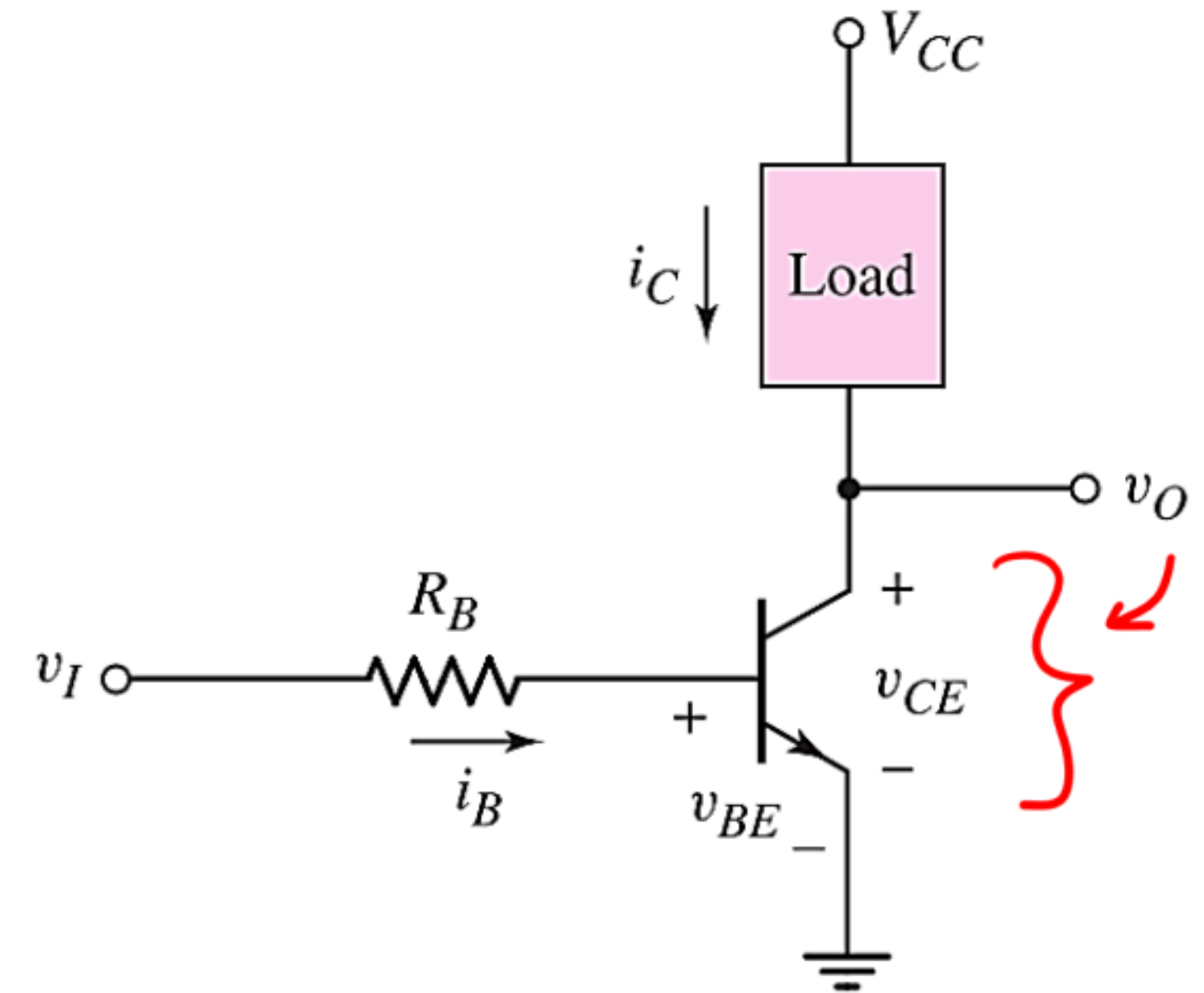
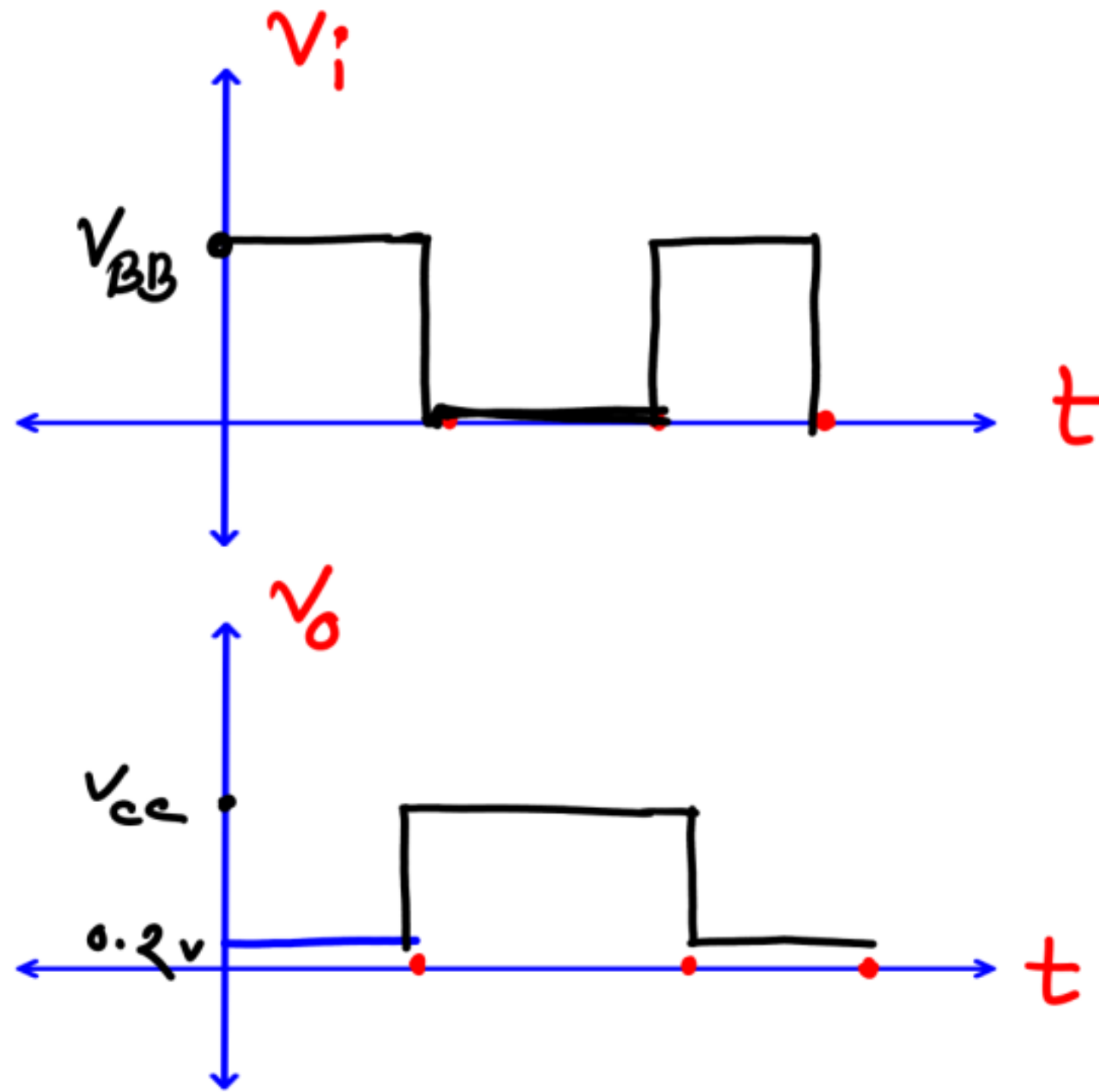


$\{ V_{CE} > 0.4V \}$
saturation

V_{BB}	V_{CE}
0V	V_{CC}
high	$0.2 \approx 0$



Inverter circuit (NOT gate)



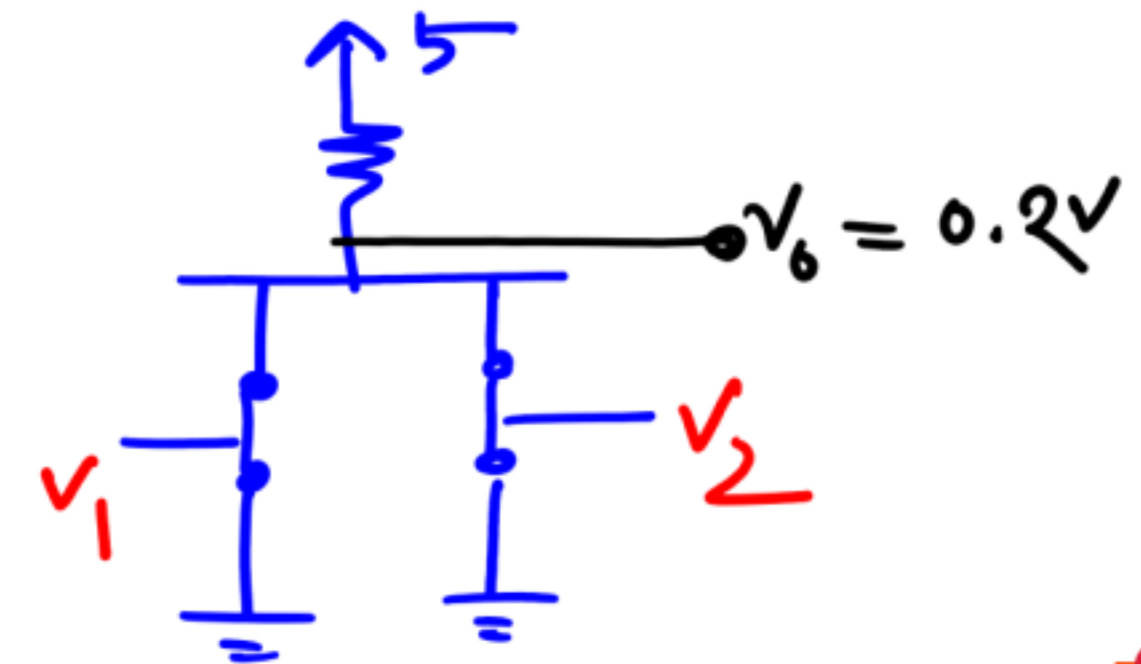
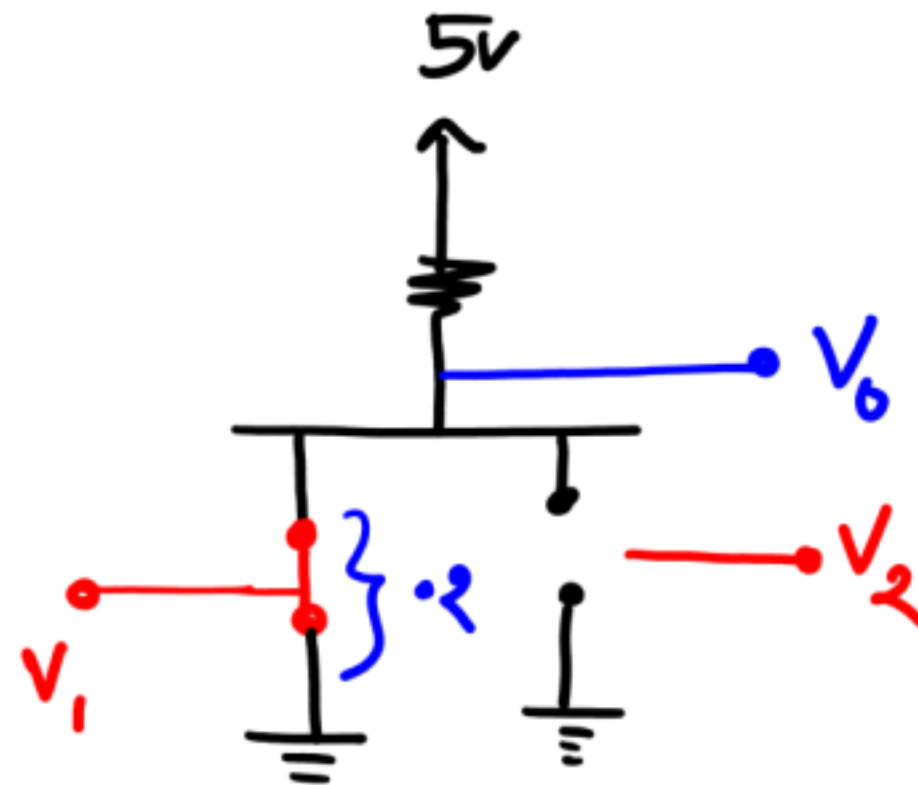
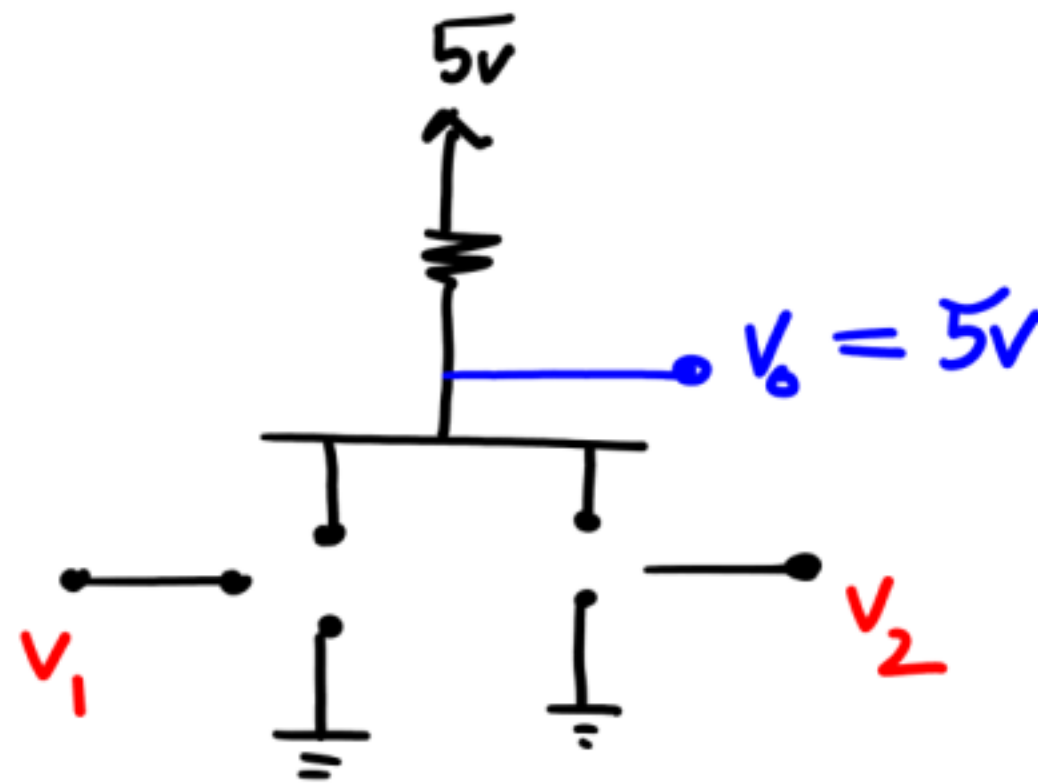
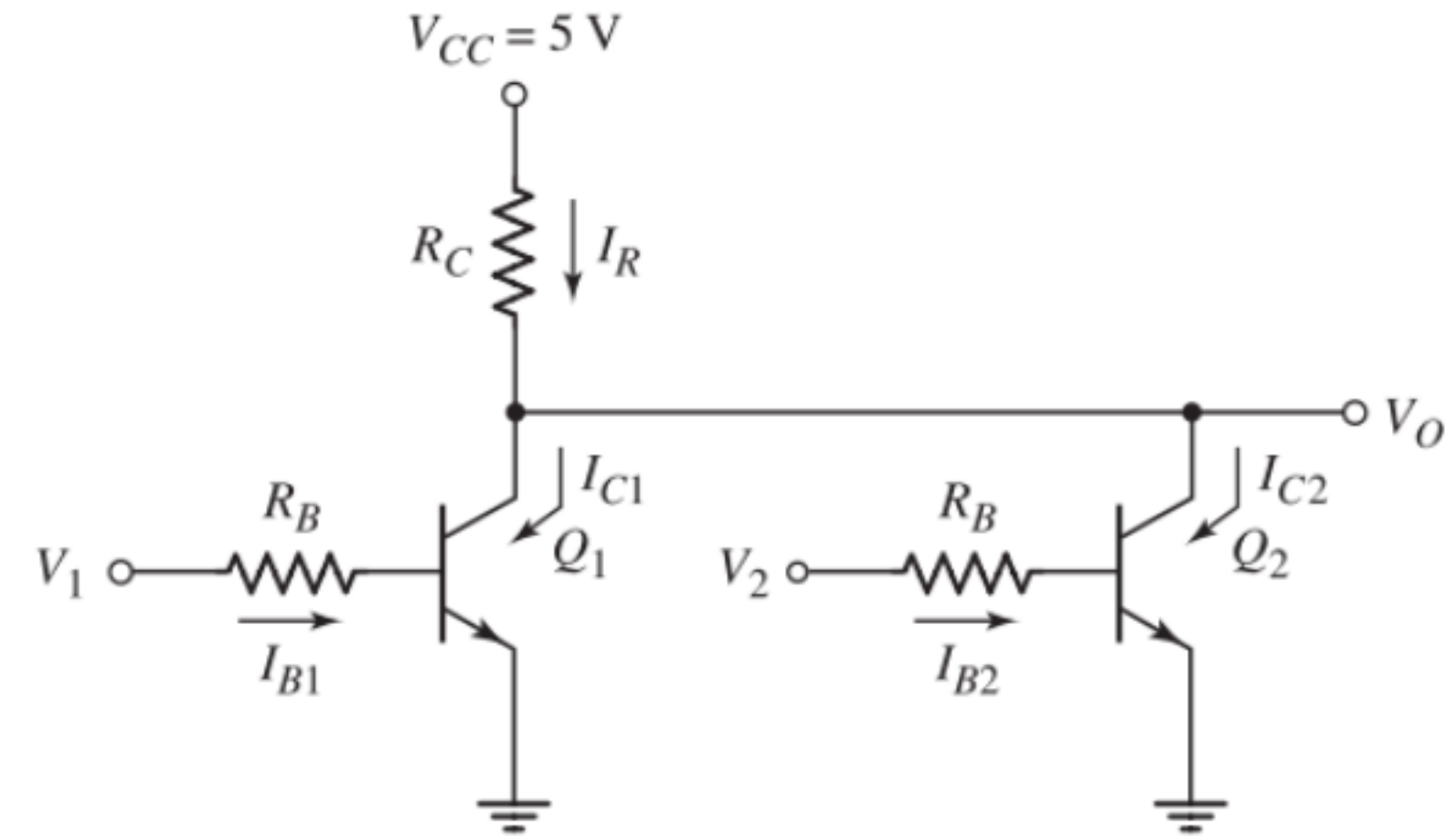
NOR Gate implementation



Table 5.2 The bipolar NOR logic circuit response

V_1 (V)	V_2 (V)	V_O (V)
0	0	5v
5	0	0.2v
0	5	0.2v
5	5	0.2v

} $\equiv 0$



Minimum base current to enter saturation stage



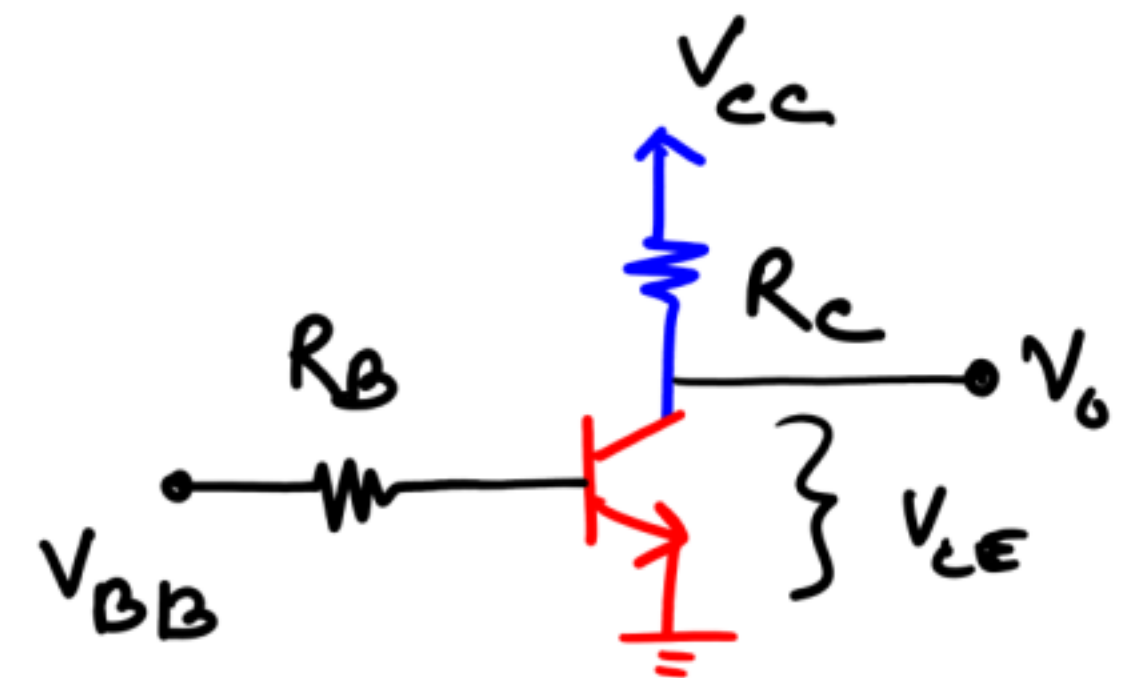
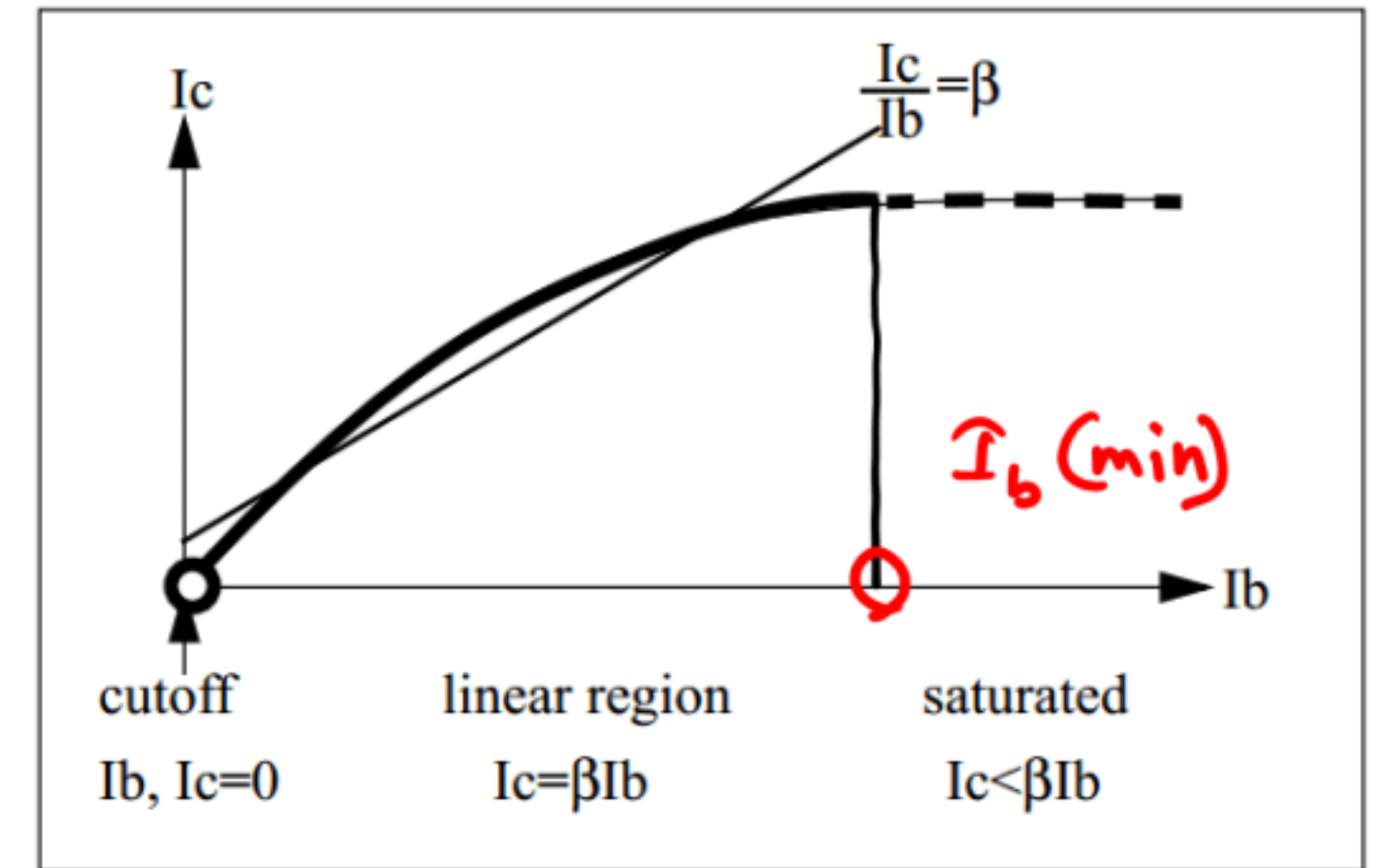
- As I_B increases, I_C also increases.
- Finally, when the transistor enters saturation, I_C saturates & becomes constant.

$$I_C(sat) = \frac{(V_{CC} - V_{CE})}{R_C} = \frac{V_{CC} - 0.2}{R_C}$$

- The minimum base current required to enter saturation,

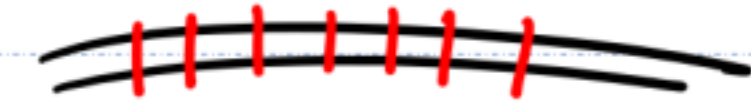
$$I_B(min) = \frac{I_C(sat)}{\beta_{dc}}$$

$$I_C = \frac{V_{CC} - V_{CE}}{R_C} \quad \text{sat.} \quad (V_{BE} > 0.4V)$$



What minimum value of I_B is required to saturate this transistor if $\beta_{dc} = 200$? Neglect $V_{CE}(sat)$

Calculate the maximum value of R_B when $V_{in} = 5V$



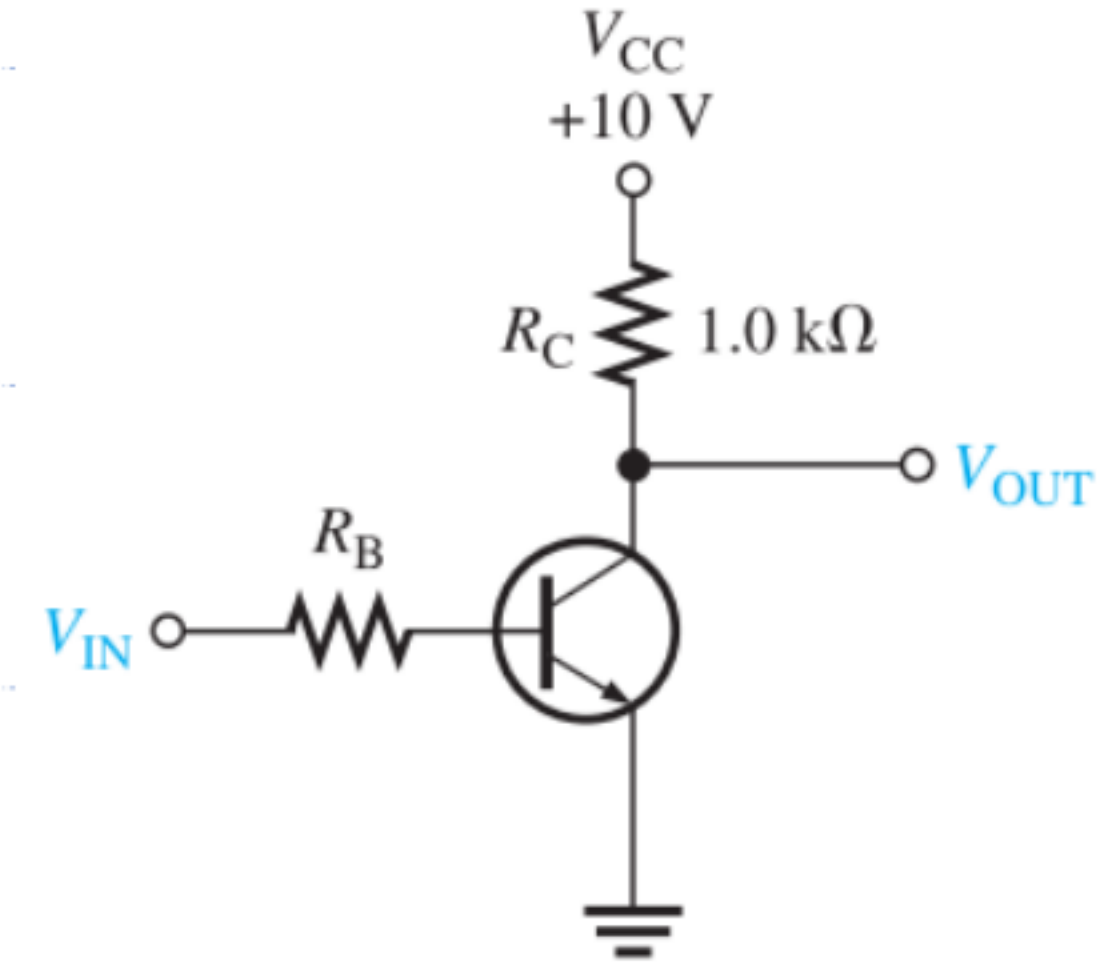
$$I_B(\min) = \frac{I_c}{\beta_{dc}} = \frac{V_{cc} - V_{ce}}{R_c \beta_{dc}}$$

$$= \frac{10 - 0}{1 \times 200}$$

$$R_B(\max) = \frac{V_{in} - V_{BE}}{I_B(\min)} = 0.05 \text{ mA}$$

$$= \frac{5 - 0.7}{0.05}$$

$$= 86 \text{ k}\Omega$$



The LED in this circuit requires 30mA to emit sufficient light. So, I_C should be apx. 30mA. For the following circuit values, determine the amplitude of the square wave input voltage necessary to make sure that the transistor saturates. Use double the minimum value of base current as safety margin to ensure saturation.

Given: $V_{CC} = 9V$, $V_{CE}(sat) = 0.3V$, $R_C = 220\Omega$, $R_B = 3.3\text{ k}\Omega$, $\beta_{DC} = 50$, $V_{LED} = 1.6V$

$$V_P = V_{BE} + I_B R_B = 0.7 + 3.3 I_B (\text{actual}) = 4.96V \approx 5V$$

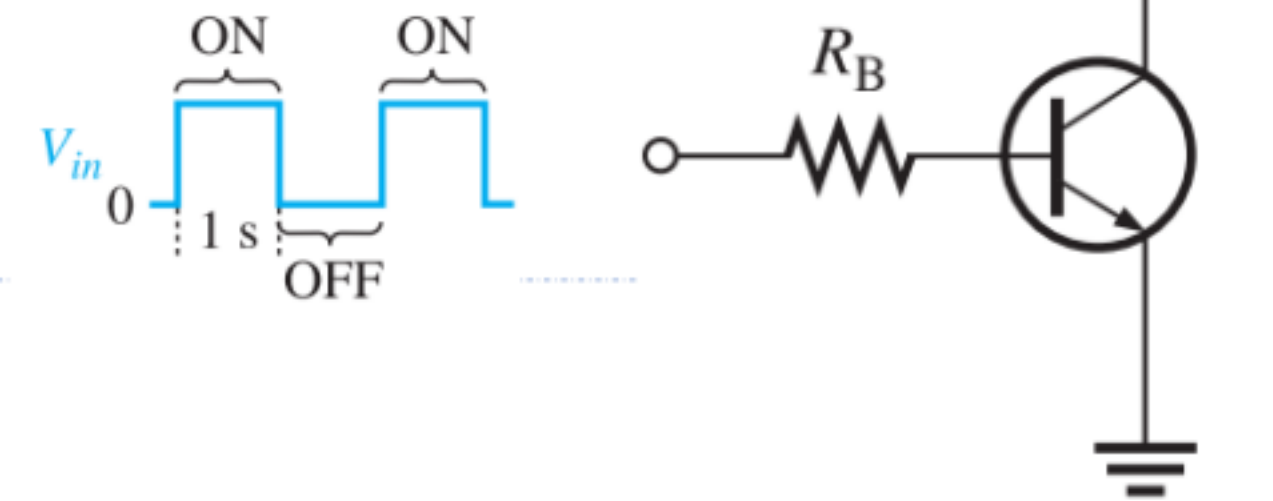
$$I_B (\text{actual}) = 2 I_B (\text{min})$$

$$= 2 \frac{I_C (sat)}{\beta_{DC}}$$

$$= 2 \left(\frac{V_{CC} - V_{CE} - V_L}{R_C} \right) \times \frac{1}{\beta_{DC}}$$

$$= 2 \left(\frac{9 - 0.2 - 1.6}{220} \right) \times \frac{1}{50} = 1.29\text{ mA}$$

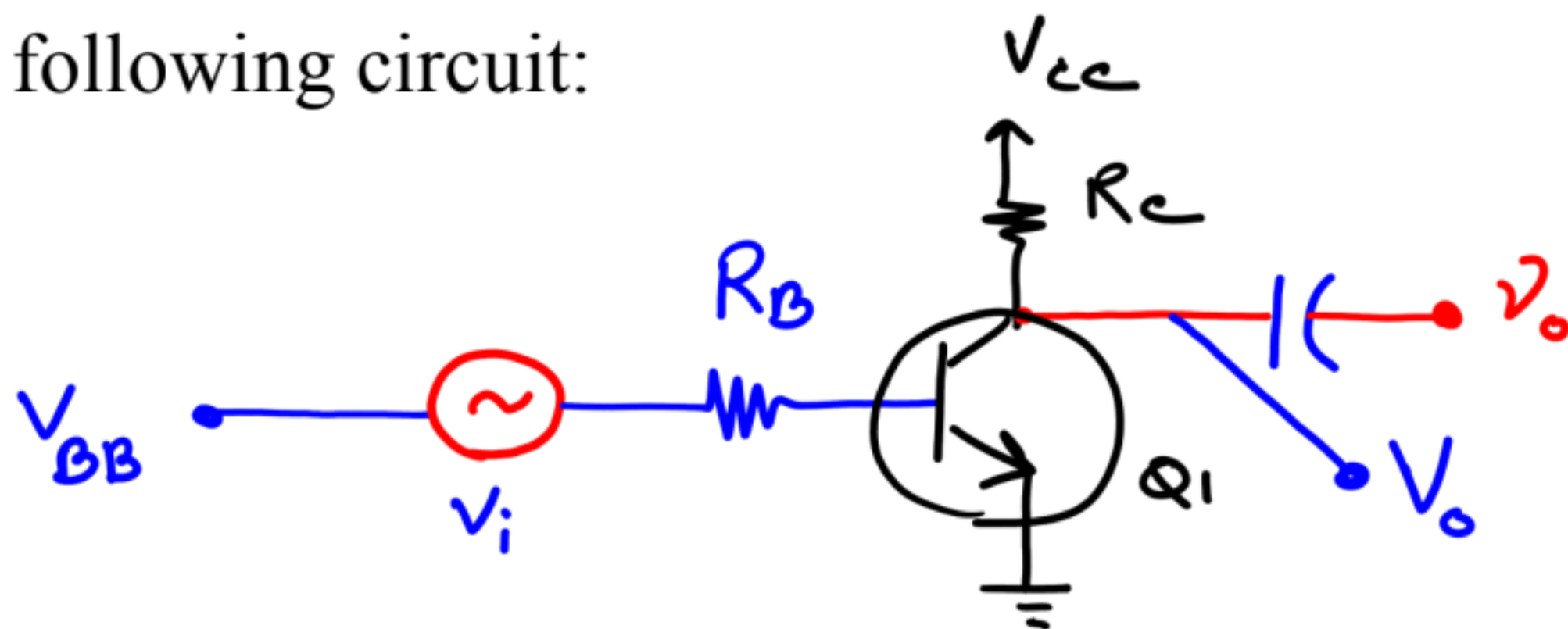
Ans. 5V



Necessity of active mode for BJT Amplifier circuit



- Consider the following circuit:

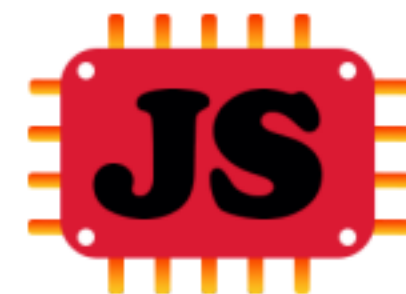


Case 01: V_{BB} high enough to make Q1 saturated

$$\begin{aligned}v_i &= 1 \angle 0^\circ \\V_o &= V_{CE(s)} = 0.2 \text{ V} \\v_o &= 0 \text{ V}\end{aligned}$$

Case 02: V_{BB} so low that Q1 is off

$$\begin{aligned}v_i &= 1 \angle 0^\circ \\V_o &= V_{CC} \\v_o &= 0 \text{ V}\end{aligned}$$

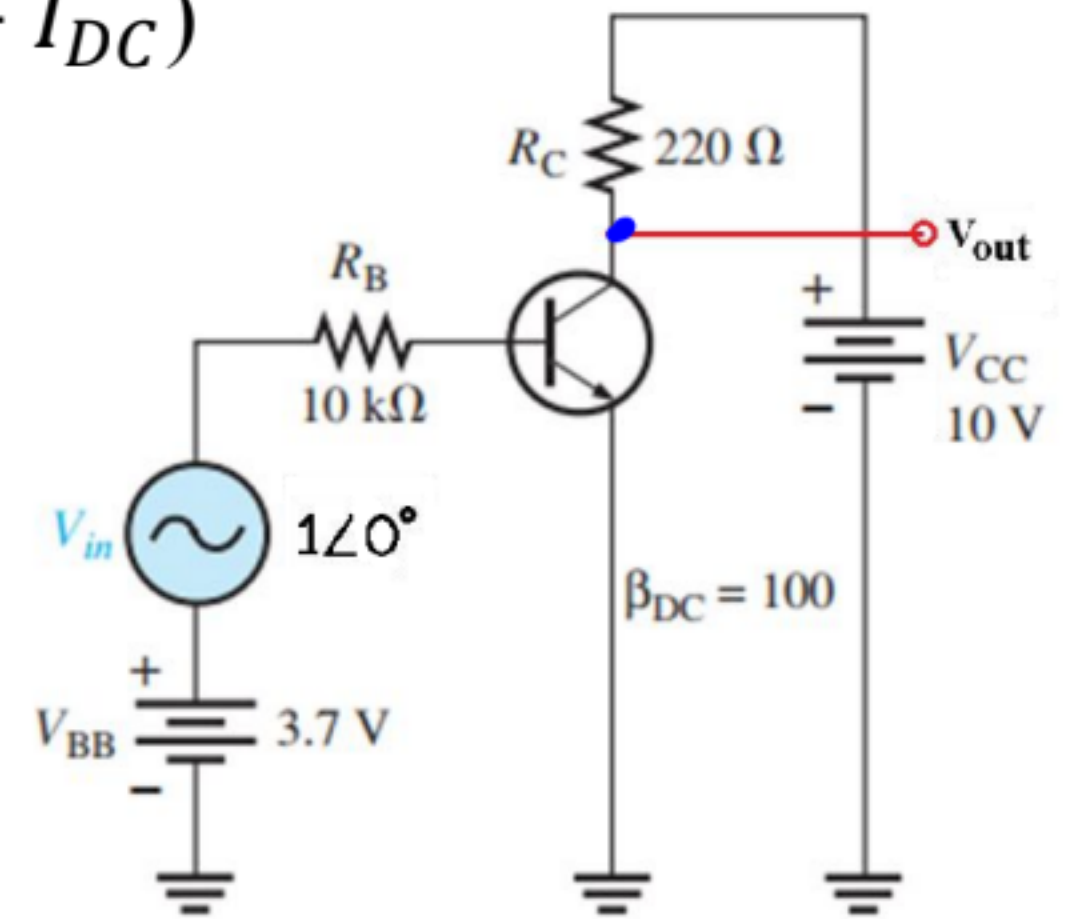


BJT as an Amplifier in Active mode

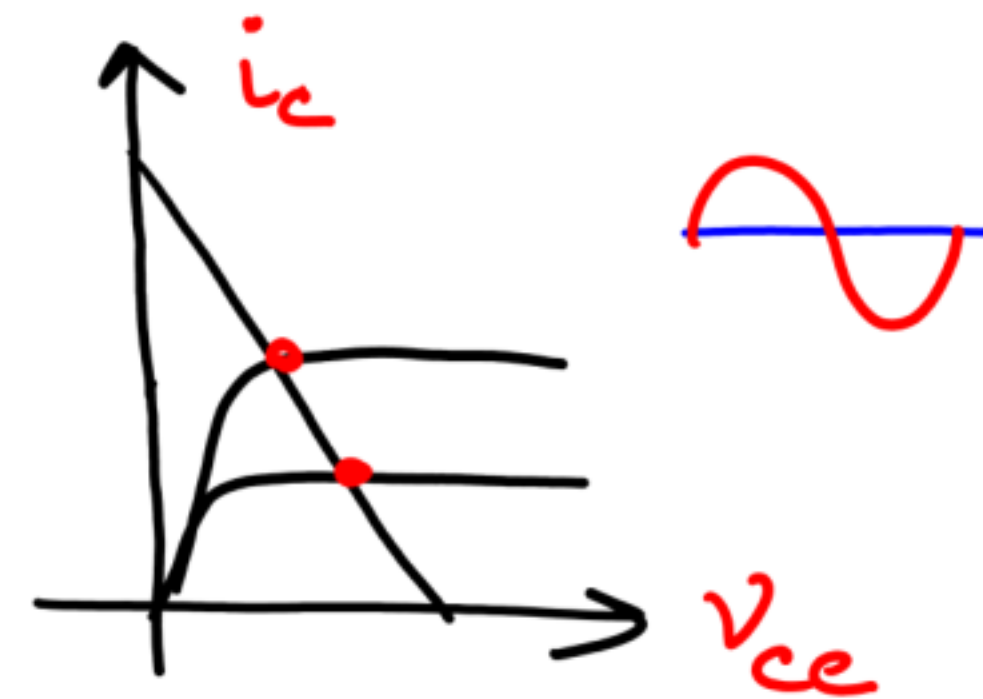


- In the given circuit, both AC & DC signals are present. ($I_{total} = i_{ac} + I_{DC}$)
- Input AC voltage, $v_{in} = 1 \sin(\omega t) = 1 \angle 0$
- Base current, $I_b = \frac{(V_{BB} + v_{in}) - V_{BE}}{R_B}$

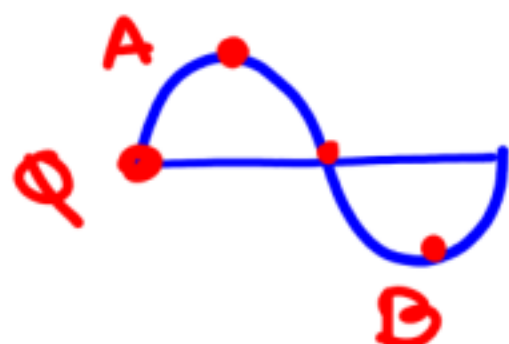
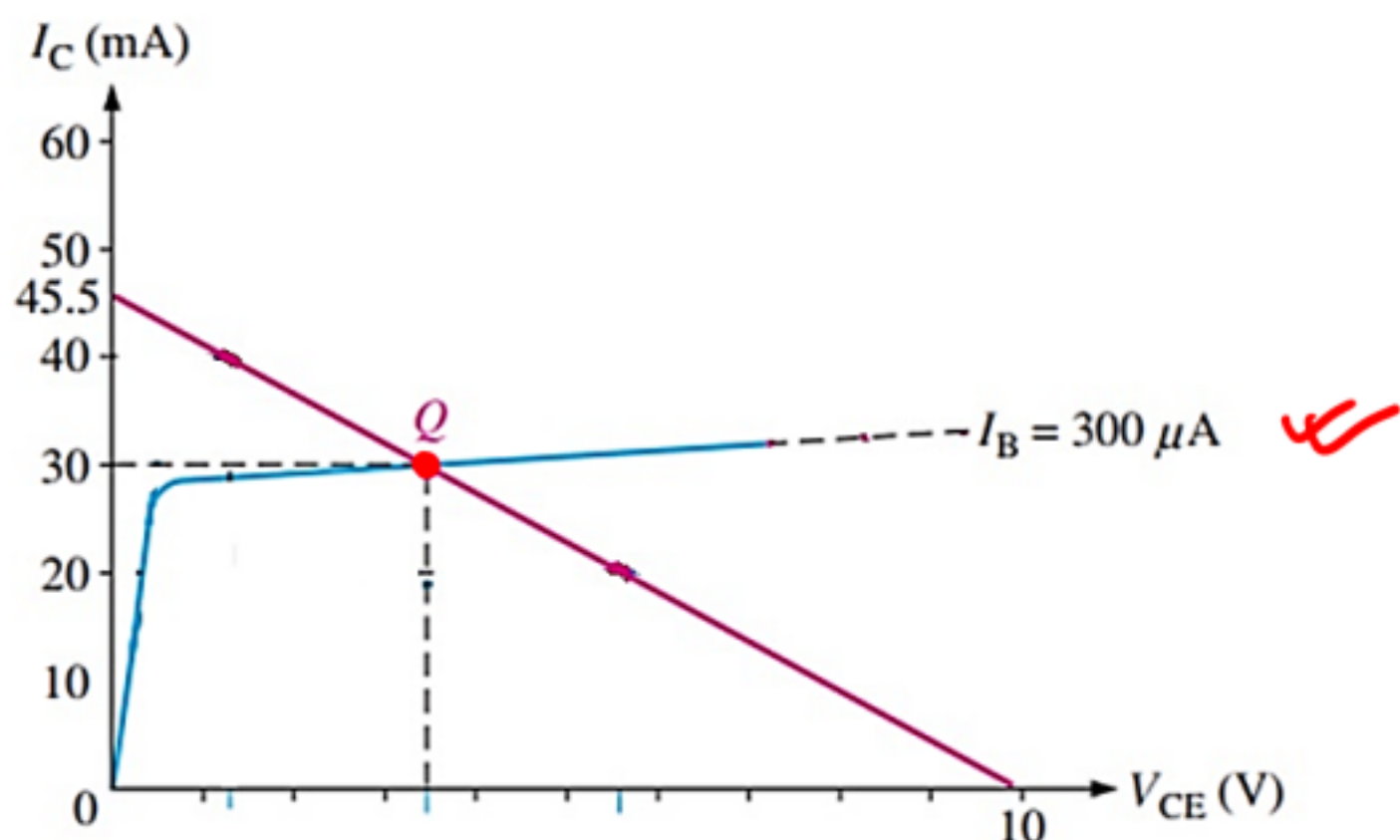
$$I_b(\max) = \frac{3.7 + 1 - 0.7}{10} = 400 \mu A \quad I_b(\min) = 200 \mu A$$



- Due to the voltage swing at base, base current varies
- Thus, Operating point is not fixed!
- DC load-line: $V_{CE} = V_{CC} - I_C R_C$
 $= 10 - 220 I_C$



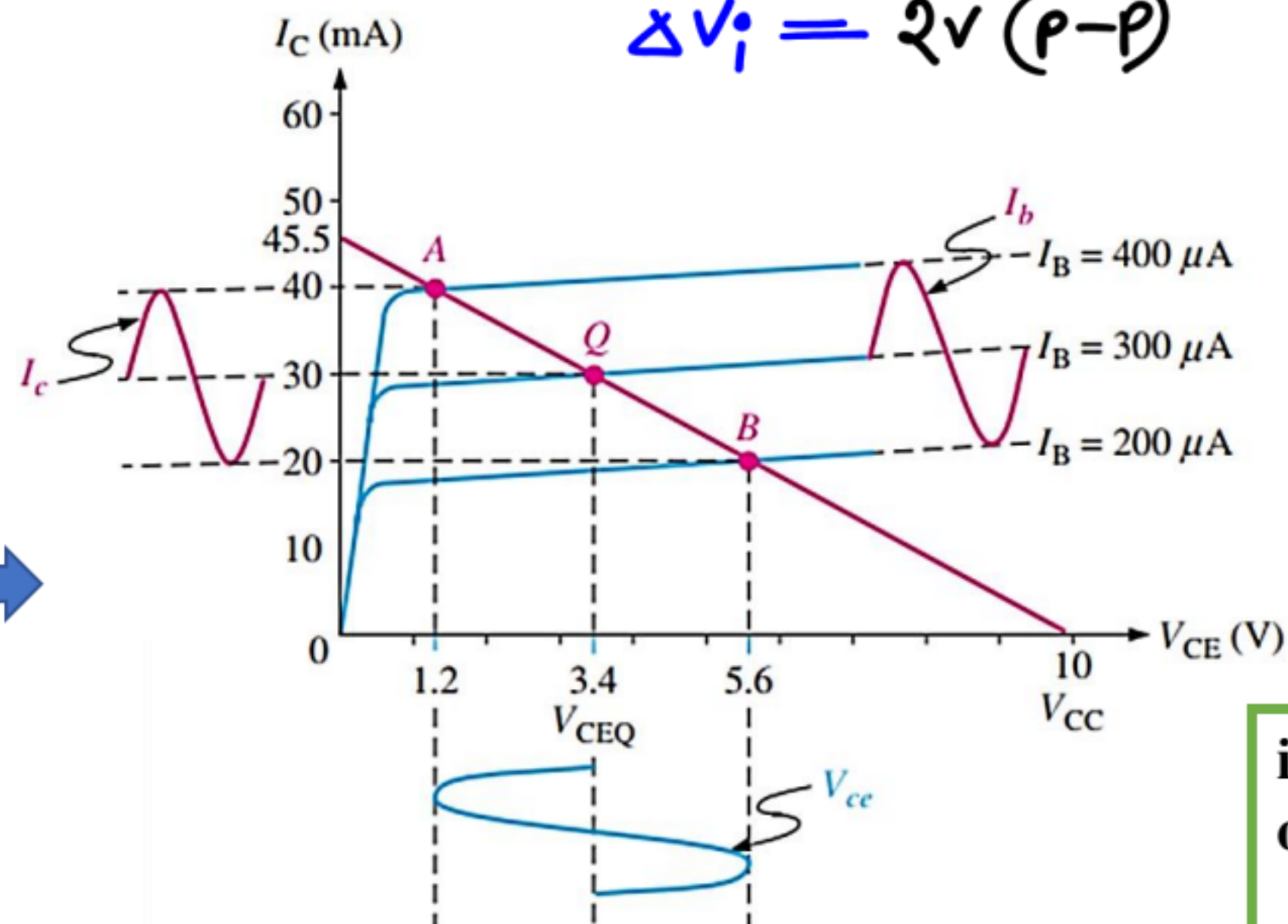
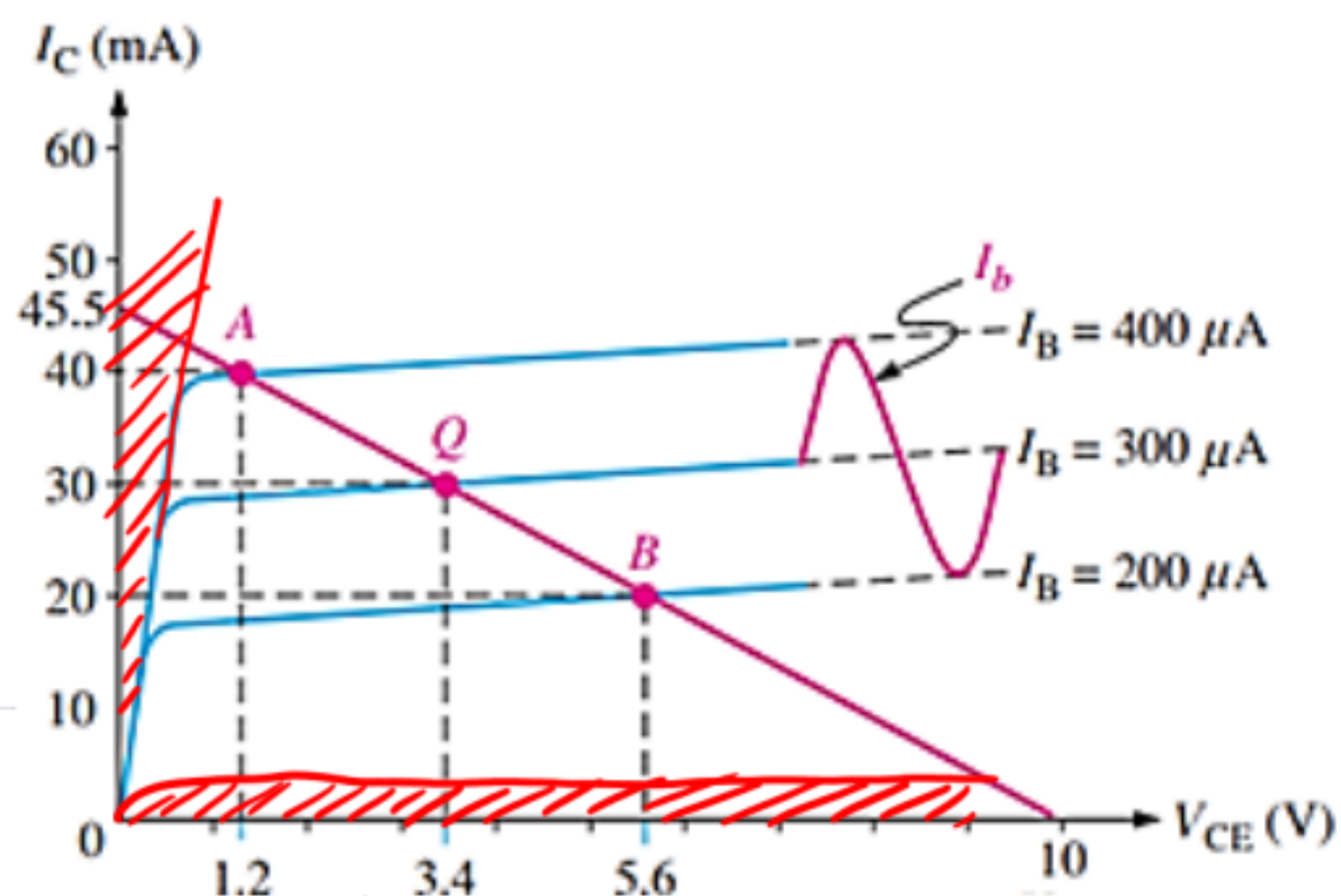
Let us fill-up the following table & draw the DC-load line on the characteristic curve



Operating point	v_{in}	Base current, I_b	Collector current, I_c	V_{CE}
A	1v	400 μA	40 mA	1.2v
Q	0v	300 μA	30 mA	3.4v
B	-1v	200 μA	20 mA	5.6v

$\downarrow$
 V_o

$$\Delta V_i = 2v \text{ (p-p)}$$



$$\Delta V_o = 5.6 - 1.2 = 4.4v \text{ (p-p)}$$

$$G = \frac{4.4}{2} = 2.2$$

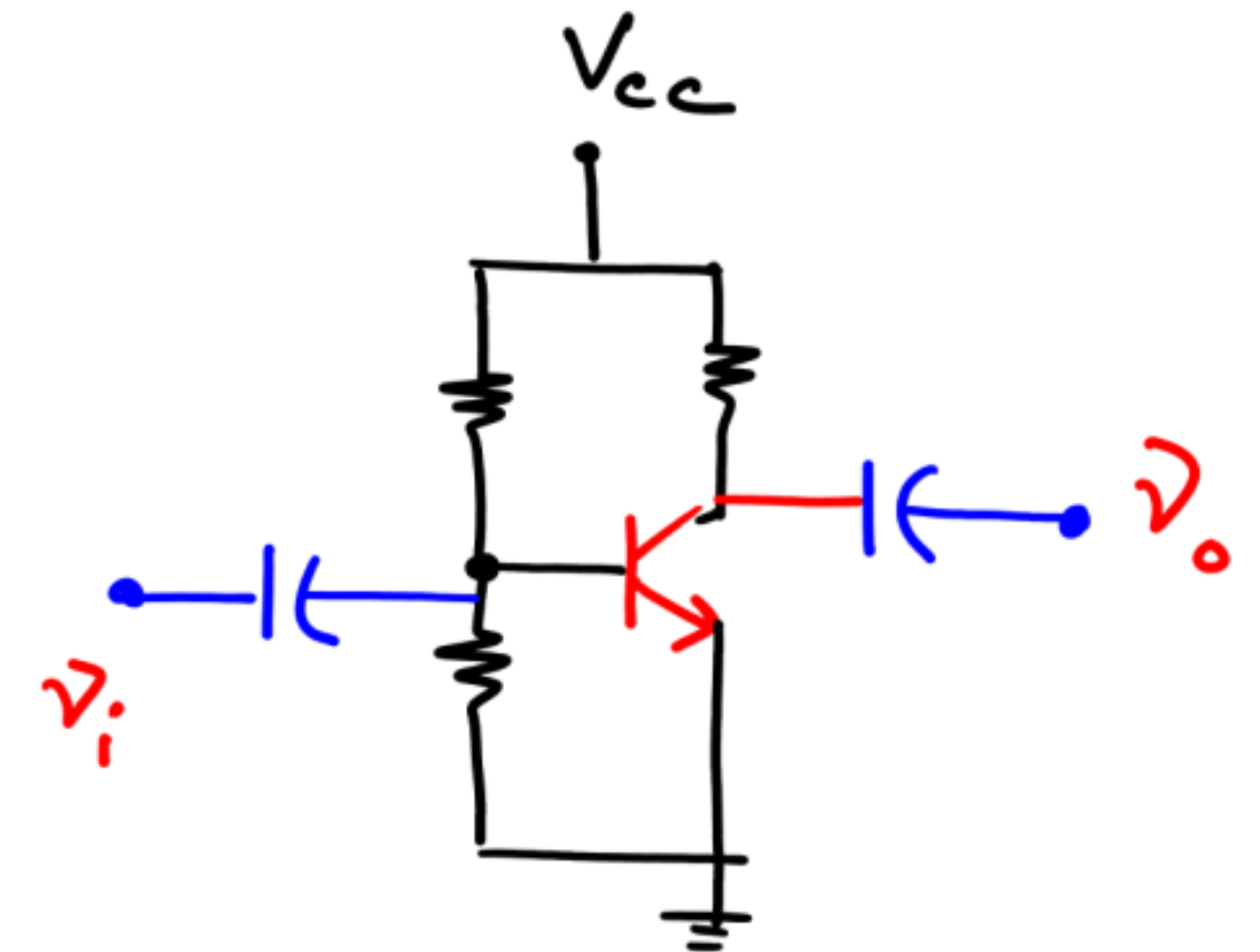
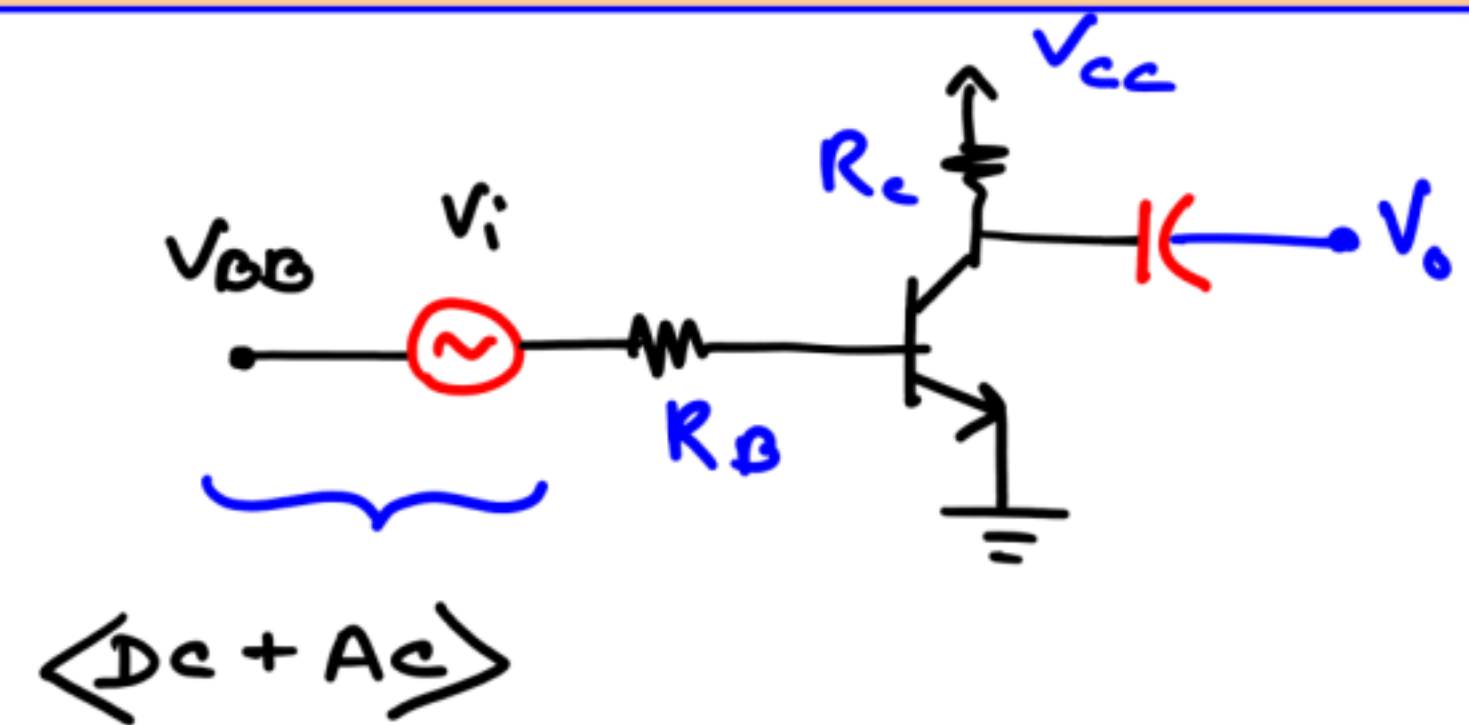
input swing = 2v p-p,
output swing = 4.4v p-p

Thus, signal amplified

Practical BJT Amplifier Circuit



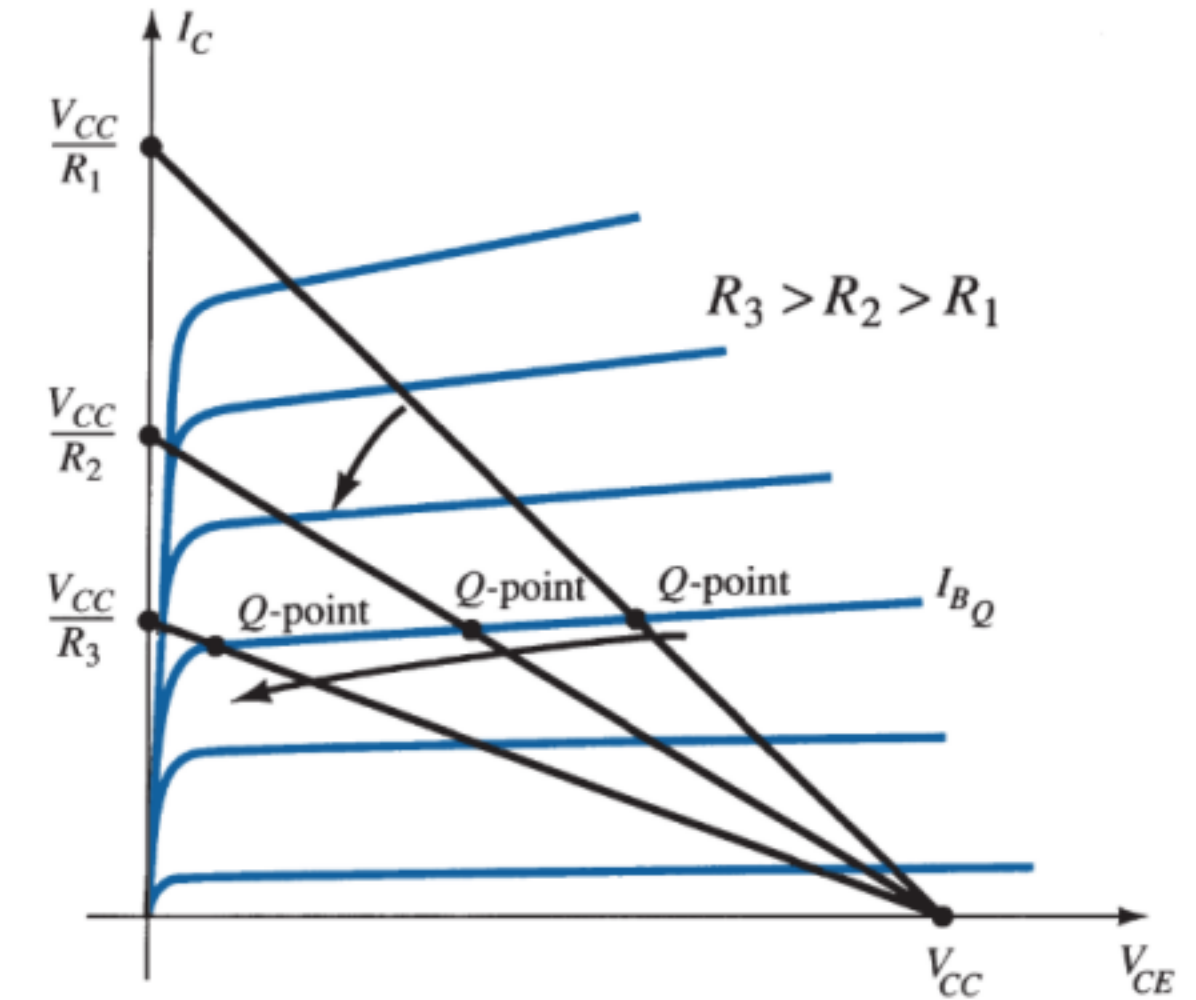
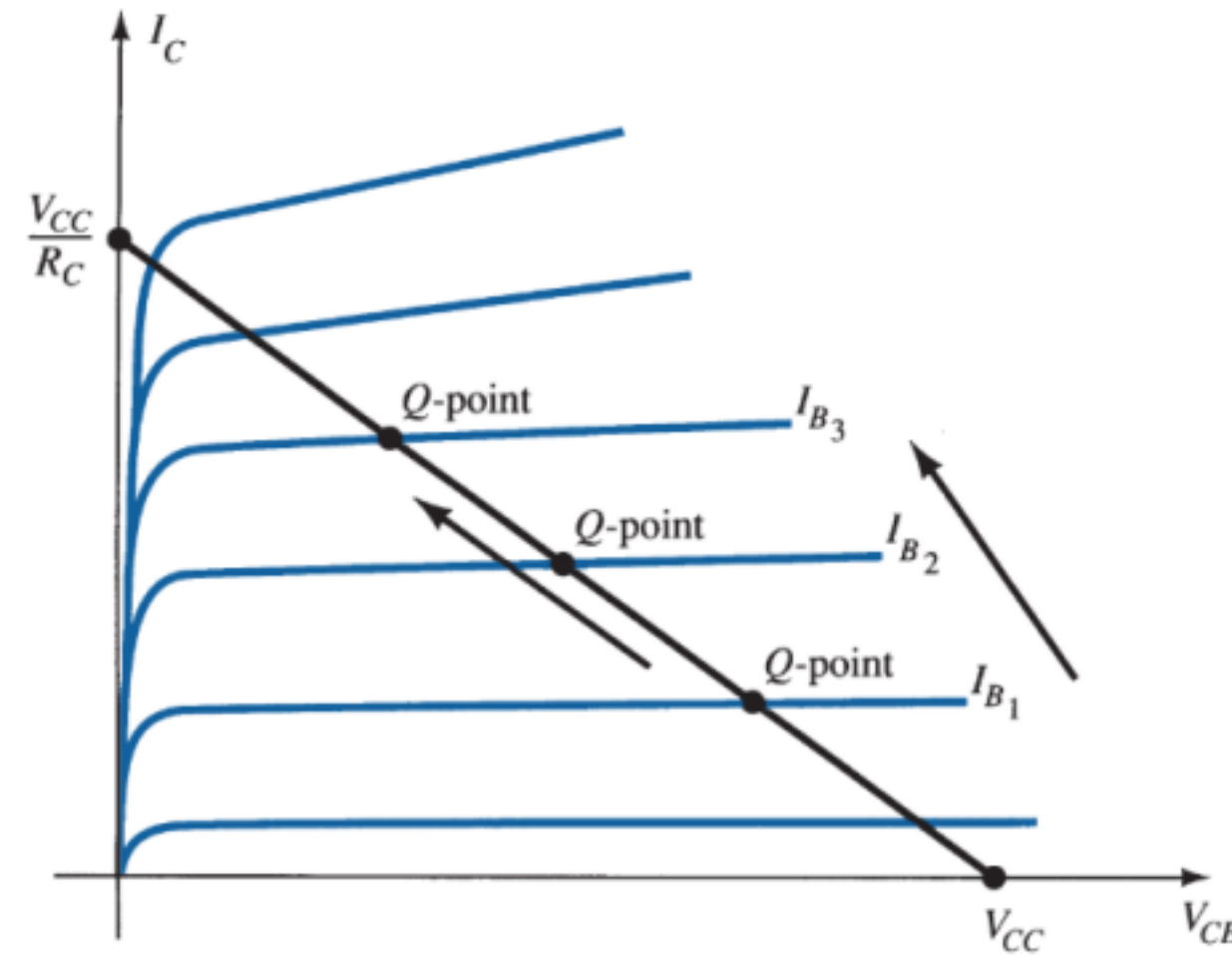
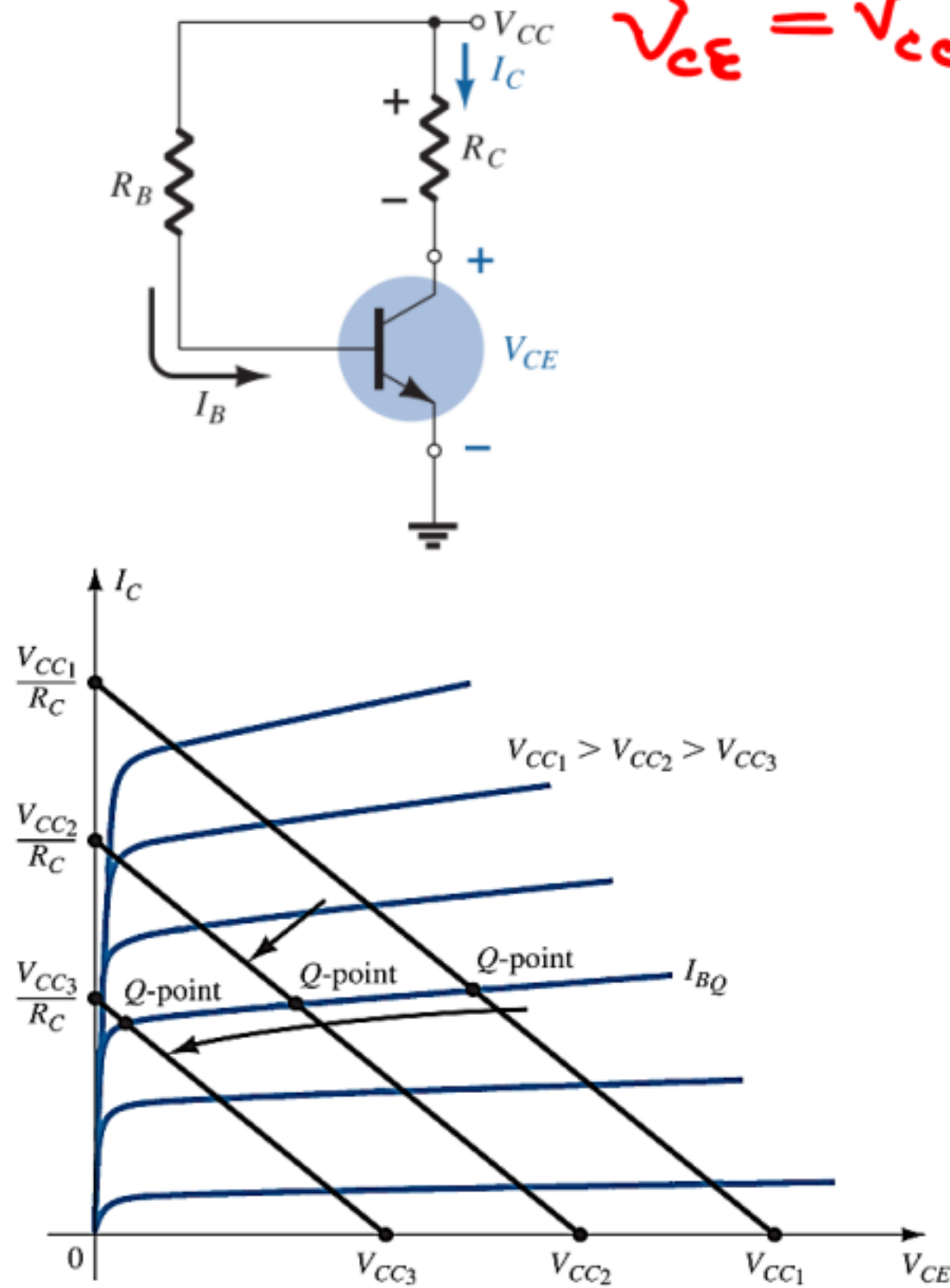
- In our assumed circuit, AC signal was superimposed with DC biasing voltage. This isn't practically done!
- AC and DC voltages are kept isolated, but they are placed in such a way that, both DC and AC voltages are found at the BASE.
- The Base makes use of the DC for Biasing in the active mode. And the Base also uses the AC signal as input for amplification!
- We only want amplified AC output at the Collector node of an amplifier circuit!
- Thus, DC blocking (also called coupling) capacitors are used again to block the DC part in the output!



Variation of Q due to change in circuit parameters



$$V_{CE} = V_{CC} - I_C R_C$$



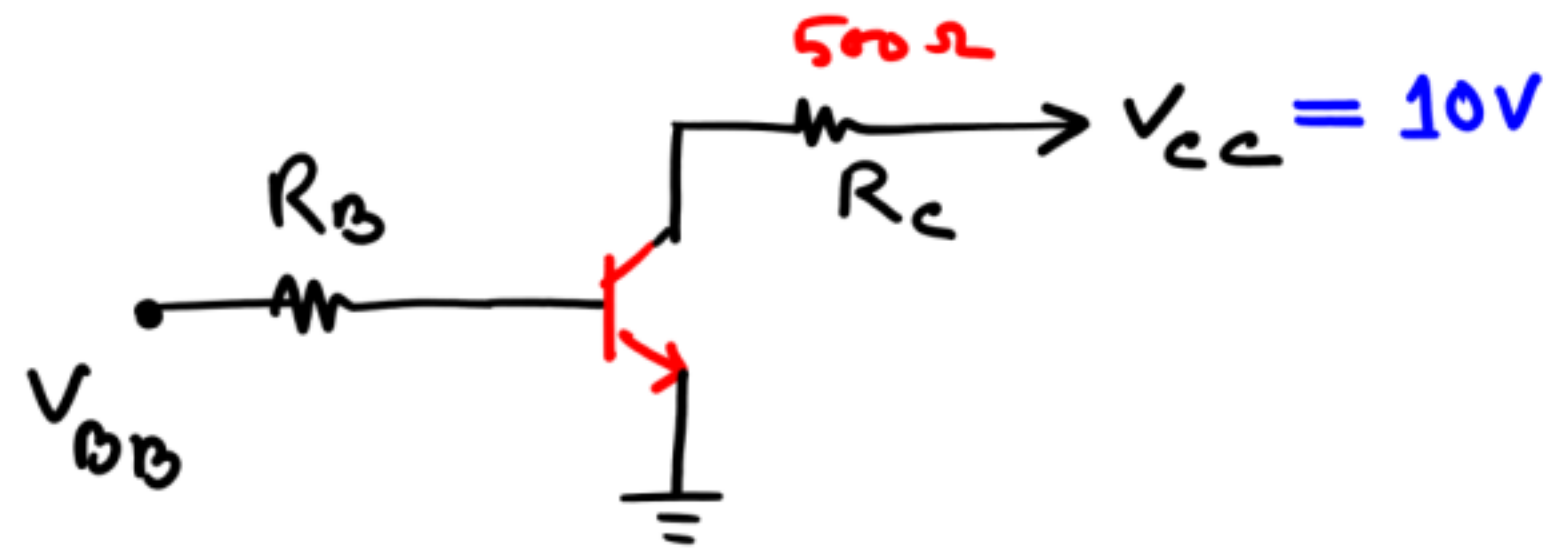
Q-point can change due to:

1. Change in base current
2. Change in R_C
3. Change in supply voltage, V_{CC}

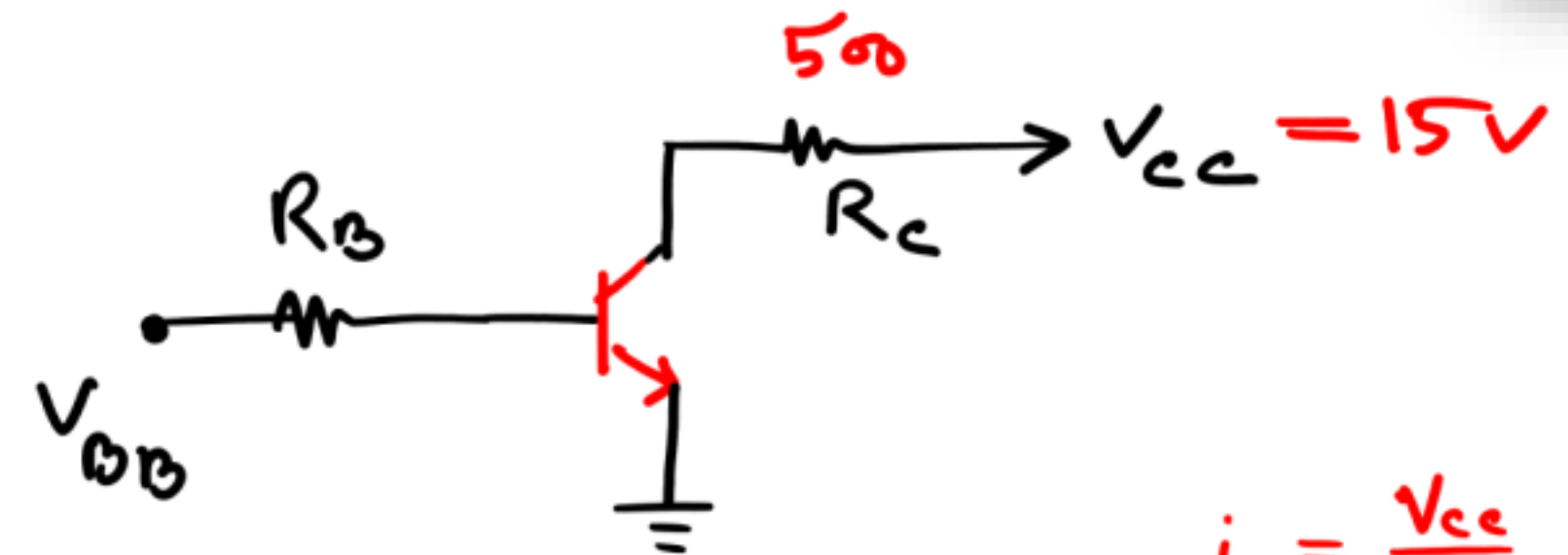
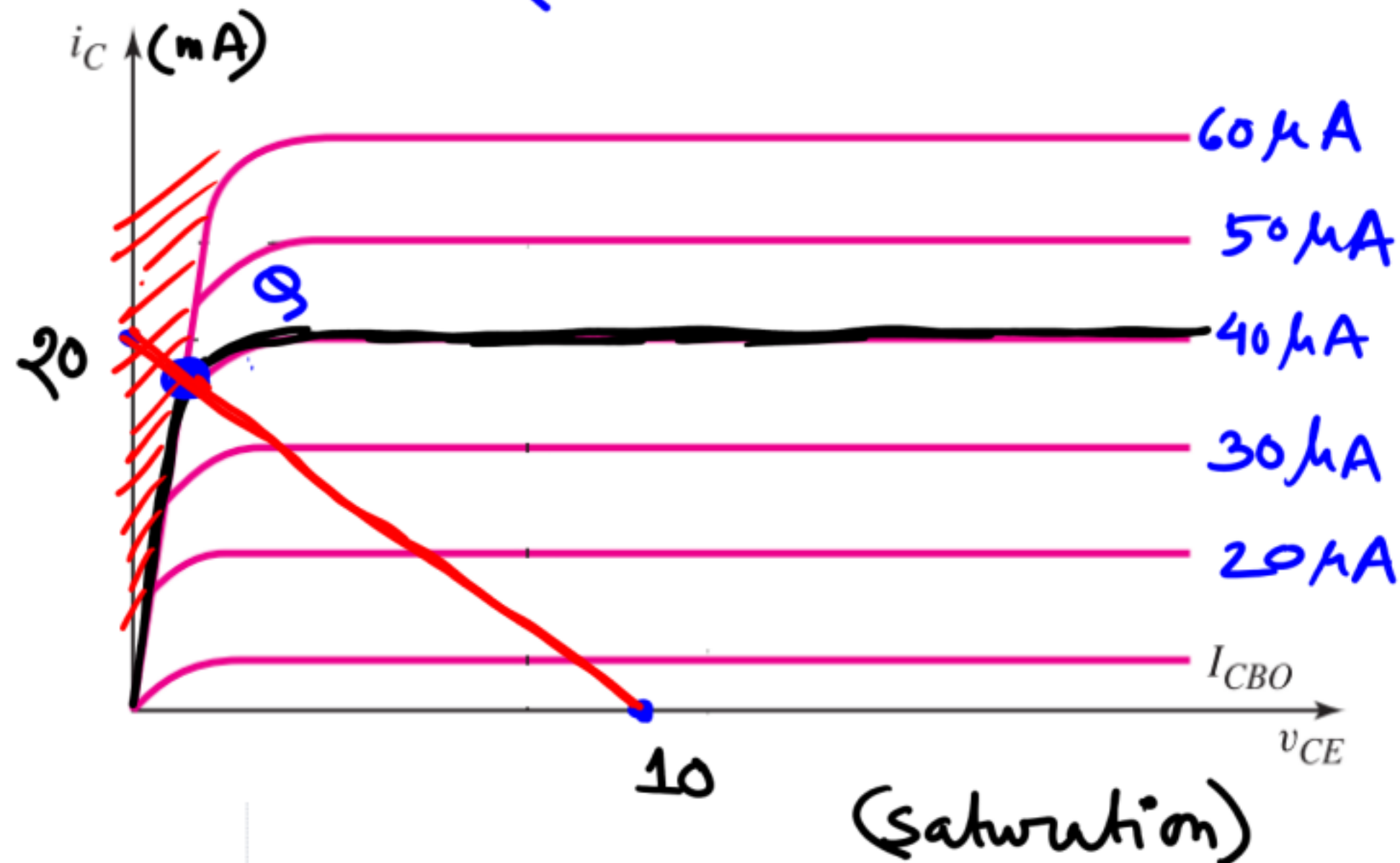
$$I_C = \frac{V_{CE}}{R_C}$$



Example of Q-point selection by setting different bias voltages

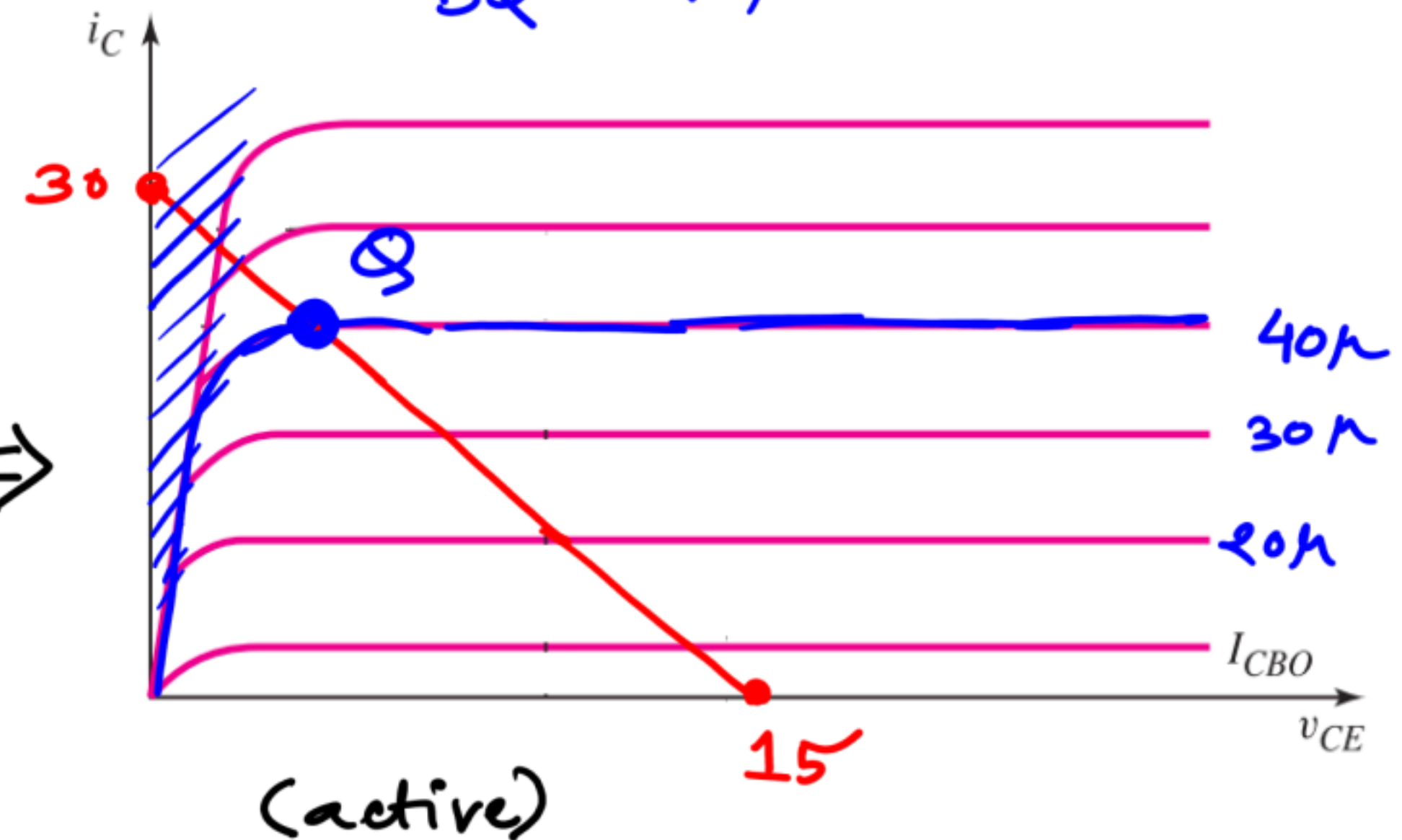


$$I_{BQ} = 40\mu A \text{ (given)}$$

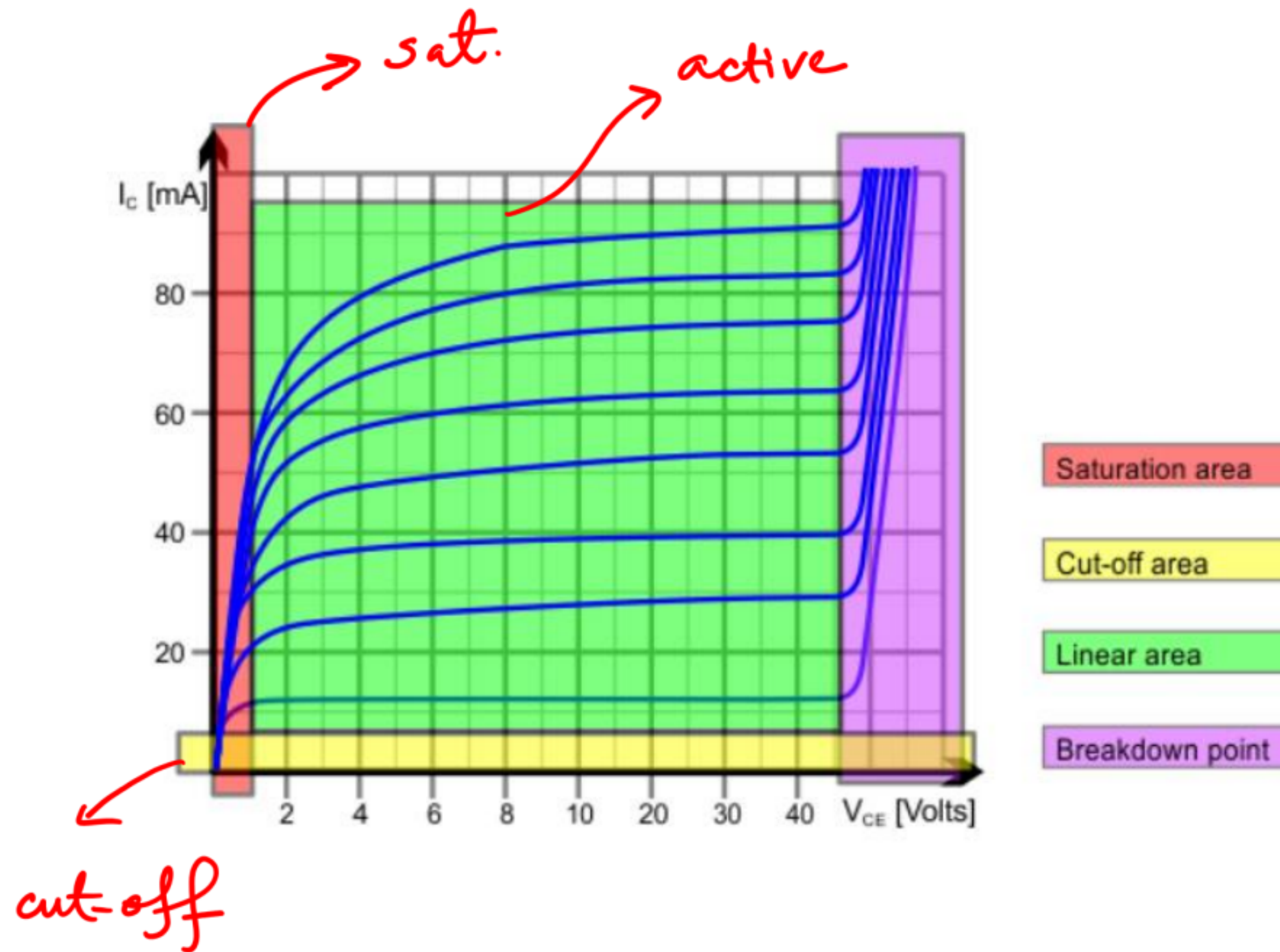


$$I_{BQ} = 40\mu A$$

$$i_C = \frac{V_{CE}}{R_C}$$



Different Regions of BJT in the characteristic curve

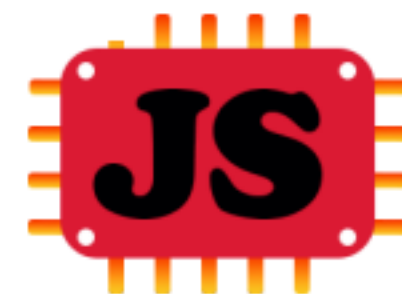
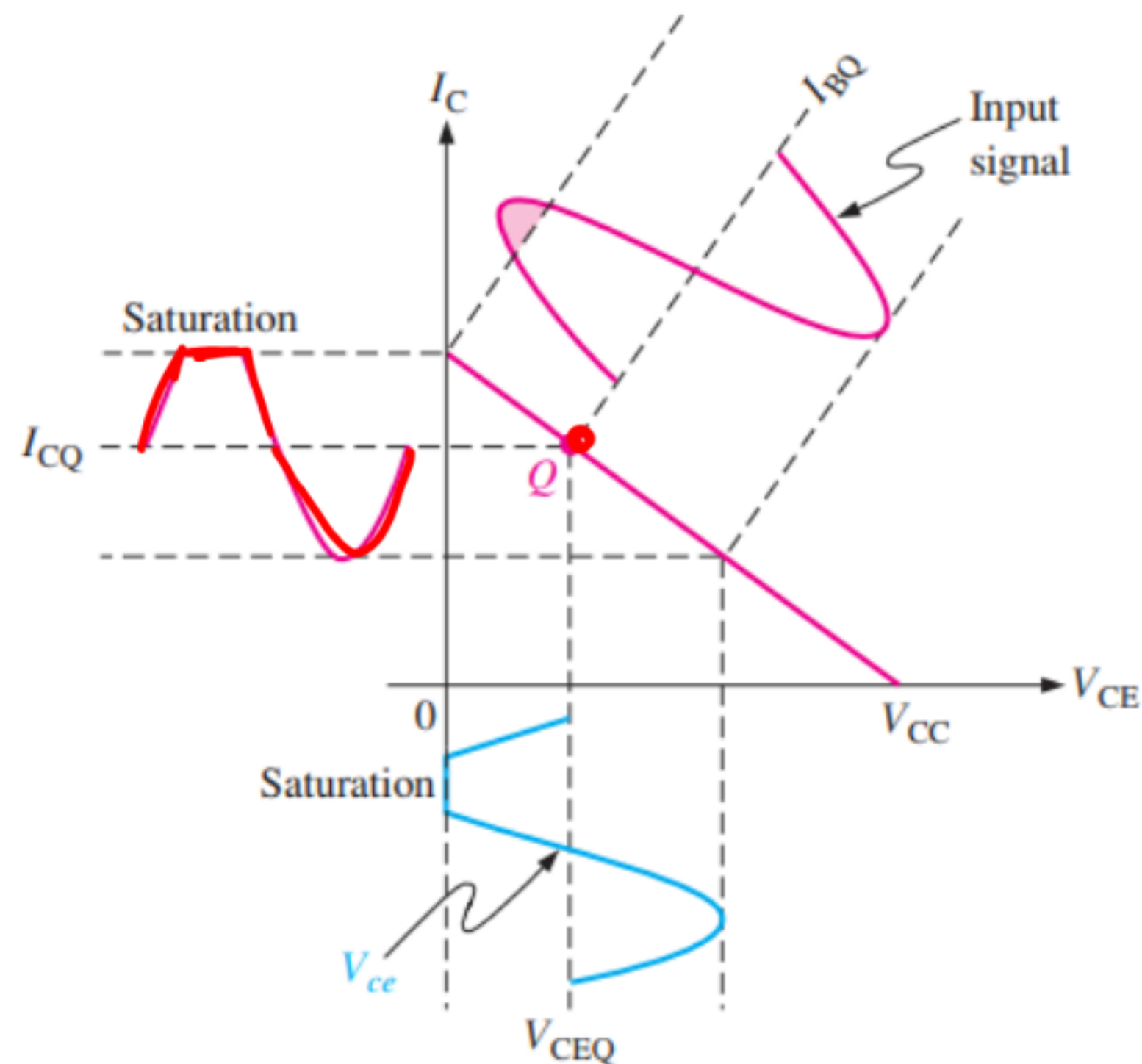
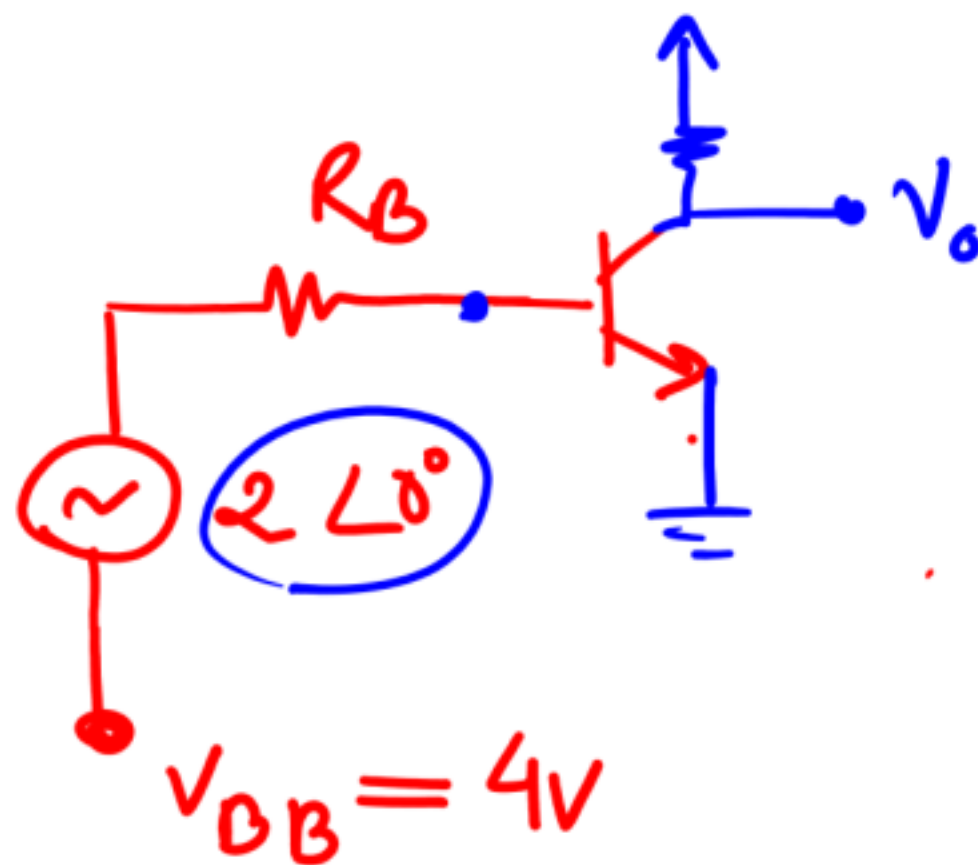


Waveform distortion due to wrong operating-point Q (1)



Case 01 :

- Q point is above the mid-point of DC-load line
- Transistor is driven into saturation due to high positive input voltage

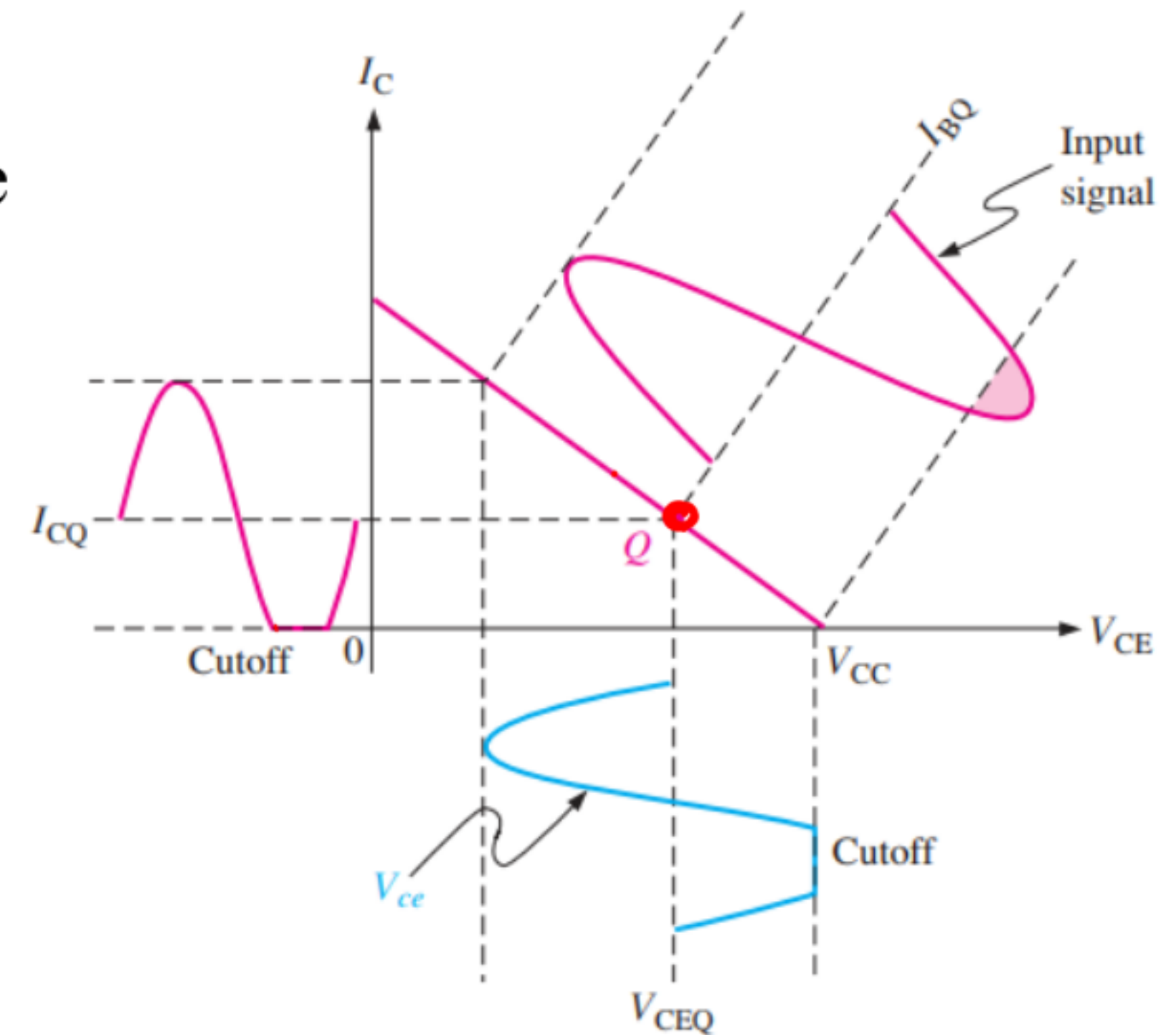
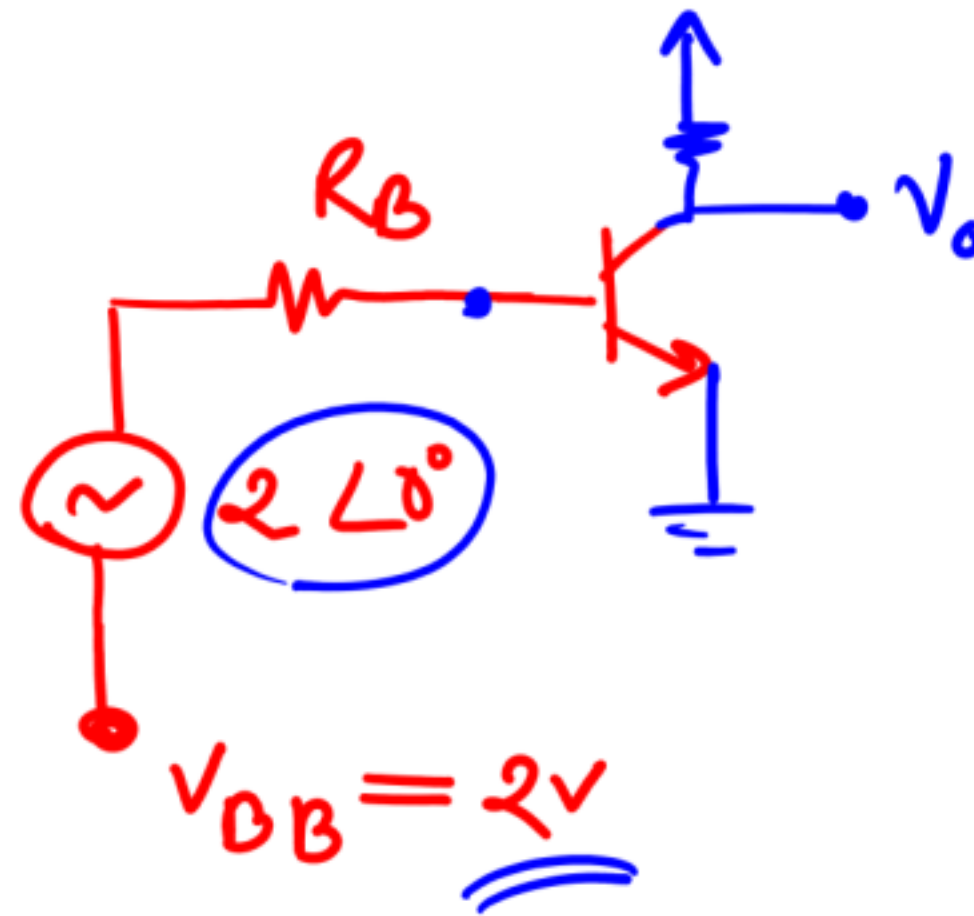


Waveform distortion due to wrong operating-point Q (2)



Case 02:

- Q point is below the mid-point of DC-load line
- Transistor is driven into cut-off due to high negative input voltage

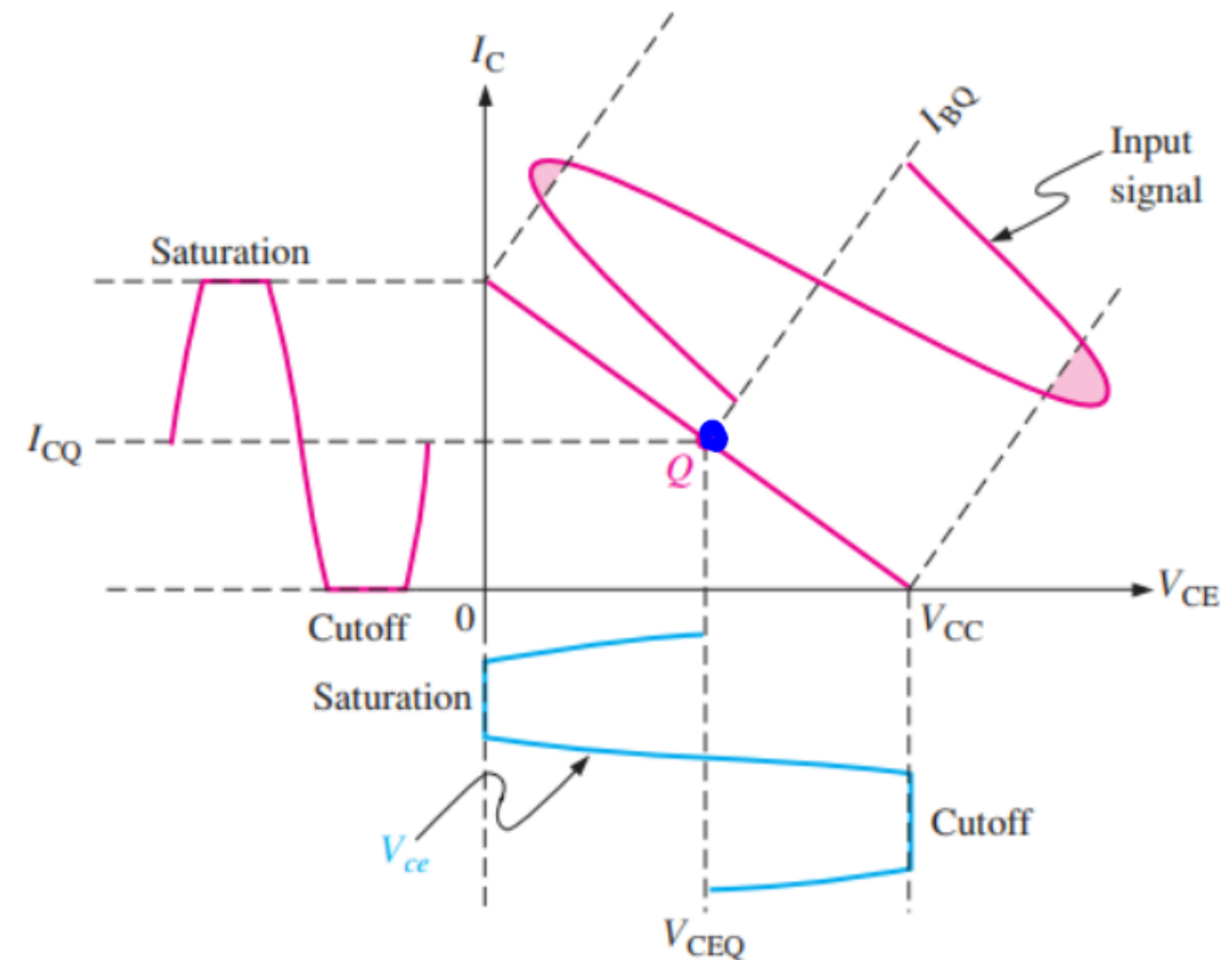
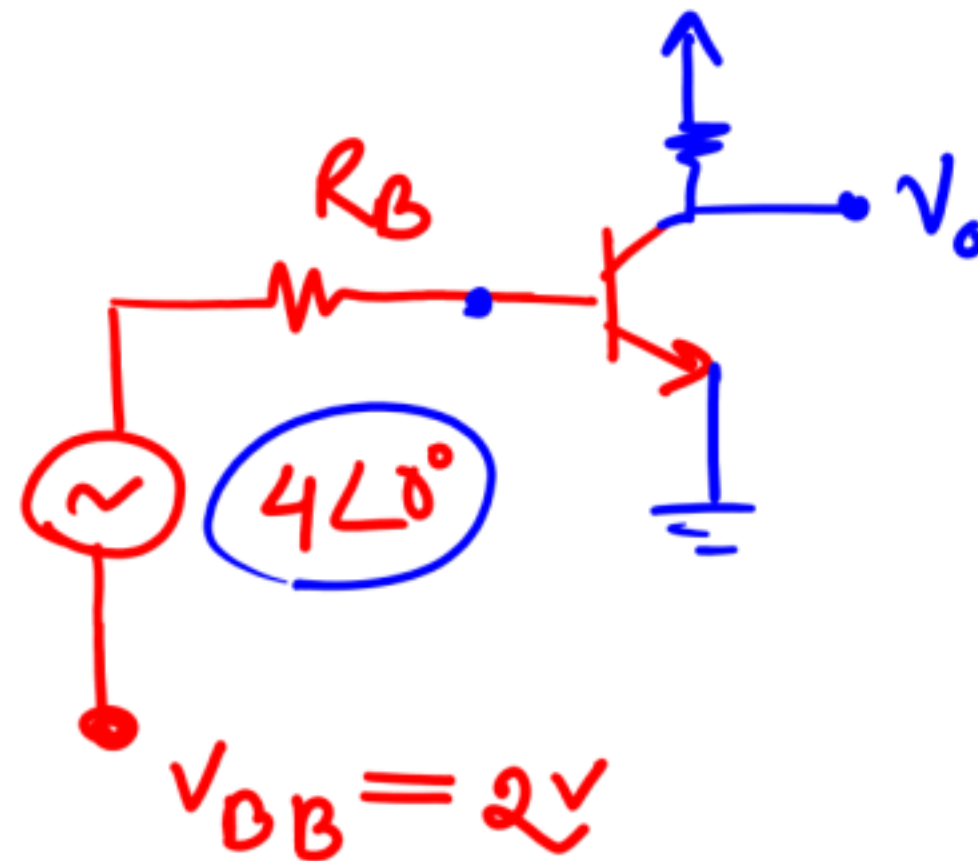


Waveform distortion due to large input AC voltage



Case 03 :

- Q point is at the mid-point of DC-load line
- Transistor is driven into both saturation & cut-off due to high positive input voltage.



Necessity of choosing proper Q-point



- **Linear Operation:** Correct Q-point keeps the transistor in the active region, ensuring linear signal amplification.
- **Avoids Distortion:** Prevents the transistor from entering saturation or cutoff, reducing signal clipping and distortion.
- **Maximizes Signal Range:** Allows maximum output signal swing without hitting power limits.
- **Stability:** Reduces sensitivity to temperature and parameter variations, ensuring consistent performance.
- **Efficiency:** Minimizes power dissipation, improving efficiency and reducing heat.
- **Longer Lifespan:** Maintains safe operating conditions, reducing thermal stress on the transistor.

